



AQ Software

Operation Manual

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Software version: AQ 2.4.13 or higher
Revision: 0
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Read the instructions in this manual carefully before installing or starting the system. SEAL Analytical can accept no liability for damages due to non-observance of this manual.

Read the software license agreement in the Appendix of this manual before you open the software package.

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Contents

AQ Software.....	I
Operation Manual	I
Contents.....	III
1 Introduction	1
1.1 Welcome	2
1.2 Conventions used in this Manual	3
2 Installation	5
2.1 System Requirements.....	6
2.2 Installation of USB Drivers for AQ2, AQ300 or AQ400.....	6
2.3 AQ Software Installation	7
2.4 Establishing USB Communications with the Analyzer.....	8
2.5 Software Upgrade: From Within AQ Software Series 2.x.x	8
2.6 Software Upgrade: From AQ2 Legacy Software	9
2.7 Starting AQ Software	10
3 AQ Software Guide.....	11
3.1 Introduction	12
3.2 AQ Software Features	12
3.2.1 AQ2, AQ300 and AQ400 Operation	Error! Bookmark not defined.
3.2.2 Fresh Look.....	12
3.2.3 Timing Optimization.....	12
3.2.4 Improvements to Reporting	12
3.2.5 Improvements to QC Configuration and Scheduling	13
3.2.6 Microsoft Windows® 7, 8, and 10.....	13
3.2.7 Turn Lamp Off Option (AQ400 & AQ300)	13
3.3 Setting Up & Initial Configuration	13
3.3.1 Initial Mechanical Setup.....	13
4 Parameters.....	15
4.1 Introduction	16
4.2 Test Parameters	16
4.2.1 Adding a Test.....	17
4.2.2 Naming a Test	17
4.2.3 Altering an Existing Test.....	17
4.2.4 Define Test Units	17
4.2.5 Define Decimal Places	17
4.2.6 Define Test Type	18
4.3 Main Test Parameters.....	18
4.3.1 Sample and Water Volumes.....	19
4.3.2 Diluent Location	19
4.3.3 Number of Mixes.....	20
4.3.4 Cuvette Primes	20
4.3.5 Cuvette Washes	20

4.3.6	Baseline on Wash	20
4.3.7	Reduction Time	20
4.3.8	Cadmium Flush Volume	20
4.3.9	Reaction Time	21
4.3.10	Wavelength	21
4.3.11	Polynomial Order	21
4.3.12	Repeat Delta Check	21
4.3.13	Auto Dilution Factor	21
4.3.14	Manual Dilution Factor	21
4.3.15	Low and High Linearity	22
4.3.16	Low and High Alert	22
4.4	Reagent Parameters	22
4.4.1	Adding Reagents to the Test Parameters	23
4.4.2	Setting Reagent Delays	23
4.4.3	Define the Blank Reagents	23
4.4.4	Adding a Reagent to the Reagent List	25
4.4.5	Adding a Reagent to a Reagent Set	25
4.4.6	Stability	26
4.4.7	Bar Codes	26
4.4.8	Removing a Reagent	26
4.4.9	Saving the Changes	26
4.5	Calibration Standard Parameters	26
4.5.1	Standards Information	27
4.5.2	All Standards	27
4.5.3	Auto Standards	28
4.5.4	Manual Standards	29
4.6	QC Pro Parameters	29
4.6.1	Auto-Dilution Insertion and Bracketing Controls	30
4.6.2	List of QC Pro Control Abbreviations	30
4.6.3	QC Pro Preferred Order	30
4.6.4	QC Pro Limits	31
4.6.5	QC Pro Location	31
4.6.6	QC Pro Failure Actions	31
4.6.7	QC Pro Frequency	31
4.6.8	Start Before	31
4.6.9	Duplicates	32
4.6.10	Spikes	32
4.6.11	Reduction Efficiency	33
4.6.12	Controls	33
4.7	Test Parameters Toolbar	33
4.7.1	Advanced Parameters	34
4.8	Saving Test Parameters	35
5	Normal Operation	37
	Switches	38
5.1	Daily Start-up	39
5.2	Scheduling a Tray	41
5.2.1	Choosing a Name and Number	42
5.2.2	Adding to the Schedule	43
5.2.3	Sample Type	43
5.2.4	Auto Fill with Unknowns	43
5.2.5	Sample ID	44

5.2.6	Sample Details.....	44
5.2.7	Adding a Test.....	44
5.2.8	Add Comments.....	44
5.2.9	Adding a Test Dilution, Cup Dilution or Sample Blanking	44
5.2.10	Show Standards and Controls.....	46
5.2.11	ID Generator	46
5.2.12	Inserting a New Line	47
5.2.13	Removing from the Schedule	47
5.2.14	Saving the Schedule.....	48
5.2.15	Saving a Schedule Template.....	48
5.2.16	Start a New Schedule	48
5.2.17	Open Saved Schedule.....	49
5.2.18	Open Template	49
5.2.19	Deleting a Template	50
5.2.20	Merging Schedules.....	50
5.2.21	Import a Schedule.....	50
5.3	Advanced Scheduler Operation.....	52
5.3.1	Using Copy and Paste Effectively	52
5.3.2	Manually Scheduling a Tray Effectively.....	53
5.4	Running a Tray	53
5.4.1	Choosing a Tray	53
5.4.2	Auto Standard Preparation.....	55
5.4.3	Reagent Volumes	55
5.4.4	Understanding the Run Screen	56
5.4.5	Run Screen: Status Window.....	56
5.4.6	Run Screen: Message Window	57
5.4.7	Run Screen: Result, T, and W.....	57
5.4.8	Run Screen: Results Window.....	58
5.4.9	Run Screen: Results, Colors and Error Codes.....	58
5.4.10	Automatically Stopping the Tray.....	58
5.4.11	Manually Stopping the Tray.....	59
5.4.12	Resuming a Tray	60
5.5	Sample Addition Mode.....	60
5.6	Data Review.....	63
5.6.1	Print Results by Test	64
5.6.2	Print Raw Data by Test.....	64
5.6.3	Print Statistics by Test	64
5.6.4	Accepting an Entire Tray	65
5.6.5	Deleting an Entire Tray	65
5.6.6	Reworking an Entire Tray	65
5.6.7	Accepting an Individual Data Point.....	66
5.6.8	Cancel an Individual Data Point	66
5.6.9	Reschedule an Individual Data Point	66
5.6.10	Rework by Test.....	67
5.6.11	Select a New Dilution Factor	67
5.6.12	Rescheduled with Cup Dilution.....	67
5.6.13	Reschedule with Test Dilution	67
5.7	Calibration Review	68
5.7.1	Selecting a Test	68
5.7.2	Selecting a Calibration.....	68
5.7.3	Understanding the Graph	69

5.7.4	Understanding the Results	69
5.8	Understanding the Calibration Status	69
5.8.1	Print Preview Calibration Report	70
5.8.2	Reworking the Calibration with a Different Polynomial Order	70
5.8.3	The Last 100 Calibrations	70
5.9	QC Trak	71
5.9.1	Select a Test	71
5.9.2	Select a Control	71
5.9.3	Selecting a Plot Type	72
5.9.4	Understanding the QC Table	72
5.9.5	Understanding the QC Plot	72
5.9.6	Understanding the QC Data	72
5.9.7	Print Preview	73
5.10	Reporting	73
5.10.1	Select a Tray	73
5.10.2	Select Multiple Trays	74
5.10.3	Clear Tray Selections	74
5.10.4	Customize Content	74
5.10.5	Select Logo	77
5.10.6	Colour Scheme	77
5.10.7	Add Format	78
5.10.8	Remove Format	78
5.10.9	Generate Report	78
5.10.10	Generate Merged Report	79
5.10.11	Generate Sample Summary	79
5.10.12	Generate Chronological Order Report	79
5.10.13	Test List Summary	79
5.10.14	Reports for Archived Trays	80
5.11	Tray Manager	80
5.11.1	Accepting Trays	80
5.11.2	Deleting Trays	81
5.11.3	Editing CSV Format	81
5.11.4	Result View	82
5.11.5	Tray Archives	83
5.11.6	Tray Log	84
5.11.7	Un-Accepting A Tray	84
6	Advanced Settings	85
6.1	Reagent Sets	86
6.1.1	Positioning Reagents	86
6.1.2	Reagent Volume	86
6.1.3	Naming the Reagent Set	87
6.1.4	Additional Reagent Information	87
6.1.5	Saving Changes	88
6.2	Test Profiles	88
6.3	Test Order	89
6.4	Advanced Parameters	90
6.4.1	System Parameters	90
6.4.2	Edit System Error Codes	92
6.4.3	Customize Error Codes	94
6.4.4	Log-In Administration	95

7	Extended Operation	99
7.1	Tray Logs	100
7.1.1	Viewing Real-Time Logs.....	100
7.1.2	Formatting Options	100
7.1.3	Export Options	102
8	Maintenance and Set Up.....	103
8.1	Introduction	104
8.2	Regular Maintenance	104
8.2.1	Turning Off Regular Maintenance Notifications.....	105
8.2.2	Daily Shutdown Maintenance	105
8.3	Maintenance	105
8.3.1	Initialize	105
8.3.2	Cuvette Functions.....	107
8.3.3	Spectro – Water Baselines	108
8.3.4	Spectro – Filter Definition	109
8.3.5	Spectro – Measurement Setup.....	110
8.3.6	Diluter	111
8.3.7	Cadmium Coil	112
8.3.8	Set Up – Motor Setup	113
8.3.9	Set Up – Sampler Test	116
8.3.10	Set Up – Setup Reagent Check (AQ2 and AQ400, some AQ300s)	118
8.3.11	Set Up – Aspirator Test	119
8.3.12	Set Up – Tray Test.....	120
9	Operation Modes.....	123
9.1	Introduction	124
9.1.1	Standard Mode (Run)	124
9.1.2	Touch-Screen Mode (Options)	125
9.1.3	STAT Mode.....	125
9.2	Touch- Screen Mode	125
9.2.1	Daily Start-Up	126
9.2.2	Schedule Tray.....	126
9.2.3	Run Tray	127
9.2.4	Accept Tray.....	128
9.2.5	Tray Log.....	129
9.3	STAT Mode	129
9.3.1	Starting STAT Mode	129
9.3.2	Scheduling the STAT Tray	130
9.3.3	Waiting for the Running Tray to Stop	130
9.3.4	Accepting the STAT Tray	130
9.3.5	Ending STAT Mode	131
9.4	Manual Read (AQ400 Only)	131
9.4.1	Setting up a Manual Read Test	131
9.4.2	Scheduling a Manual Read Test	132
9.4.3	Starting a Manual Read Test.....	133
9.4.4	Other Functions with a Manual Read Test	134
	Appendix A – Software Licence Agreement	135
	General Terms and Conditions for the Transfer of Software	136

1 Introduction

1.1 Welcome

Welcome to the SEAL Analytical AQ software package. AQ software is designed to control all SEAL Analytical AQ2, AQ300 and AQ400 systems. It also provides comprehensive and easy-to-use tools to process and evaluate the data collected from the analyzers.

This manual provides information and instructions about using the AQ software. We assume the reader to have a basic knowledge of chemistry and laboratory procedures. We also assume basic computer knowledge and knowledge of working with Microsoft Windows®.

The manual is divided into the following sections:

Section 1	Introduction
Section 2	Installation
Section 3	AQ Software Guide
Section 4	Parameters
Section 5	Normal Operation
Section 6	Advanced Settings
Section 7	Extended Operation
Section 8	Operation Modes
Section 9	Maintenance and Set Up
Appendix A	Software License Agreement

All of us at SEAL Analytical are committed to providing our customers with a high degree of after-sale support to ensure your complete satisfaction with our products and services.

SEAL has an international network of support centers and authorized service partners. Please visit our website www.seal-analytical.com to find the one nearest to you if you have any problems or questions.

1.2 Conventions used in this Manual

The following conventions are used:

Bold type	Field labels, e.g., Statistics
Bold + italic type	Buttons and menu options you can select e.g. <i>Settings</i> . Terms like <i>Settings – Test Parameters</i> means that you should select <i>Settings</i> from the Main Menu and then select <i>Test Parameters</i> from the options.
Typewriter text	File or directory names and text to be entered via the keyboard, e.g., c:\AQ

The following safety precautions are used:



Caution:

Is used in cases where damage to the software or the data may occur if you do not follow the precautions.



Note:

Indicates notes and additional information useful for the operation of the software and/or the analyzer.

For closing a window or leaving a program you have the following options:

OK	To save the entered data or to carry out a command and close the window / screen
Cancel	To close the window / dialog box <u>without</u> saving or <u>without</u> carrying out a command

2 Installation

2.1 System Requirements

The AQ Series Software Version 2.4.4 requires the following minimum specifications to run effectively:

- Windows® 7 or above
- Type-A USB Port
- 100MB free hard drive space

2.2 Installation of USB Drivers for AQ2, AQ300 or AQ400

The USB flash-drive for AQ Software Version 2.4.4 includes automatic selection of USB drivers, according to Microsoft Windows® operating system (Windows® 7, Windows® 8, and Windows® 10) and according to computer architecture, 32-bit, or 64-bit. Installation of AQ Software (section 2.3) may follow or precede installation of USB driver software.



Note:

Please read these instructions before connecting the USB cable between the analyzer and the computer. Installation of driver software requires (temporary) administrator privileges for Windows®.

To install USB driver software for AQ2, AQ300 or AQ400:

- Switch on the computer and log-in to Windows.
- Insert the USB flash-drive for AQ Software Version 2.4.4 to your computer's USB port. If a dialog box opens in response to the flash-drive, close it without taking action.
- Connect the USB communication cable to the analyzer:
 - If your AQ2 employs a steel USB adapter, mounted on the rear chassis, plug the cable into the standard-B receptacle on the side of the steel adapter.
 - If your AQ2 employs internally-mounted USB communications hardware, plug the cable into the mini-A receptacle located on the rear chassis.
 - AQ300: Plug the USB cable into the standard-B receptacle on the rear panel.
 - AQ400: Plug the USB cable into the standard-B receptacle on the rear panel.
- Connect the USB communication cable to a USB receptacle on the PC. Usually, Windows® responds with a temporary message that new hardware or a new USB device was found. Then a Found New Hardware Wizard begins. The Wizard dialog boxes and user-instructions displayed differ slightly among Windows® 7, Windows® 8 and Windows® 10.
 - Step 1: The first screen pertains to internet search for driver software. Mark the selection that prevents the Wizard from searching the internet automatically.
 - Step 2: Locate and click the Browse button. Navigate to the AQ software flash-drive, then into the subfolder named USB Drivers. Click OK
 - Driver installation proceeds automatically. The Wizard may declare that the best driver is already installed.

- If the Wizard asks for a specific file, direct the Wizard to that file by navigating into the i386 folder (32-bit PC) or the amd64 folder (64-bit PC).
- Repeat Steps 1 and 2.
- If Found New Hardware Wizard does not start or it terminates with an abnormal message, please refer to section 2.6.

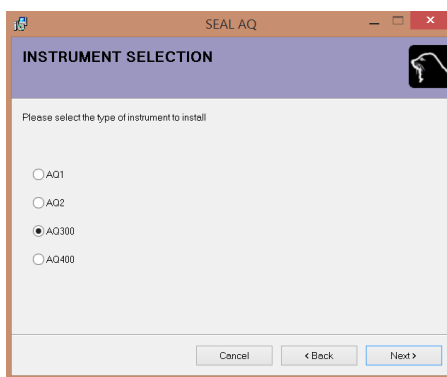
2.3 AQ Software Installation

**Note:**

If upgrading from an older version of software, please see section 2.5 or 2.6 first.

Install AQ Software Version 2.4.4 as described below:

- Insert the AQ software flash-drive into the computer's USB port. If a dialog box opens in response to the flash-drive, close it without taking action.
- Using Windows® Explorer, open the "Installer" folder on the software flash-drive. Locate the Windows® Installer Package file named "AQsetup". Right click this file and select *Install*.
- The Installer Package begins with two informative screens; read through the information and click *Next* for each one. The third screen shows C:\AQ as the default folder location for software installation to the hard drive. Accept the default location and select "Everyone" to allow all computer users to access AQ Software. The fourth screen (illustrated below) will ask which AQ analyzer will be used with the software. Select the corresponding analyzer and click *Next*.

**Note:**

If unsure of which type of analyzer, please contact SEAL Tech Support for assistance.

- The Installer Package creates the AQ software folder C:\AQ and transfers a copy of the AQ Application file, an icon file, HTML-type files, and PDF document files from the flash-drive into the C:\AQ folder.
- Once the software installation has finished, create a shortcut of the software on the desktop. To do this, use the Windows® Explorer and navigate to C:\AQ. Right click the new application file in the C:\AQ and select "Create Shortcut" and move the shortcut to the Desktop.

2.4 Establishing USB Communications with the Analyzer

During USB driver software installation, section 2.2, Windows® assigns a port number, typically 3 or 4, for USB communications with the analyzer. The port number assigned is found within Windows® Device Manager. The Windows® port number will then need to be entered within AQ Software system settings. This section presumes that the USB communication cable is connected between the analyzer and the computer, and that the analyzer is powered on.

- Depending on your Windows® operating system, open the Device Manager utility
 - Windows® 7: Left-click the Start menu, then locate My Computer in the right-hand column. Right-click on Computer and select Properties to display Control Panel → System and Security → System. In the left column of System, click Device Manager to open this utility.
 - Windows® 8: Move the mouse to the top right of the screen and select Settings. Open the Control Panel → System. On the left column, click Device Manager to open this utility.
 - Windows® 10: Go to the bottom left corner and click search. Search for Device Manager and open this utility.
- In Device Manager, locate and expand Ports. The control device for the AQ2 is named “USB Serial Port (COMx)”, and “TivaWare USB Serial Port (COMx)” for the AQ300 and AQ400. Record the value for x.
- Use the Desktop shortcut to launch the AQ Software. If an error message appears for the COM Port, ignore it and click OK to close. This brings up the AQ Software log-in screen. Enter username as password (case sensitive). Click OK to log-in.
- In the main screen for AQ Software, locate **Maintenance and Setup** tab. Select **Advanced Parameters—System Parameters**. To the first field, Com Port, select COMx from the dropdown menu, where the value of x was recorded using Windows® Device Manager. Click OK to save this setting.

2.5 Software Upgrade: From Within AQ Software Series 2.X.X

Upgrade AQ version 2.0 to AQ version 2.X:

Upgrade of AQ Software within AQ version 2 requires a swap of the AQ application file. Archived data, test and reagent parameters, user accounts and calibrations are preserved.

- Log-in to AQ Software. Archive or delete all trays that appear in New and Current Trays, as listed in Tray Manager of AQ software. Close AQ software.
- Navigate to the AQ Software folder on the computer, usually found at C:\AQ.
- Copy the AQ folder on the C: Drive and paste the AQ folder into a different location on the computer (i.e., in documents or on the Desktop) and rename as “AQ_backup + the current date” or something similar.

- Inside AQ Software folder are numerous subfolders, including \trays. Delete all contents of the subfolder \trays, but not the folder itself.
- Delete or rename the current AQ Application file. Delete any shortcuts to the current AQ software, from within the software folder, C:\AQ, and from the Windows® Desktop.
- Copy the new AQ Application file to the software folder, C:\AQ.
- Create a shortcut to the new AQ application file by right clicking the new application file in the C:\AQ and select "Create Shortcut" and move the shortcut to the Desktop.

2.6 Software Upgrade: From AQ2 Legacy Software

Upgrading from AQ Software version 4.x.x or version 1.x.

Depending on length of service for your AQ and rate of archiving data from AQ trays, the AQ Software folder, C:\AQ, will have grown from a few gigabytes (GB) to tens of GB. Therefore, SEAL Analytical recommends upgrade to AQ Software as a fresh software installation, leaving the previous installation of AQ2 Software undisturbed with prior test parameters and archived data left intact.

Upgrade to AQ Software version 2 from AQ2 version 4.x.x or AQ version 1.x requires uninstalling the existing software prior to installing the new software.



Note:

Test Parameters and Log-in information will need to be entered manually into the AQ version 2 software.

- Power up the analyzer and log-in to AQ2 Software version 4.x.x or AQ version 1.x. Verify USB communications by selecting **Settings—Maintenance** to initialize the analyzer.
- While in the software, go to **Settings—Advanced—System Settings**. Then select Run in Development Mode.
- Close AQ2 version 4.x.x software or AQ version 1.x and delete the AQ shortcut from the Desktop.
- Copy the existing AQ folder found on the C:\ drive.
- Paste the AQ folder into a different location on the computer (i.e., in documents or on the Desktop) and rename as "AQ_backup + the current date" or something similar.
- Delete the AQ folder from the C:\ drive.
- Uninstall the AQ Software file from "Add or Remove Programs"
 - For example, in Windows® 10, go to Control Panel, then Programs and Features. Click on AQ2 Software or AQ software and select Uninstall
- Ensure the USB communication cable from the instrument is connected to the PC.
- In Device Manager, locate and expand Ports. The control device for the AQ2 is named "USB Serial Port (COMx)" and "TivaWare USB Serial Port (COMx)" for the AQ300 and AQ400.
- Right click on the COMx and select Update Driver. Then follow the subsequent steps:

- Step 1: The first screen pertains to internet search for driver software. Mark the selection that prevents the Wizard from searching the internet automatically.
- Step 2: Locate and click the Browse button. Navigate to the AQ software flash-drive, then into the subfolder named USB Drivers. Click OK
 - Driver installation proceeds automatically. The Wizard may declare that the best driver is already installed.
 - If the Wizard asks for a specific file, direct the Wizard to that file by navigating into the i386 folder (32-bit PC) or the amd64 folder (64-bit PC).
- Repeat Steps 1 and 2.
- Complete AQ Software installation according to sections 2.3 and 2.4.

2.7 Starting AQ Software

The installation application creates a Windows® Desktop shortcut named 'AQ.' Double-click this shortcut to launch the application and to reach the AQ Software log-in screen.

3 AQ Software Guide

3.1 Introduction

This software guide is for AQ Series Software Version 2.4.4 and above. It provides a complete view of the software operation of the AQ2, AQ300, and AQ400 Discrete Analyzers. For help on hardware issues, see the Operator Handbook.

3.2 AQ Software Features

There are some significant changes from AQ Software version 1.x to AQ 2.x Software. This section will give you a brief overview of the differences. If you need further information please refer to the appropriate section in the main volume of the manual.

3.2.1 Fresh Look

Even though the operation of AQ Software is very much the same as the previous AQ software, the icons and screens have been updated to a modern look to better help identify the functionality of each feature.

3.2.2 Timing Optimization

The software has optimized the timing of instrument operation, allowing for faster test times. It will also provide an accurate time for completion of the test.

3.2.3 Improvements to Reporting

- Split Screen preview of the report before it is generated
 - % Recovery and %RPD can now appear on the report
 - Manual Dilutions now appear on the report.
 - Comments can be added to the report
-

3.2.4 Improvements to QC Configuration and Scheduling

- **Spikes (Auto and Manual)** and **Spike Duplicates** calculations can be done.
- Cadmium Coil Efficiency check can now be scheduled and calculated.
- QC Names can now be edited in the Test Parameters page.
- Cups can be scheduled multiple times for duplicates, spikes, and spike duplicates.
- QC order for automatically populated QCs can now be controlled

3.2.5 Microsoft Windows® 7, 8, and 10

AQ software is compatible with Microsoft Windows® 7, Windows® 8, Windows® 10, and Window® 11.

3.2.6 Turn Lamp Off Option (AQ400 & AQ300)

The AQ400 and AQ300 also have the capability to turn the lamp off while the instrument is on and after a run has been completed.

3.3 Setting Up & Initial Configuration

When an analyzer is first run with a PC, some initial settings may need to be adjusted. Some of these may have already been set during production.

3.3.1 Initial Mechanical Setup

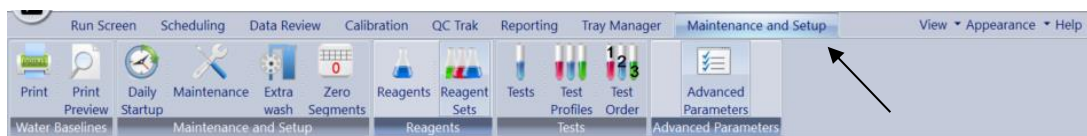
The AQ2, AQ300, and AQ400 analyzers store settings on-board its EEPROM. These settings will have been matched to the analyzer at the factory. The settings relate to the various hardware modifications that can be performed on an analyzer.

The settings can all be found on the Maintenance dialog box. For more information on the other options available, see the Maintenance section.

4 Parameters

4.1 Introduction

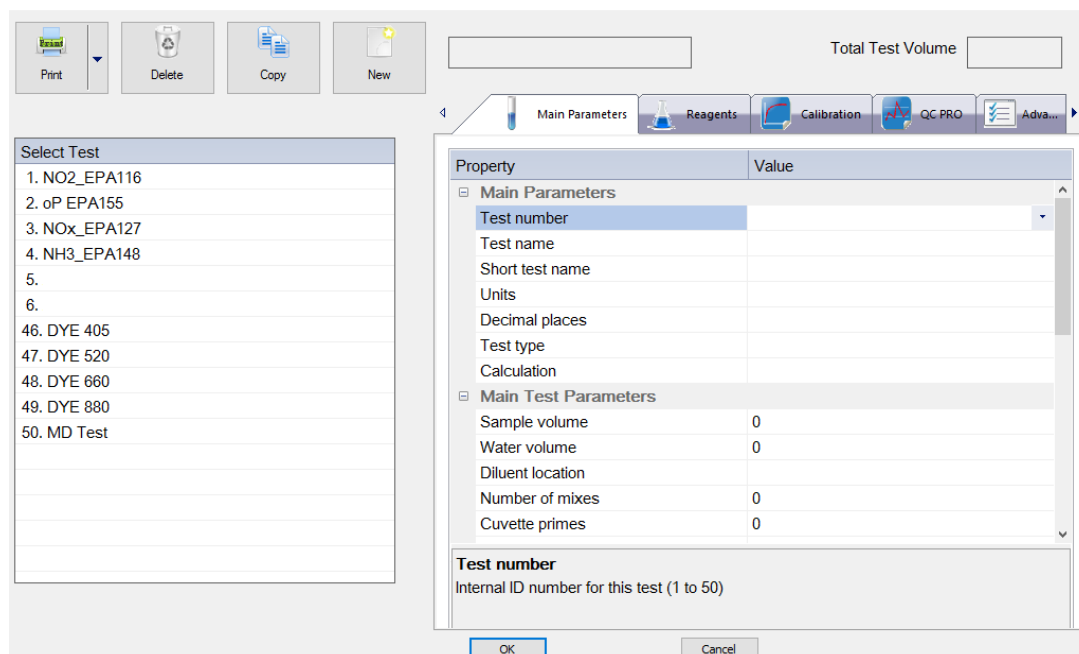
All the parameters required for the analyzer to perform a chemistry test are configured from the utilities available under the **Maintenance and Setup** tab.



Some of the parameters in this section are used for method definition and cannot be edited while the system is running those methods. **Maintenance** and **Daily Start-up** both require components of the analyzer to move and cannot be accessed at all while the analyzer is working.

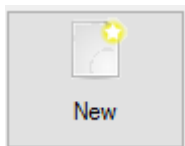
4.2 Test Parameters

This section details adding and configuring the tests that can be performed on the analyzer. To open **Test Parameters**, select the **Tests** icon from the **Maintenance and Setup** tab. The necessary test parameters for a specific method can be found in its designated SEAL method document.



4.2.1 Adding a Test

To define a new test, click on the **New** icon. A window with a generated **Test Number** appears.



4.2.2 Naming a Test

Once a test has been created, a **Test name** must be assigned. A **Short test name** of seven characters or less is also required.

The **Short test name** is used in Scheduling to avoid long lists of test names.

4.2.3 Altering an Existing Test

To edit an existing test, select the test from the **Test Selection** list. Once selected, the fields can be altered but **Test name** and still remain unique.

4.2.4 Define Test Units

The units that the test is reported in can be entered in the **Units** box. These do not directly affect any operation performed by the analyzer but will be output on reports and calibrations to provide context for the calculated results.

4.2.5 Define Decimal Places

The number of decimal places that the results are reported to should be selected in the **Decimal Places** box. This can be a number between 0 and 4. This affects the output of results for the current test across the entire software system including real-time result output and HTML reports.

4.2.6 Define Test Type

The type of test should be selected from the drop-down **Test Type** list. The choices are End Point, Cadmium Reduction, Sulphate Kinetic, Sample Blank, and Calculated.

- If the test type selected is **End Point**, the reaction solution comes to full color formation before the absorbance is read.
- If the test type selected is **Cadmium Reduction**, the reaction solution reduces Nitrates to Nitrites via the open tubular cadmium coil installed on the analyzer. The reaction solution then comes to full color formation before the absorbance is read.
- If the test type selected is **Sulphate Kinetic**, the reaction solution allows the sulfate ion to be converted to barium sulfate suspension under controlled conditions in the reaction cuvette.
- If the test type selected is **Sample Blank**, will automatically assign sample blanking to every sample. This is not recommended for most applications. See section 4.4.3 for recommended sample blanking procedure.
- If the test type selected is **Calculated Test**, the calculation should be entered in the **Calculation** box.

**For example, if Nitrite is test 10, Total Oxidized Nitrogen is
test 15, and the calculation is for Nitrate, enter:**

$$T15 - T10$$

When viewing this field in the future, the calculation will be shown in Reverse Polish notation (T15 T10 -).

The AQ400 also has a test type called **Manual Read**. This test type is selected if the user is setting up a test for the optional Manual Read Cuvette add-on available only on the AQ400. This allows users to insert their own standard cuvette into the detector on the AQ400 and track their colorimetric results in the AQ software. For more information on setting up a Manual Read test, see section 9.4.

4.3 Main Test Parameters

On the **Main Parameters** tab, the test information can be entered following the Test Parameters section of the SEAL method document for the desired method. The following sections detail the specifics of the currently selected test. The analyzer will use this information to conduct the test.

Property	Value
Main Parameters	
Test Number	1
Test Name	Nitrite_1.2
Short Test Name	NO2
Units	mg N/L
Decimal Places	4
Test Type	1. End Point
Calculation	
Main Test Parameters	
Sample Volume	160
Water Volume	0
Diluent Location	Reagent 15
Number of Mixes	2
Cuvette Primes	1
Cuvette Washes	2
☐ Baseline on Wash	
Cadmium Reduction Time	0
Cadmium Flush Volume	0
Measurement / Math	

4.3.1 Sample and Water Volumes

On the Main Parameters page, the user can enter the **Sample Volume** (2 –500 µl) and the **Water Volume** in the appropriate boxes.



Note:

The maximum **Sample Volume** plus **Water Volume** is 500 µl.

4.3.2 Diluent Location

Diluent Location enables the user to specify a particular diluent location for analyzer-prepared dilutions and for automated standard curve preparation. This would normally be placed in a reagent wedge. Selecting “From Bottle” uses water from the water reservoir, which is not recommended. This field is tied to the **Blank location** in the **Calibration** tab. Changing the value in one of these two fields will also change it in the other field.

4.3.3 Number of Mixes

The **Number of Mixes** is the number of times the reaction solution is mixed after the addition of each reagent and can be 0, 1, or 2 for the AQ2 and between 0 and 4 for the AQ400 or AQ300.

4.3.4 Cuvette Primes

Cuvette Primes are used to pre-condition the Measurement System Cuvette with the reagents prior to reading actual tests and can be set from 0 to 5.

4.3.5 Cuvette Washes

Cuvette Washes are used to draw in DI water from the water reservoir to rinse the Measurement Cuvette between Test measurements. This can be set to 0, 1, or 2.

4.3.6 Baseline on Wash

The option **Baseline on Wash** can be ticked to enable the system to re-zero itself between samples. To determine if this should be used for the chemistry being entered, please refer to the Method Document for SEAL's recommendation.

4.3.7 Reduction Time

CADMIUM REDUCTION CHEMISTRY ONLY!

Reduction Time is the time that a sample will be passed through the Cadmium Coil during the Cadmium reduction chemistry. This is measured in seconds.

4.3.8 Cadmium Flush Volume

CADMIUM REDUCTION CHEMISTRY ONLY!

Cadmium Flush Volume is the volume (in microliters) of liquid that fills the system between the tip of the sample probe and the start of the Cadmium Coil. The precise value will depend on the length of tubing used in the system and will normally be determined empirically.

4.3.9 Reaction Time

Reaction Time is the time in seconds between final reagent addition and measurement of the color produced.

4.3.10 Wavelength

Select the appropriate wavelength needed to measure the colorimetric reaction from the **Wavelength** drop down list.

4.3.11 Polynomial Order

The **Polynomial Order** of the test is the math model that will be applied to calibrations. This ranges from 1 to 7 with 1 being linear. This may be changed later following standardization of the test if appropriate.

4.3.12 Repeat Delta Check

The **Repeat Delta Check** is a decimal percentage value of the allowable difference between a result generated for a sample and the result of that same sample if it is subsequently rescheduled.

4.3.13 Auto Dilution Factor

The **Auto Dilution Factor** is the default factor that the system will use when automatically diluting over range samples. This can be set up to x100. An Auto Dilution Factor can also be applied to individual samples in **Scheduling** and **Data Review**. If this dilution factor fails to bring the sample within range, a second attempt is made by squaring the initial auto dilution factor.

4.3.14 Manual Dilution Factor

The **Manual Dilution Factor** is the default factor that the system will use if the operator schedules a sample as manually pre-diluted. This can be over-ridden when dilution factors are assigned to specific samples in **Scheduling**.

4.3.15 Low and High Linearity

The **Low and High linearity** are absorbance values automatically updated following calibration and require no operator input. Should the absorbance values be exceeded, the sample will be flagged for operator attention. These samples will be labelled as either ELL (Exceed Linearity Low) or ELH (Exceed Linearity High).

4.3.16 Low and High Alert

The **Low and High Alert** is optional. They are expressed in the defined units of concentration and can be used to flag a warning code to the operator should a sample exceed the given limits. This error code will be ALT. To activate this alert, tick the **Use alert limits** tick box.

4.4 Reagent Parameters

Click the **Reagents** tab to enter the Reagents section of Test Parameters. This section allows the user to set the reagents that are used for this test.

Test: Nitrite_1.2 Total Test Volume: 820

Print Delete Copy Reagents

Select Test

1. Nitrite_1.2
2. Ammonia 10
3. Ammonia_2_3R-
4. TURB. SULFATE
5. Ammonia_2_3R-
6. Ammonia_2_3R-
7. Nitrate+Nitri
8. Nitrite
9. Nitrate Only
24. Hardness
46. DYE 405
47. DYE 660
48. DYE 520
49. MEAS Diag
50. DYE 880

Property	Value
Number of Reagents	2
Reagent 1	
Reagent Name	NO2 Sulf-NEDD
Reagent Volume (ul)	550
☐ Replaced by the blank reagent	
Reagent 2	
Reagent Name	NO2 Buffer
Reagent Volume (ul)	110
Reagent delay (secs)	0
☐ Replaced by the blank reagent	
Reagent 3	
Reagent Name	
Reagent Volume (ul)	0
Reagent delay (secs)	0
☐ Replaced by the blank reagent	
Reagent 4	
Reagent Name	
Reagent Volume (ul)	0
Reagent delay (secs)	0

Reagent 4

OK Cancel

4.4.1 Adding Reagents to the Test Parameters

To add a reagent to the test, first select the **Number of Reagents** that the test will use. This can be from 1 to 4.

For each reagent, select the **Reagent Name** from the drop-down lists and enter the **Volume** of reagent that is to be used in the test.



Note:

If a reagent to be used has not yet been defined in Reagent Parameters, click the “Reagents” button to open the Reagent Parameters dialog box. Please see section Error! Reference source not found. below for further details.

4.4.2 Setting Reagent Delays

For reagents other than the first, the user can enter a **Delay** (in seconds) before addition of a subsequent reagent.

This delay should be no less than 25 seconds as only one reagent can be added per cycle of the analyzer and the cycle is 25 seconds. Leaving this at zero means the subsequent reagent will be added during the next cycle.

4.4.3 Define the Blank Reagents

The **Blank Reagent** is used when the user wants to gather the background absorbance of the samples + reagents without the color production of the reaction. If this is not required, the user can ignore this section and leave the default settings.

The user must design an appropriate reagent system such that substituting one or more test reagents with the same volume of Blank Reagent will produce the desired Blank Test in the reaction well.

Sample + Modified reagents for Sample Blanking = All interfering sample color and none of the analyte color.

Test: Nitrite_1.2 Total Test Volume: 820

Print Delete Copy Reagents

Select Test

1. Nitrite_1.2
2. Ammonia 10
3. Ammonia_2_3R-
4. TURB. SULFATE
5. Ammonia_2_3R-
6. Ammonia_2_3R-
7. Nitrate+Nitri
8. Nitrite
9. Nitrate Only
24. Hardness
46. DYE 405
47. DYE 660
48. DYE 520
49. MEAS Diag
50. DYE 880

Property	Value
Number of Reagents	2
☒ Reagent 1	
Reagent Name	NO2 Sulf-NEDD
Reagent Volume (ul)	550
☒ Replaced by the blank reagent	
☐ Reagent 2	
Reagent Name	NO2 Buffer
Reagent Volume (ul)	110
Reagent delay (secs)	0
☐ Replaced by the blank reagent	
☐ Reagent 3	
☐ Reagent 4	
☐ Blank reagent	
Reagent Name	Water Bottle
Reagent Volume (ul)	550

Replaced by the blank reagent
Reagent will be replaced by Blank reagent if sample blanking

OK Cancel

On the Reagents tab, in addition to entering reagent information as listed per the Test Parameters section of the SEAL method document, please perform the following modifications:

1. Choose the reagent to replace the color reagent by setting it as the **Reagent name** under the section labelled **Blank reagent** in the fifth row of the Reagent Parameters tab.



Note:

This Blank Reagent MUST be defined under Edit Reagent List, just like any other reagent.

2. Each of the 4 Test Reagents has a tick box labelled **Replace by the Blank Reagent**. Tick the box for the reagent that will be replaced with the Blank Reagent.
3. For the Blank Reagent **Volume**, the same volume as the reagent chosen will be entered automatically.

For any regular End Point test, Sample Blanking mode can be invoked at will for any sample. To do so, go to the **Scheduling** screen of the AQ software; enter **Sample Type** and **Sample ID** for all samples as usual. In the **Requested Test** column choose the desired test to run on the samples.

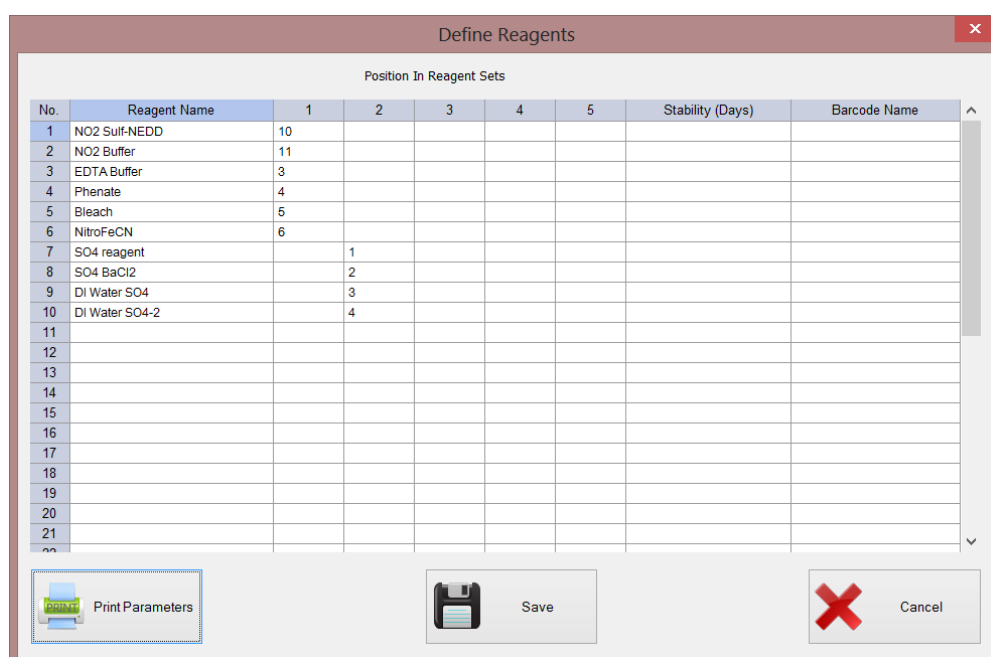
Select the samples which you wish to apply Sample Blanking mode, point to the desired test in the **Available Tests** box, and right-click on the Short test name. The scheduled test will appear blue in **Available Tests** and will also display a (B) next to it under **Requested Tests**, indicating Sample Blanking mode has been applied to that sample and test only. The user saves and runs the scheduled tray per typical AQ protocol.

4.4.4 Adding a Reagent to the Reagent List

When adding a new test to the system, it is necessary to define the reagents that the test will use. This can be done by selecting the **Reagents** icon from the **Maintenance and Setup** tab or by selecting the **Reagents** icon in Test Parameters.

This utility will allow the operator to define up to forty different Reagents on the system and these will be available for use in any defined test.

The numbers on the left-most column are ID numbers for the reagent and are not related to reagent positioning on the instrument. Reagents can be entered in any order in this window. Enter a name in the **Reagent Name** column by double-clicking in the field. This adds the Reagent to the system.



No.	Reagent Name	1	2	3	4	5	Stability (Days)	Barcode Name
1	NO2 Sulf-NEDD	10						
2	NO2 Buffer	11						
3	EDTA Buffer	3						
4	Phenate	4						
5	Bleach	5						
6	NitroFeCN	6						
7	SO4 reagent		1					
8	SO4 BaCl2		2					
9	DI Water SO4		3					
10	DI Water SO4-2		4					
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								
21								

Buttons: Print Parameters, Save, Cancel

4.4.5 Adding a Reagent to a Reagent Set

The AQ2 can hold up to 15 Reagents at a time, the AQ300 up to 18, and the AQ400 up to 26. These defined reagents can be grouped into any of five Reagent Sets.

The **Position in Reagent Sets** is the actual reagent wedge position on the analyzer (i.e. '8' refers to Reagent position 8 on the Reagent Tray). A defined Reagent can be placed in as many of these Reagent Sets as is desired. Additionally, if more than one wedge of a reagent is needed for a run, the reagent can be assigned multiple positions. See section 6.1.1 for further information.



Note:

If all the Reagents required for a test are not placed in the same Reagent Set, the test will not be available to schedule.

For more information, see the subsequent section on Reagent Sets (section 6.1).

4.4.6 Stability

The number of days the Reagent can be used for can be entered in the **Stability** column. This input is optional.

**Note:**

Tests that attempt to run expired reagents will be flagged for operator attention. This feature will be triggered if **Stability** is entered in the **Reagents** window along with **Preparation Date** in the **Reagent Sets** window. It will also be triggered if **Expiration Date** is entered in the **Reagent Sets** window.

4.4.7 Bar Codes

If the reagent has a bar code, the exact name as printed on the barcode label should be typed into the **Barcode Name** field. If you have a bar code reader, then the bar code will be automatically inserted in the active field when scanned. Leave this blank if not using barcodes.

4.4.8 Removing a Reagent

To remove a reagent, simply clear the name of the reagent and clear all the related fields. Then click Save to save the updated Reagents.

4.4.9 Saving the Changes

When complete, click **Save** to save the parameters. Clicking **Cancel** will reject all changes.

4.5 Calibration Standard Parameters

Click the **Calibration** tab to enter the calibration standards section of Test Parameters. This section allows the user to define the configuration of the current test's calibration.

Property	Value
All Standards	
Correlation coefficient limit	0.0000
Failed correlation actions	Continue to end of tray
☐ Exclude the blank (S1)	
Carryover limit %	0.0000
Failed carryover action	Continue to end of tray
Auto Standards	
Number of auto standards	7
Auto standard concentration	1.0000
Auto standard position	Cup 1
Blank position (S1)	Reagent 18
☐ High to low	
☒ Run carryover check	

Correlation coefficient limit
The lower limit of correlation at which the selected action is applied

OK Cancel

4.5.1 Standards Information

The system has two types of standards that can be defined. These are manually-prepared standards (**Manual Standards**) and automatically-prepared standards (**Auto Standard**). The standards screen is therefore also divided into two parts. The calibration must be comprised of either all automatically-prepared standards or all manually-prepared standards. A combination of both automatically-prepared standards and manually-prepared standards cannot be used.

4.5.2 All Standards

The **All Standards** section on this screen applies to both types of standardization and includes options to give a lower limit for the correlation coefficient (**Correlation Coefficient Limit**) and the carryover check (**Carryover limit %**). If the correlation coefficient or carryover check is below the specified limit, three actions (**Failed Correlation Actions**) are given which may be selected. When **Exclude the Blank (S1)** from the standardization calculation is selected, there will not be a blank in the calibration curve.

4.5.3 Auto Standards

No. of Stds	The number of standard points desired for the calibration curve (not including an initial blank) should be entered here.
Auto Std Conc	This is the top standard concentration (in the defined units for the test) of the single standard solution which the analyzer will use to prepare a standard curve.
Auto Standard Position	Select the location of the bulk top standard solution. This can be in a reagent bottle or in a sample cup.
Blank Pos.	Select the location of the standardization blank and diluent. This field is tied to the Diluent location in the Main Parameters tab. Changing the value in one of these two fields will also change it in the other field.
High to Low	Ticking this box will cause the auto-calibration to run starting from the highest standard in concentration and ending with the lowest standard in concentration.
Run carryover check	By default, the AQ Auto-Calibration feature automatically brackets the calibration standards with blanks (S1 and S0). A carryover check is then calculated from these values to detect an increase in absorbance values over the course of the calibration. Unticking this box will remove the S0 from the calibration and cancel the carryover check calculation.
Auto Standard s#	The desired name and percentage of the top standard should then be entered in the auto standard boxes provided for each calibration point on the calibration curve. The software may update these percentages automatically if the exact dilution cannot be made. The Standard Concentration will appear in the box below. <i>These auto-standards should always increase from lowest to highest concentration.</i> To run the calibration from highest to lowest, tick the High to Low box listed above.



Note:

These auto-standards should increase from lowest to highest concentration **IN ALL CASES** (Standard 90 as the lowest, increasing with each sequential standard). To run the calibration from highest to lowest, tick the **High to Low** box listed above.

4.5.4 Manual Standards

Standards 1 to 20 are the manually prepared standards and the value of each must be entered into the corresponding box. **Standard 1** is reserved for a blank and should always be zero.

The Standard names and concentrations can be entered for each standard solution. These must then be scheduled and poured into sample cups for use (See section 5.3.2 for more details on scheduling a manual calibration).

4.6 QC Pro Parameters

Click the **QC PRO** tab to enter the QC PRO section of Test Parameters. This section allows the user to define the configuration of the current test's controls. QC PRO consists of a series of controls for which ranges can be defined and failure actions specified.

Property	Value
Auto Bracketing	
First auto bracketing control	CCV
Second auto bracketing control	CCB
☒ Insert for auto reschedules	
☒ Insert for manual reschedules	
QC PRO	
Preferred order	
1	ICV
2	CCV
3	LCS
4	
5	
6	
CCV	

Insert for manual reschedules
Insert the auto bracket controls when manually rescheduling or diluting from Data Review

OK Cancel

4.6.1 Auto-Dilution Insertion and Bracketing Controls

Two **Auto Bracketing Controls** can be selected to run with the over-range automatic dilution feature. These can be either Controls (Control 1, Control 2, etc...) or QC PRO controls (CCV, CCB, LCS, etc...). These will run at the frequency set for the particular control selected.

These will also be run at the end of a chemistry to bracket the samples. Select the first and second **Auto bracketing control** by choosing the control from the drop down menu. To Auto bracket samples which have been rescheduled automatically, tick the box for **Insert for auto reschedules**. To bracket samples rescheduled manually, tick the box for **Insert for manual reschedules**.

4.6.2 List of QC Pro Control Abbreviations

Below is a list of the QC Pro Control Abbreviations. There is an option to change the name that appears in test parameters by typing a new name into **Control Name**.

ICB:	Initial Calibration Blank
ICV:	Initial Calibration Verification
CCB:	Continuous Calibration Blank
CCV:	Continuous Calibration Verification
MB:	Method Blank
LCS:	Laboratory Certified Standard (alternately, ERA)
DUP:	Duplicate of the Previous/Same Cup
SPKA:	Spike of the Previous/Same Cup performed by the instrument
SPKM:	Spike of the Previous Cup manually
SPKD:	Spike Duplicate
REFF:	Reduction Efficiency

4.6.3 QC Pro Preferred Order

QCs that have been scheduled to run at a specific frequency (See section 4.6.7) can be set to run in a specific order. The control set to position 1 will be run first, followed by position 2, and so on.

4.6.4 QC Pro Limits

Each control requires a **Low Value** and **High Value** limit expressed in terms of concentration of the sample. If you do not intend to use a particular control, or do not wish for the software to mark a control as passing/failing, these can both be left as zero.

4.6.5 QC Pro Location

The **Location** field enables the operator to choose where the control will be sampled from. This can be a designated cup, a reagent position, or in cups "As Scheduled" on a tray. If "As Scheduled" is selected, the software will not follow the **Frequency** or **Start Before** fields and will only run the control as directed by the schedule created by the user.

4.6.6 QC Pro Failure Actions

For all controls on the QC Pro page, an action can be set that will be triggered by a Control failure. The **Control Out Actions** include "Continue to end of tray" which assumes results will be manually assessed and rescheduled, "Stop sampling and wait for operator", "Resample from last good control", "Resample from start of test", and "Resample from last CCV/CCB pair." These all sound an audible warning if the PC has a sound card.

4.6.7 QC Pro Frequency

The **Frequency** determines how often a control is run. For example, if set to 10, the control will run every 10 samples. Frequency cannot be used if location is set to "As Scheduled."

4.6.8 Start Before

If the control is set to run on a frequency, the **Start Before** setting must also be set. This field contains the sample number which will follow the first sampling of the control. For example, if it is set to 1, the control will be run prior to sample 1, and thereafter at a rate defined by the Frequency setting. If this is set to "Off" the control will not run automatically.

4.6.9 Duplicates

Duplicates have a percentage box instead of a low and high range. The AQ Software will calculate the Relative Percent Difference (RPD) for duplicate samples. For example, a value of 10 will allow the duplicate to have $\pm 10\%$ difference between the two values.

4.6.10 Spikes

There are 3 different spike options in the AQ Software: **Spike Auto**, **Spike Manual**, and **Spike Duplicate**. In each of these boxes, there is a percentage instead of a low and high range. For the **Auto** and **Manual Spike** the software will calculate the Percent Recovery. For the **Spike Duplicate**, the RPD value will be calculated for the two spikes in the same way as a **Duplicate** (Section 4.6.9)

For example, a value of 10 will calculate the spike recovery and it will need to be between 90% and 110% recovery of the target value.

Spike Manual (SPKM): Parameters for manually spiked samples can be set by typing the spike concentration that will be manually added to the sample in the **Spike concentration added** box. Passing parameters for manually spiked samples can be set by typing the desired allowable percent recovery in the **% (+/-)** field. Set the **Control out action** field to the desired action that the instrument should perform if the spike fails.

Spike Auto (SPKA): Samples can also be set up to be spiked by the AQ instrument, rather than spiking the sample by hand. A sample cup or reagent wedge of a stock spiking standard will need to be placed on the instrument. Set the location for this standard in the **Location** field. Next, set the **Stock spike concentration** value to the concentration of the stock spiking standard.



Note:

The concentration of the stock spiking standard must follow certain requirements in order for the instrument to spike properly. For guidance on choosing a proper concentration, please see the Calculate Spike Stock Concentration in Section 8 Technical Tips of the Customer Support Manual.

In the **Spike concentration added** field, type the desired concentration amount to be added to the sample being spiked by the instrument. Passing parameters for automatically spiked samples can be set by typing the desired allowable percent recovery into the **% (+/-)** field. Set the **Control out action** field to the desired action that the instrument should perform if the spike fails.



Note:

The AQ Software is calculating the reverse chemical equation $C_1V_1=C_2V_2$ to calculate the volume taken from the Spiking Stock Concentration cup to add to the spiked samples.

Spike Duplicate (SPKD): Passing parameters for spike duplicates can be set by typing the desired allowable RPD into the **% (+/-)** field. Set the **Control out action** field to the desired action that the instrument should perform if the spike fails.

4.6.11 Reduction Efficiency

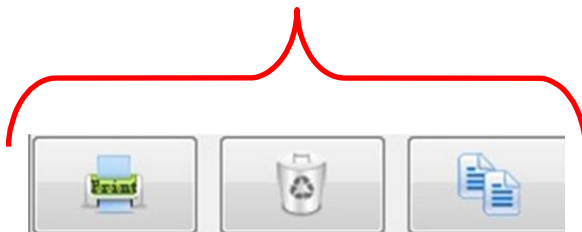
In addition to spiking and duplicates, there is an option for a **Reduction Efficiency** control in QC Pro to automatically calculate the efficiency of the cadmium coil when running Nitrate + Nitrite. The **Reduction Efficiency** has a percentage box used to identify the acceptable reduction efficiency. For example, if a value of 15 is entered, then the reduction efficiency calculated will need to be between 85% and 115% recovery of the target value to pass.

4.6.12 Controls

This section operates similarly to the **QC Pro** section. More controls are listed here and can be set up in the same manner as the other QC Pro controls (CCV, CCB, LCS, etc.). The only difference is no **Control out action** is available for these controls.

4.7 Test Parameters Toolbar

There are three icons on the **Test Parameters** tool bar:



The **Printer** icon will open an html copy of the test parameters for the currently selected test. From here the user can save or print the file.

The **Trash Can** icon will delete the currently selected test.

The **Copy** icon will allow the currently selected test to be copied to another test number. This is useful when setting up a different range of the same chemistry but it is important to alter the Test Name and Short Test Name to maintain unique Test identities.

4.7.1 Advanced Parameters

The final tab is **Advanced Parameters**. This tab will allow the user to select the following options:

Property	Value
☒ Set diagnostic slope & intercept	
☐ Eliminate air from transfer	
☐ Eliminate air from cuvette wash	
☐ Extra debubble action	
Extra aspiration steps	0
☒ Original preparation order (AQ400)	
☐ Original preparation order (AQ400)	

Set Diagnostic Slope and Intercept

This setting is only used for the Measurement Diagnostic (MD) test where no calibration is required. It forces a linear fit where the slope is 1.0 and the intercept is 0.

Eliminate air from test transfer

Air will not be aspirated when transferring reaction liquid through the aspiration probe into the cuvette.

Eliminate air from cuvette wash

Air will not be aspirated when transferring water from the aspiration wash bath through the cuvette.

Extra debubble action

The aspiration peristaltic pump will perform an extra pump step after the reaction sample has been aspirated into the cuvette in order to dislodge any possible stuck bubbles in the cuvette.

Extra aspiration pump steps

Allows modification to the number of pump steps which will adjust the position of the aspirated liquid within the cuvette.

Original preparation order (AQ400)

Allows the user to select if the diluent or Reagent 1 is picked up first by the sample probe.



Note:

“Set diagnostic slope and intercept,” “Eliminate air from test transfer” and “Extra debubble action” will automatically be selected when entering a new test.

4.8 Saving Test Parameters

When finished editing tests, click **OK** to save the parameters or click **Cancel** to undo all changes made.

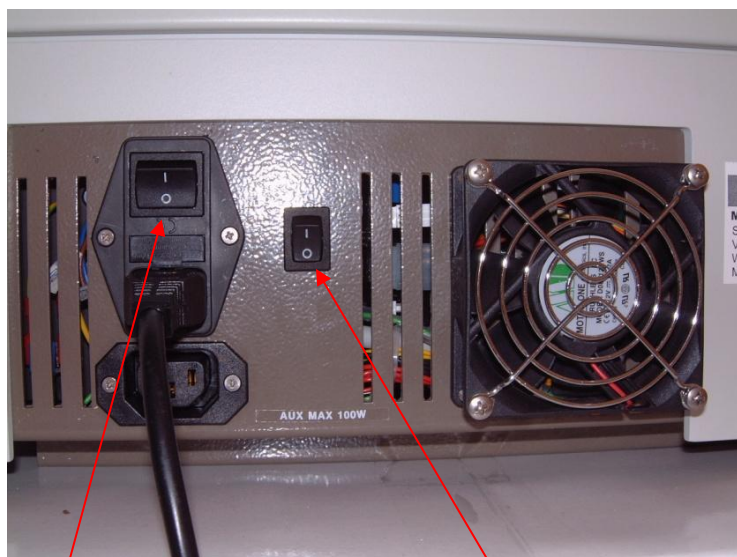
Test Parameters can be archived as html reports through ***Maintenance and Setup – Advanced Parameters – System Parameters*** and ticking the **Archive Test Parameters** box.

5 Normal Operation

Switches

AQ2

Power on the AQ2 analyzer by switching on the Reagent Cooling first (small switch) and then the Main Power (big switch) at the rear. Reverse this order to power the instrument off.

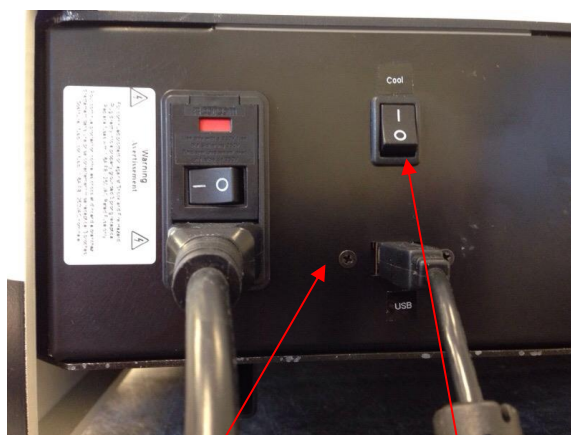


Main Power On/Off Switch

Reagent Cooling On/Off Switch

AQ400

Power on the AQ400 analyzer by switching the Main Power (switch near power cable) at the rear. If reagent cooling is to be used, switch it on as well (switch near USB cable).



Main Power On/Off Switch

Reagent Cooling On/Off Switch

AQ300

Power on the AQ300 analyzer by switching the Main Power (switch near power cable) at the rear. If reagent cooling is to be used, switch it on as well (switch near USB cable).



Reagent Cooling On/Off Switch

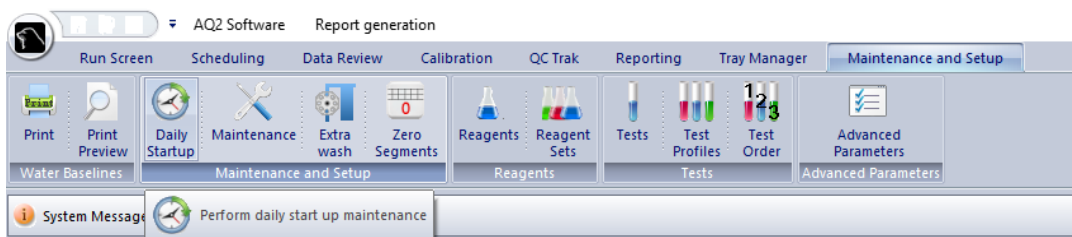
Main Power On/Off Switch

5.1 Daily Start-up

This option begins a simple maintenance procedure that should be performed each day before the analyzer is used.

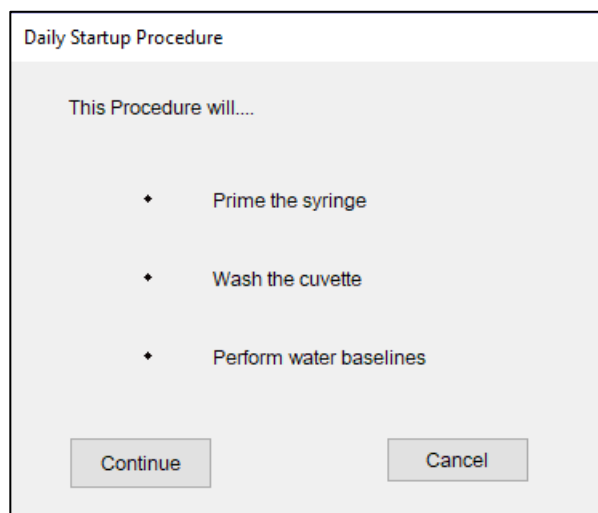
Allow the analyzer to warm up for 30 minutes before doing the Daily Start-up.

Go to the **Maintenance and Setup** tab and select **Daily Start-up**. Then click **Continue** to initiate.



During **Daily Start-Up**, the analyzer will perform the following procedure:

- 1) Prime the glass body syringe with fresh DI water from the DI water reservoir.
- 2) Flush the cuvette with fresh DI water from the DI water reservoir.
- 3) Perform Water Baselines.



The results of the **Water Baselines** will be posted in the **Message** window on the Run screen.

- AQ2:
 - Adjustable Offset – The voltages of all seven filters must fall within the range **3.5 to 9.5 Volts**. The highest energy filter should be approximately 8.9V. The Dark Current must be in the range **0.0030 to 0.0040** and typically reads 0.0030. When filters are within in range, it will say “OK” and the instrument is ready. If the filters are out of range, please see the AQ Operator Manual for further instruction.
 - Fixed Offset – The highest voltage must be 4.85 ± 0.05 volts for filters 1 through 9. The Dark Current (Filter 10) must be 0.0275. If the filters are out of range, please see the AQ Operator Manual for further instruction.
- AQ300:
 - Fixed Offset – The highest voltage must be 4.85 ± 0.05 volts for filters 1 through 9. The Dark Current (Filter 10) must be 0.0275. If the filters are out of range, please see the AQ Operator Manual for further instruction.
 - Auto-Gain – The voltages should be between 4.0 – 4.85 V for filters 1 through 9. The values are automatically set when running the Water Baselines function.
- AQ400:
 - Adjustable Offset – The highest voltage must be 3.8 ± 0.1 volts for filters 1 through 9. The Dark Current (Filter 10) must be 0.003 ± 0.001 volts. If the filters are out of range, please see the AQ Operator Manual for further instruction.
 - Fixed Offset – The highest voltage must be 4.85 ± 0.05 volts for filters 1 through 9. The Dark Current (Filter 10) must be 0.0275. If the filters are out of range, please see the AQ Operator Manual for further instruction.

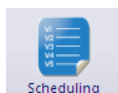
- Auto-Gain – The voltages should be between 4.0 – 4.85 V for filters 1 through 9. The values are automatically set when running the Water Baselines function.



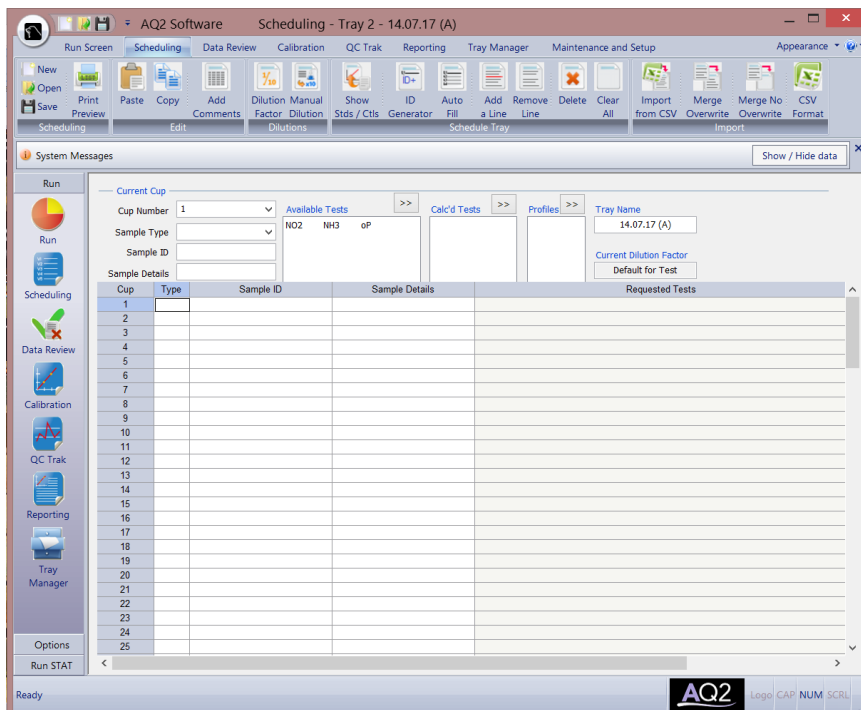
Note:

The gain value of the Auto-gain baselines should not change by more than 4 steps over the course of 24 hours. If a change of larger than 4, please see Customer Support Manual.

5.2 Scheduling a Tray



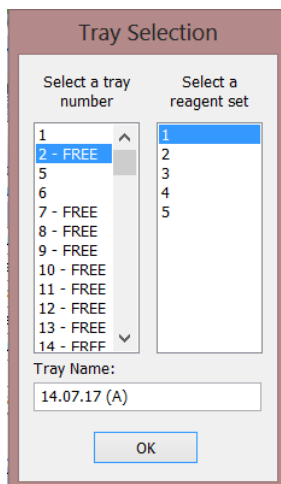
The AQ analyzers run on a concept of **Trays**. A Tray on the analyzer corresponds directly to a tray of sample cups that is placed on the instrument. There are a variety of ways in which a tray can be scheduled: directly through AQ software, on an external source and imported via a Comma Separated Value file, or cut and pasted from a spreadsheet application.



Schedules created in any of these ways can be manipulated directly in the AQ software prior to the running of the tray.

5.2.1 Choosing a Name and Number

Click on the **Scheduling** icon to enter the scheduler.



If a tray name and number have not been previously chosen this session, the **Tray Selection** box will open. If a **Tray Number** is not in use by any new or current trays, the number will appear with the word FREE after it in the list.

Tray numbers that have already been used to schedule a tray, but have not been run, will be available to select; these are the tray numbers that do not have the word FREE displayed after them. Selecting them will open the tray stored with this number.

If a tray has been scheduled and the tray has been run but not accepted, the tray number will not be present on this list. As soon as this tray is accepted or deleted, the number it occupies will be returned to the system as a FREE tray.

The first FREE tray number is automatically selected for you, as shown above. However, you may choose any FREE tray from numbers 1 to 99.

You will also need to choose the **Reagent Set** for this tray. If you do not select the proper reagent set, then your desired test(s) will not be shown in the Available Tests box on the Scheduling screen. For more information on setting up your Reagent Sets, please see section 6.1.

The default **Tray Name** is the current date (YY.MM.DD format) followed by a letter (A, B, C) giving unique names to each run. SEAL recommends adding additional information to the name to easily distinguish from other trays. The Tray Name can be edited now or at any point during the scheduling process.

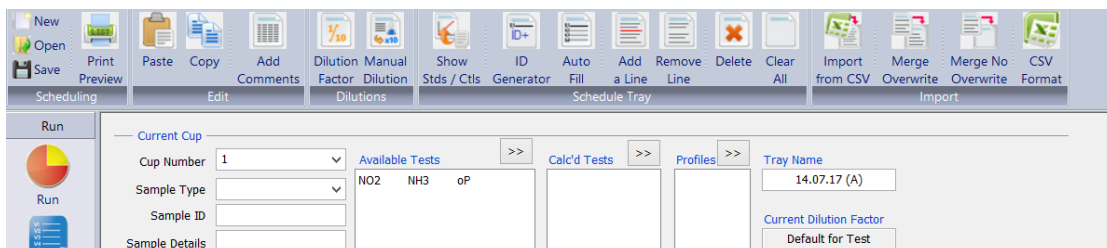
Select **OK** to complete the Tray Selection and open the scheduling screen.



Note:

Trays in the AQ Software can exist in one of three states: new, active, or accepted. Only one tray with a given number can be in the running state at any time.

5.2.2 Adding to the Schedule



5.2.3 Sample Type

The AQ Software allows the user to specify the type of sample to be run. There are three main types to choose from: **Standard**, **Control** and **Unknown**.

Standards are used to calibrate the analyzer for the scheduled tests. Only Scheduled Standards (S1, S2, S3, etc.) should be entered in a Tray. If the Auto-Standardize feature will be used, do not enter any standards in the Schedule.

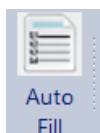
Controls are used to verify the integrity of the analyzer by comparing the measured results against the controls' known concentrations. These include Duplicates, Spikes, Spike Duplicates, CCVs, CCBs, etc.

Unknowns are the samples whose concentration needs to be measured (U1, U2, U3...).

To add a sample to the schedule, enter the **Sample Type** in the **Type** column. The **Sample Type** can be selected from the drop-down menu box at the top of the page or entered manually. Be sure to coordinate the **Cup** number with the sample's position on the sample tray. When you save a schedule, any samples without a Sample Type will be removed.

Additionally, you can schedule spikes and duplicates to be pulled from the previous cup. To do this, change the **Cup** number from the drop down menu to be the previous cup number.

5.2.4 Auto Fill with Unknowns



To quickly fill all empty cup positions with unknowns, simply click the **Auto Fill** button on the scheduling toolbar. If you highlight a set of rows and then click this button, only those empty cup positions you have selected will be filled with unknowns.

5.2.5 Sample ID

Enter a name for any unknown samples (up to 19 characters) in the **Sample ID** column.

5.2.6 Sample Details

Enter more-detailed information about any sample (up to 48 characters) in the **Sample Details** column.

5.2.7 Adding a Test

To add a test to a sample, simply click the test's short name in the **Available Tests** box. You may also add a **Calculated Test** (section 4.2.6) or a **Test Profile** (section 6.2). The former adds a test where the result is calculated from other tests scheduled on the same sample. The latter adds a pre-defined list of tests for faster scheduling.

When you select an Available Test or a Calculated Test, that short test name is highlighted in black in the Available Tests box and then added to the list of **Requested Tests** for that sample.

When you save a schedule, any samples without **Requested Tests** scheduled on them will be removed.

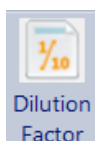
5.2.8 Add Comments

The **Add Comments** icon is used to add a Heading Text or a Trailing Text that can be shown in the report. The heading text will be added to the report before any data and the Trailing text will be added at the end.

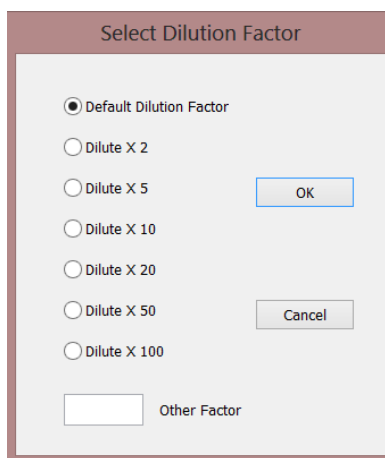
5.2.9 Adding a Test Dilution, Cup Dilution or Sample Blanking

A **Test Dilution** is a dilution that the analyzer performs. A **Manual Dilution** is a dilution performed by the user before analysis.

To add a dilution, first the dilution factor to be applied must be set. Click on the **Dilution Factor** icon.



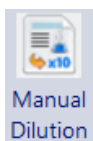
When the window appears, select an available dilution factor or specify any other factor by entering a dilution factor in the **Other Factor** box at the bottom and then click **OK**.



The dilution factor selected will now be displayed in the **Current Dilution Factor** field at the top of the screen.

To add a **Test Dilution** to a sample, highlight the sample line that will have the dilution applied, and then click on the test name again in the **Available Tests** box. The test name will turn **red** and the dilution factor will be appended to the test name in the **Requested Tests** column.

To add a **Manual Dilution** to a sample, highlight the sample line the dilution factor will be applied to and then click on the **Manual Dilution** button on the scheduling toolbar.



The **Manual Dilution** factor applied will appear next to the **Cup** number of the selected sample. The result of the sample will be multiplied by this factor and the final result of the manually-diluted sample will be displayed.

If a different dilution factor for a different sample is required, the dilution factor can be altered by clicking on **Dilution Factor** button on the scheduling toolbar and repeat the steps above.

To apply **Sample Blanking** for a given sample, right-click the test name in **Available Test** box. The test name will turn **blue**. The test name in the main display will have the letter "B" appended in parentheses. Sample Blanking must be set up appropriately in the test's **Test Parameters** in order for sample blanking to work effectively. See section 4.4.3 for more details.

If both **Test Dilution** and **Sample Blanking** are selected, the test name will be highlighted in **purple**.

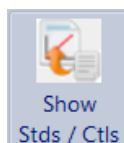
These test modifications can be removed by repeating the action taken to schedule them.



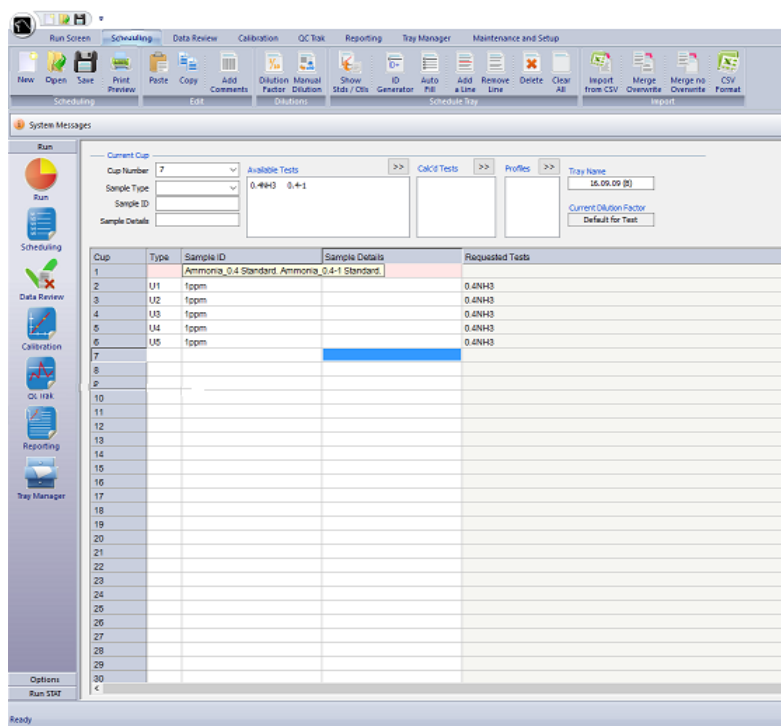
Note:

The Dilution Factor may be slightly modified automatically from the value chosen to ensure that the analyzer will only draw a sample volume in integer values and above the minimum value permitted.

5.2.10 Show Standards and Controls



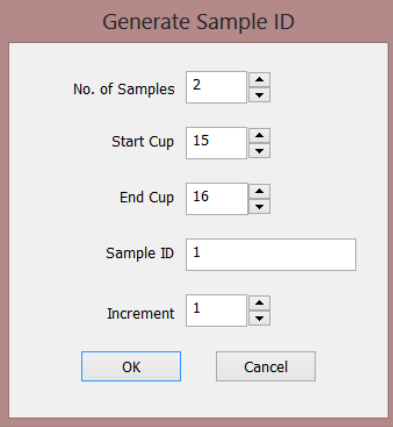
Selecting **Show Stds/Ctls** will highlight cups pink or blue if they are already designated in test parameters for standards or controls: pink for standards and blue for controls. Samples can still be scheduled in the highlighted cups, but it is to bring attention to the location of controls and standards setup in **Test Parameters**. To see what is already designated in test parameters, hover the mouse over the highlighted cups.



5.2.11 ID Generator



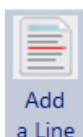
AQ Software can generate alpha-numeric sample ID's. Click on the **ID Generator** icon at the top of the screen. The user then specifies the number of samples, what cup to start and end the auto-generation with, the sample ID, and what increment to increase by. Click **OK** to generate the sample IDs on the schedule screen.



The 'Generate Sample ID' dialog box contains the following fields and controls:

- No. of Samples: 2 (with up/down arrows)
- Start Cup: 15 (with up/down arrows)
- End Cup: 16 (with up/down arrows)
- Sample ID: 1 (text input field)
- Increment: 1 (with up/down arrows)
- OK button
- Cancel button

5.2.12 Inserting a New Line

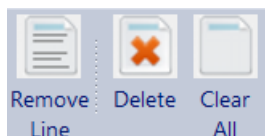


A new line can be inserted into the schedule by clicking the **Add a Line** button on the Scheduling toolbar. All subsequent lines will be moved down to accommodate the new line. A new line cannot be inserted if all tray positions are already assigned a Sample Type or a Test.

5.2.13 Removing from the Schedule

Removing information from a schedule can be done in three ways.

- 1) Click the **Remove Line** button on the scheduling toolbar to remove a single line.
- 2) Alternatively, highlight a section of the schedule and tap the delete key on the keyboard or **Delete** on the Scheduling toolbar to remove an arbitrary section of the schedule.
- 3) Clear All will remove everything currently setup in the schedule.



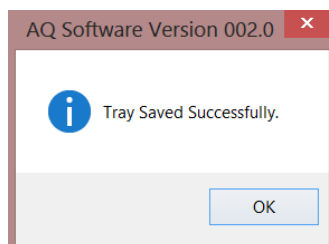
Note:

The Remove Line button is used, the subsequent lines will all move up one place. If you wish to remove a complete line but keep the other lines in place, you should highlight the entire line and tap the delete key or the toolbar Delete button.

5.2.14 Saving the Schedule



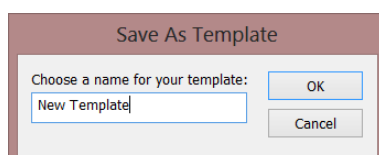
To save the schedule you have created, click on the **Save** icon on the main toolbar. A dialog box will open to confirm the tray has saved.



5.2.15 Saving a Schedule Template

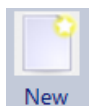
Templates are used for rapid scheduling of regular Tray configurations. The schedule created can be saved as a template with any alpha-numeric name.

To save the template, click the **SEAL Logo** (top-left corner of AQ window) and choose **Save as Template**. This will display the **Save As Template** dialog box. Simply enter the name for the template and then click **OK**.



5.2.16 Start a New Schedule

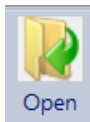
Click on the **New** icon on the main toolbar to create a new Schedule.



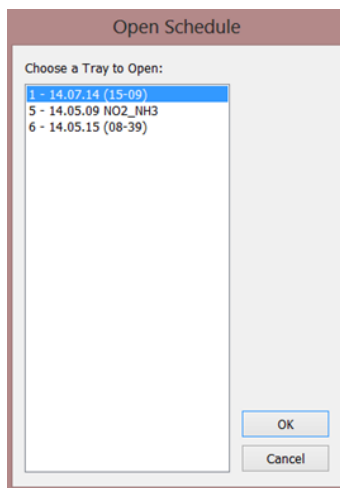
A Tray Name and Number, as well as a Reagent Set will need to be selected again.

5.2.17 Open Saved Schedule

To open a previously created schedule, click on the **Open** icon on the main toolbar.

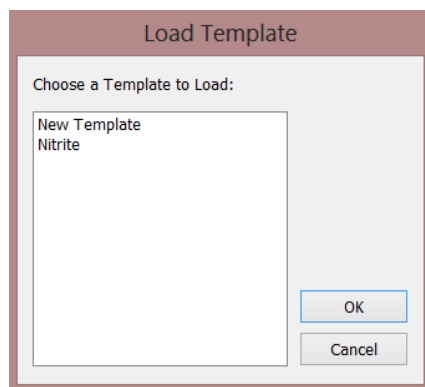


This will display a dialog box of all the previously created trays that have not been run yet.



5.2.18 Open Template

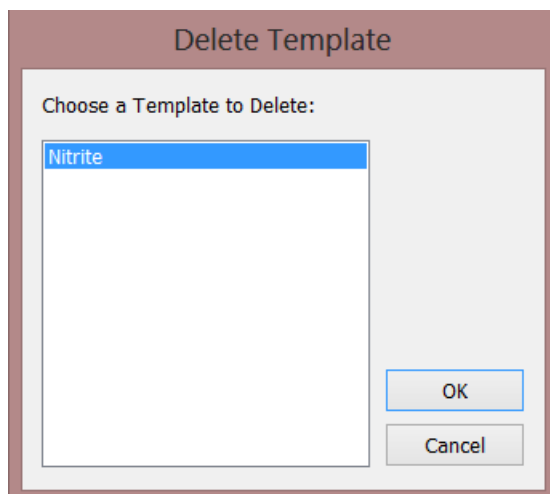
To use a previously created template, open a new tray as shown above. Click the **SEAL Logo** (top-left corner of AQ window) and choose **Open Template**. This will display the **Load Template** dialog box.



Select the template you wish to use and the contents will be loaded into the new tray. Anything already scheduled here will be lost.

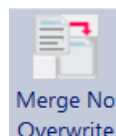
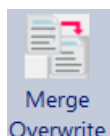
5.2.19 Deleting a Template

Old or unused templates can be deleted by clicking on the **SEAL Logo** (top-left corner of AQ window) and choose **Delete Templates**. This will display the **Delete Template** dialog box. Highlight the template you want to permanently delete from the AQ software by clicking **OK**.



5.2.20 Merging Schedules

The current schedule can be merged with one previously saved. There are two options available when merging schedules: **Merge with Overwrite** or **Merge without Overwrite**.



With **Merge with Overwrite**, any clashes of cup numbers between the two trays will be resolved in favor of the merging tray. That is, if two samples with the same cup number are scheduled, the merging tray that is selected will overwrite that sample position with its own data.

With **Merge without Overwrite**, any clashes between trays will be resolved in favor of the currently opened schedule.

5.2.21 Import a Schedule

If a schedule is created in another application, there are two ways to import the schedule.

First, if the application has an MS Excel compatible clipboard format, one can cut and paste directly from the application provided the data layout conforms exactly to that of the AQ Schedule (i.e., if copying a 2x8 Section, then a 2x8 sections in the software needs to be highlighted: then paste).

Second, if an Excel spreadsheet is created, you can import the information into the AQ software through a .CSV file.

To create a schedule in excel, type each parameter (sample type, dilutions, test, etc.) into a column (see below). Then the samples can be built into the different rows. Once all the samples have been entered, save the spreadsheet as a .CSV (Comma Delimited) file.

Sample Type	Sample I.D.	Sample Details	Test Dilution	Manual dilution	Test
u1	1	High conc.	1	1	1;3
SPK	SPK		0	0	1
u2	2		0	2	2;3
u3	3		0	1	1;2
u4	4		4	0	1;2

To import the schedule into the AQ software, the import format must first be set. Open the **Scheduling** menu and choose **CSV Import Format** as shown here:



A dialog box will appear which allows the user to determine which fields to import and the order (location) in the .CSV file.

Import Setup

Schedule Field	Order
Sample I.D.	2
☐ Cup Number	0
☒ Sample Type	1
☒ Auto Test Dilution Factor	4
☒ Manual Dilution Factor	5
☒ Sample Details	3

Delimiter Character: ,

☒ Need to Exclude Header

Tests

☒ Test Number ☐ Short Test name ☐ Long Test Name

Order: 6 Delimiter: ;

OK Cancel

In addition to entering the **Order** of the columns in the .CSV file, the **Delimiter Character** will need to be entered. In this case, Excel uses a comma as the main delimiter; therefore, enter a comma in the **Delimiter Character** field.

If using column headers in the .CSV file, be sure to select the **Need to Exclude Header** to ignore the column headings during importation.

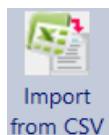
In the Tests section, there are three choices that can be used to assign a test: test number, short test name, and long test name. Any of the three can be used, but needs to be input

exactly as it appears in the software. If multiple tests are going to be assigned, a **Delimiter Character** needs to be assigned. This can be any punctuation other than a comma (which is already in use). Click **OK** to save the settings.

Below is an example of a schedule set up to be imported into the software:

Sample Type	Sample I.D.	Sample Details	Test Dilution	Manual dilution	Test
u1	1	High conc.	1	1	1;3
SPK	SPK		0	0	1
u2	2		0	2	2;3
u3	3		0	1	1;2
u4	4		4	0	1;2
u4	5		1	0	1;3
DUP	DUP		0	0	1;3
CCV	CCV		0	0	1
MB	MB		0	0	1;2;3

To import the schedule, choose **Import Schedule from CSV file** from the **Scheduling** menu.



A dialog box will open allowing the user to select the file to import. By clicking **OK**, the schedule will be imported into the AQ Software. The Schedule is then free to be edited as necessary before it is saved and used for analysis.

5.3 Advanced Scheduler Operation

5.3.1 Using Copy and Paste Effectively

The AQ Scheduler allows the user to copy data from tables and paste them into another application and vice versa. When used in this fashion the AQ Software is interacting according to the Windows® standard. That is, the data is copied in a one-to-one relationship and preserves the rows and columns of the data exactly. What this means is a schedule can be created in Excel and pasted into the Scheduler as long as the format created in Excel directly mirrors the layout of the Scheduler. For example, if the Excel table is 4x15 cells, the user must highlight 4x15 cells in the AQ Software to accurately paste the table.

5.3.2 Manually Scheduling a Tray Effectively

Scheduling a tray manually can take a long time. To schedule a tray quickly follow these instructions:

- If **Manual Standards** or **Controls** are to be scheduled, click in the **Sample Type** box for Cup 1. Type **S1** and tap the down arrow. Continue typing in the Standard and Control types that you require in the correct Cup Positions. Use the down arrow to quickly jump between rows. Do not type in any unknown types.
- Highlight all the **Cup** Positions in use on the tray (including unknowns) and click the **Auto-Fill** button on the toolbar.
- Leaving these rows selected in the main display use the mouse to select the tests that you require for the tray (including dilutions and blanking).
- Now enter the **Sample IDs** and **Details**. Alternatively, run the tray and enter the Sample IDs in **Data Review** by simply clicking on the Sample ID and typing it in.

5.4 Running a Tray



After the tests you wish to conduct on the current tray are scheduled, that tray can be run, and the analyzer will perform those tests.

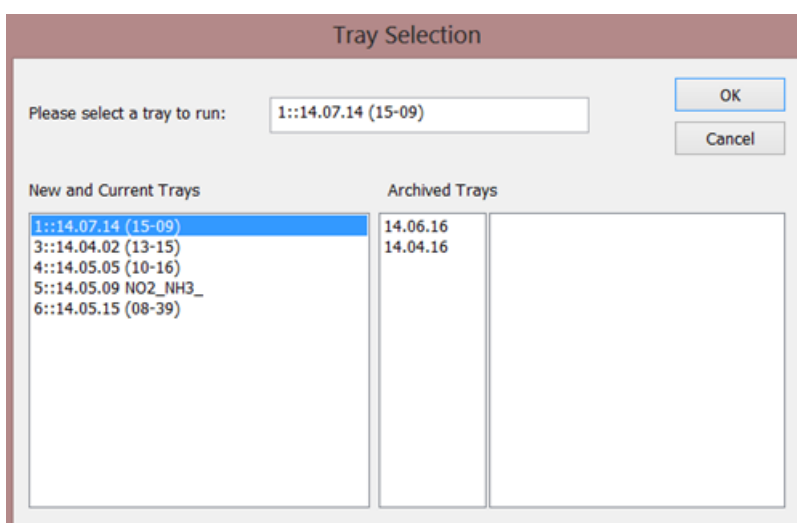


Note: The AQ2, AQ300 and AQ400 have full stop/start functionality. A tray can be stopped and restarted at any point during its run.

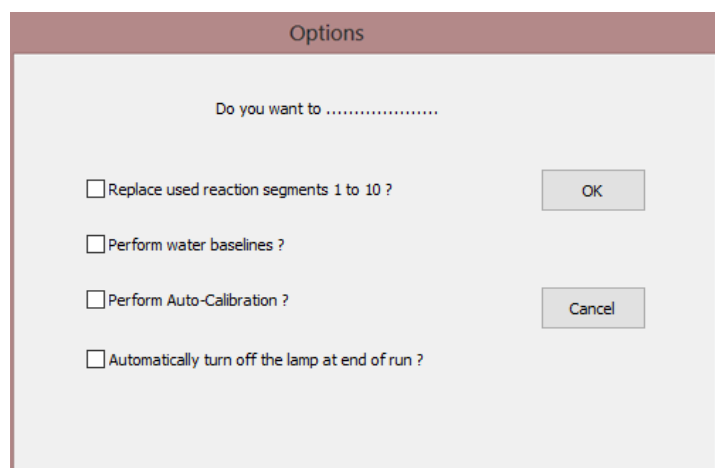
5.4.1 Choosing a Tray

Double-click on the **Run** icon, or you can also click on the **New** icon on the main toolbar if already on the Run screen. You will be presented with the **Tray Selection** dialog box.

There are three columns here. The first column shows **New and Current Trays**. The other two columns show the **Archived Trays**. The second column shows a list of dates (in reverse international format). Clicking on a date will show all the trays that were accepted and archived on that day in the third column.



Once a tray has been selected, click on the **OK** button to proceed. The **Options** dialog box will appear.



The four options available on this dialog box are:

- | | |
|---|--|
| Replace used reaction segments 1 to x? | The tube counter should be set to zero if all of the reaction segments have been replaced. |
| Perform water baselines? | If this box is ticked, the instrument will run water baselines before starting the run. |
| Perform Auto-Calibration? | This box must be ticked in order for the instrument to generate a calibration curve automatically. |
| Automatically turn off the lamp at end of the run? | If this box is ticked, the instrument will turn off the lamp at the end of the run. (AQ400 and AQ300 only) |

After the options for the run are selected, click **OK**.

Depending on the options set here and elsewhere in the software, the following dialog boxes may or may not appear.

5.4.2 Auto Standard Preparation

Ticking the **Auto Standardize** box on the previous page will show the **Auto Standard Preparation** dialog box next. There will be a list of all the available tests. Those that are present on the tray will be automatically scheduled to run an auto-calibration. This screen allows the user to double check that the top standard and diluent have been placed in the correct location.

Auto Calibration

PLEASE ENSURE ALL MATERIALS ARE POSITIONED AS SHOWN

Test Name	Blank Location	100% Standard Location	Date of Last Calibration	Calibrate ?
Nitrite_1,2	Reagent Bottle 15	Sample cup 1		☒
Ammonia 10	Reagent Bottle 15	Sample cup 1		☒
Ammonia_2_3R	Reagent Bottle 15	Sample cup 1		☒
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐
				☐

OK

Cancel

The **Date of Last Standardization** for each test is also shown to help determine if the standardization need to be run again. If the auto standardization for a specific test is not needed, the **Standardize** box for that test can be unticked. Click **OK** to proceed.

5.4.3 Reagent Volumes

If the reagent level sensor is not set to AUTO, or if the instrument is not set up with a reagent-level detector, the following dialog box will appear. To set the reagent level sensor to AUTO, go to the **Maintenance** section of the AQ software (section 8.3.1).

Pos	Reagent Name	Volume	Batch Number	Expiry Date	Operator	Preparation Date
1						
2						
3	EDTA Buffer	19.00				
4	Phenale	19.00			seal	
5	Bleach	15.00				
6	NitroFeCN	40.00				
7						
8						
9						
10	NO2 Sulf-NEDO	0.00				
11	NO2 Buffer	4.00				
12						
13						
14						
15						

Verify that the value in the **Volume** column is correct for each reagent and that there is enough volume for all scheduled tests. If there is insufficient volume the instrument will not run. Fill the required reagent wedges and enter the correct volume. Click **OK** to start the run. NOTE: A full reagent wedge holds 45mL of liquid.

5.4.4 Understanding the Run Screen

The run screen is made up of three main components.

The top section of the screen contains the **Status** and **Message** windows that inform the user of important events in the run.

The lower portion of the screen contains the main **Results** output.

5.4.5 Run Screen: Status Window

The current state of the analyzer is reflected in this window. Information on the current user, which tray is running (if any) and which cups are being sampled and aspirated is shown here. This is updated once every cycle of the analyzer.

Status					
Status		PREPARATION		ASPIRATION	
Current User:	seal	Cup:	33	Cup:	29
Name:	15.02.18 Dye 2	Sample ID:		Sample ID:	
Tray:	3	Test:	DYE 660	Test:	DYE 660
Reagent Set:	5	Well:	19	Well:	15
Current cycle:	19	Reagent:	40th Reagent		
Status	Running	Elapsed:	11:05	Remaining:	47:15
Lamp	ON	Front shield:	Closed	Temperature:	Normal



Note: The run screen for the AQ2 does not have the Lamp, Front Shield, and Temperature notifications.

5.4.6 Run Screen: Message Window

Any non-essential messages that are generated during the run of the analyzer are presented here. Messages that could be displayed here include the results of a water baseline procedure, notification of a Quality Control failure, or communication error messages.

5.4.7 Run Screen: Result, T, and W

If baseline on wash is ticked, this box will show the absorbance values for the test (sample) in the *T* column and for the wash solution in the *W* column. The software then subtracts the *W* value from the *T* value and uses the calibration curve to calculate the final *Result* value (value shown in concentration units, as assigned in **Test Parameters – Main Parameters**). When applicable, the cup number is shown on the far-left gray column.

		Result	T	W	
		0.0252	0.02605	0.00081	Reagent 6 Volu
		0.0437	0.04471	0.00098	Reagent 7 Volu
		0.0654	0.06643	0.00104	
		0.0823	0.08345	0.00111	
		0.0035	0.00422	0.00069	
		0.0005	0.00488	0.00066	
3		-0.0743	0.00396	0.00059	
		0.0252	0.00510	0.00060	

5.4.8 Run Screen: Results Window

This window shows a real-time color-coded representation of the data the analyzer is generating. The information is updated every cycle of the analyzer.

Results					
Cup	Type	Sample ID	NO3+NO2	oP	
	S1	Standard 1	0.003	0.000 PND	
	S90	Standard 90	0.019	0.000 PND	
	S91	Standard 91	0.040	0.000 PND	
	S92	Standard 92	0.078	0.000 PND	
	S93	Standard 93	0.000 PND	0.000 PND	
	S94	Standard 94	0.000 PND	0.000 PND	
	S95	Standard 95	0.000 PND	0.000 PND	
	S96	Standard 96	0.000 PND	0.000 PND	
	S0	Standard 0	0.000 PND	0.000 PND	
3	REF1	Nitrate Standard	0.000 PND		
4	REF2	Nitrite Standard	0.000 PND		
	CCV	CCV	0.000 PND	0.000 PND	

The first column shows the **Cup** number (if any) of the row. The second and third column show the **Sample Type** and **Sample ID** respectively. The remaining columns display the results for the tests you have scheduled.

5.4.9 Run Screen: Results, Colors and Error Codes

The analyzer will update the results window as the analyzer is running. In addition to the numerical data, the results are highlighted in the following three ways: color coded backgrounds, text emphasis, and three letter error codes.

Calibration standard results are emphasized with a **bold text**. These results are reported in absorbance units rather than concentration like the rest of the results.

The background color of a cell can be one of five different colors: Black, Green, Blue, Red, or Orange. Black denotes a cell is pending. Green denotes a cell is complete and free of errors. Blue denotes a cell has a Test Dilution applied to it. Red denotes an error has occurred. Orange denotes the cell will be approved pending some control to be run later in the tray.

If an error occurs, the result will be flagged with a 3 letter code. For example, if the sample's absorbance value exceeds the highest standard of the current calibration, it will be flagged with ELH. For further details on specific error codes, refer to the section *System Error Codes* (section 6.4.2).

5.4.10 Automatically Stopping the Tray

As the analyzer can resume any prematurely aborted run, there may be times when the user wants the analyzer to halt operation automatically. This action can be triggered by a number of user-configurable variables. For example, if the Correlation Coefficient check has failed, the analyzer can automatically stop and wait for user intervention.

The analyzer can be stopped automatically from the following places:

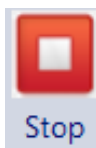
- **Test Parameters – QC Pro**
- **Test Parameters – Calibration**

Once the problem has been corrected, you can resume the tray. Refer to the section 5.4.12 *Resuming a Tray* for details.

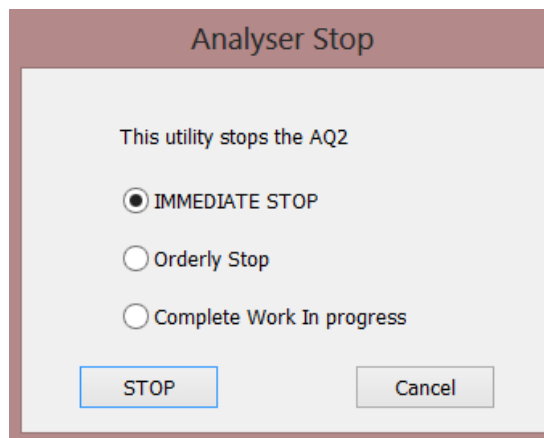
5.4.11 Manually Stopping the Tray

At any time during the running of a tray, the analyzer can be stopped manually. Any tray that is stopped prematurely can be resumed from the place it left off. Refer to the section 5.4.12 *Resuming a Tray* for details.

To manually stop a tray, click on the **Stop** icon on the main toolbar.



This icon is only available while the tray is running. Three **Analyzer Stop** options are available: **Immediate Stop**, **Orderly Stop**, and **Complete Work in Progress**.



IMMEDIATE STOP

Will stop the analyzer as soon as it is safe to do so. This will lose all samples currently incubating and being aspirated. If the tray is resumed, the samples will be prepared again.



Caution:

If the probe is in the down position, it will remain so. Only use this if you have a need for the analyzer to stop immediately. If possible, re-initialize before proceeding!!

Orderly Stop

Will take slightly longer to complete than the **Immediate Stop** option, but will save currently aspirating results, though incubating samples will be lost. The analyzer will go through a wash cycle before stopping.

Complete Work in Progress

Will stop the analyzer sampling but will continue until all currently incubating samples are aspirated. This can take up to the entire incubation time for the currently running test. Once all prepared samples have been aspirated, the analyzer will go through a wash cycle before stopping.

As with an automatic stop, the tray can be resumed later. Any position still marked as **Pending (PND)** will be resampled at that time.

5.4.12 Resuming a Tray

The process for **Resuming a Tray** is the same as running a tray. Either click on the **New** Document icon on the main toolbar (while in the Run Screen) or double-click on the Run icon. Follow the options as though this is a new tray.

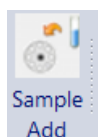
**Note:**

When resuming a tray, you may add an Auto Standardization. This will only affect the results reported after the restart. To have this new calibration model applied to previously reported results, a rework in Data Review will have to be performed.

5.5 Sample Addition Mode

Sample Addition Mode allows the user to stop the run in progress, add samples, and restart the run.

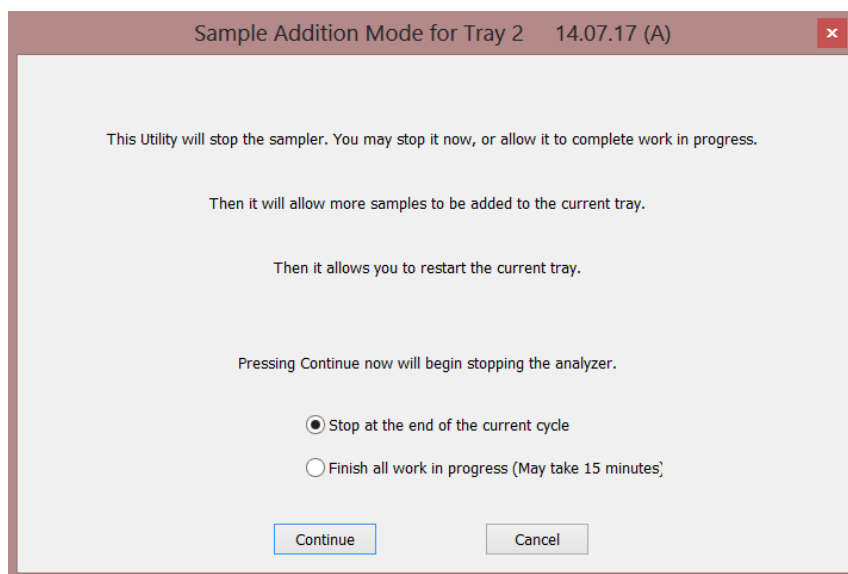
The **Sample Addition** icon is located at the top of the screen. Clicking this will initiate Sample Addition Mode.



The screen shown below then appears, allowing the user to choose between the following two options: **Stop at the end of the current cycle** or **Finish all work in progress**.

If **Stop at the end of the current cycle** is selected, the analyzer will finish the current cycle, immediately go through its wash cycle, and stop. The samples and/or standards that were prepared but not aspirated will remain in the reaction wells and will be re-prepared after restarting the run.

If **Finish all work in progress** is selected, the analyzer will continue to aspirate and read all samples and/or standards that have been prepared and are incubating, but it will stop preparing additional samples or standards. Once the final sample or standard has been reported, the analyzer will then go through its wash cycle and stop.



After clicking **Continue**, the Scheduling screen appears which allows the user to add samples. Both options (i.e. **Finish work in progress** and **Stop at the end of current cycle**) allow samples to be added to the schedule once the analyzer has stopped.

Notes:

- New samples can only be added using this feature. AQ software does not allow the user to edit or add tests to previously scheduled samples!
- If running multiple tests on samples and if scheduling multiple tests for the samples that are added on, please note:

If Sample Addition Mode is initiated while the analyzer is still performing **Test 1** and has not started Test 2, the work will be completed in the following order:

- a) Remaining “PND” samples for Test 1
- b) Samples added on for Test 1
- c) All samples for Test 2

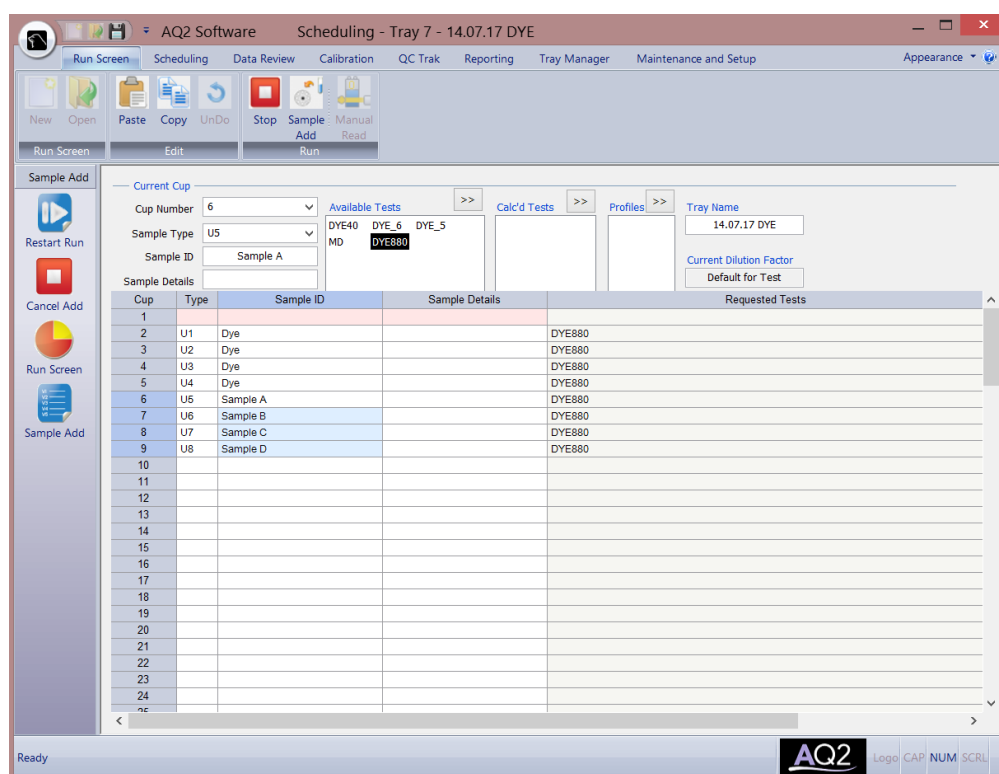
If Sample Addition Mode is initiated while the analyzer is performing **Test 2**, after having completed Test 1, the work will be completed in the following order:

- a) Samples added on for Test 1
- b) Remaining “PND” samples for Test 2
- c) Samples added on for Test 2

Neither auto-standards nor manual standards can be added to the schedule in Sample Addition Mode.

If the frequency counting feature of QC Pro is being used for a test, the **QC Pro** settings in Test Parameters must have ‘1’ located in the **Start Before** field. This will ensure the QC is inserted prior to the first added sample.

The Scheduling screen below shows the samples that have been added in cups 6 – 9.



After the addition of necessary samples, click on the **Run** icon located on the left side on the screen to view the status of the initial run. Once the status indicator bar displays **Finished**, click the **Restart Run** icon to initiate the run with the added samples.

The **Options** screen appears where the user has the option to reset the reaction segment counter and/or perform water baselines.

Options

Do you want to

☐ Replace used reaction segments 1 to 1 ?

☐ Perform water baselines ?

☐ Perform Auto-Calibration ?

☐ Automatically turn off the lamp at end of run ?

OK

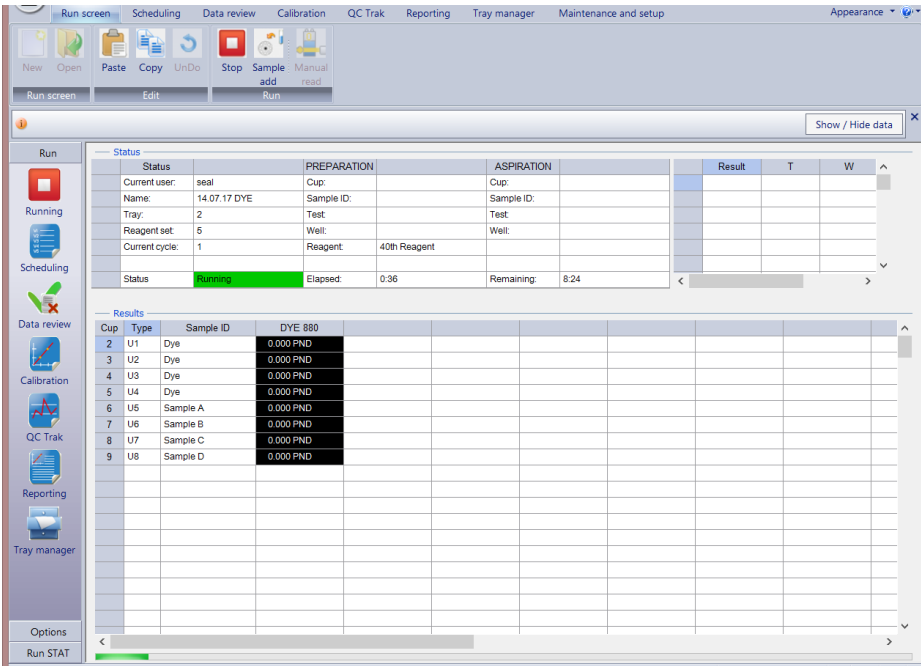
Cancel



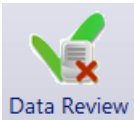
Note:

Although it is possible to perform water baselines at this point, it is not recommended.

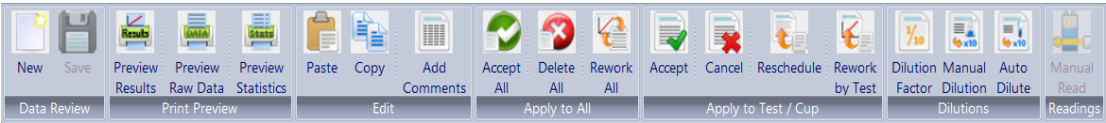
Once the user has clicked **OK**, the **Run** screen will appear with the additional samples appended to the tray. Below is the updated Run screen which includes the added samples as well as the proper insertion of QCs.



5.6 Data Review



The final stage of a tray's passage through the AQ Software is the **Data Review** screen. This is where a tray's data is accepted and added to the archives or where the tray has new conditions applied in preparation of a re-run through the system.



The conditions that can be applied in preparation of a re-run are: a simple reschedule, a reschedule with automatic test dilution, and a reschedule with a manual cup dilution.

It is also possible to cancel a result, accept individual results, rework data, and print off a variety of reports from the Data Review screen.



Note:
All of the above options can be undone from this screen as well.

5.6.1 Print Results by Test

To print a set of test results, first select a test by clicking on the test name (the column header) and then click the **Preview Results** button on the Data Review toolbar.

This will open an HTML format report in a print preview window. From here the results can be printed or simply viewed. This report can be generated at any time before the tray is accepted.

The report will contain general information about the current tray as well as a table of the results. Also included are dilution factors, error codes and time/date stamps.

For more information on the AQ Software reporting capabilities, see the *Reporting* section 5.10.

5.6.2 Print Raw Data by Test

To print a set of raw data, first select a test by clicking on the test name (the column header) and then click the **Preview Raw Data** button on the Data Review toolbar.

There are two choices of raw data that can be displayed, **Absorbance** and **Well Result**.

Selecting **Absorbance** displays the raw absorbance for each result on the report. **Well Result** displays the concentration before any sample blanking, dilution factors, or spike calculations are taken into account.

This will open an HTML format report in a print preview window. From here the results can be printed or simply viewed. This report can be generated at any time before the tray is accepted.

The report will contain general information about the current tray as well as a table of the raw data. Also included are dilution factors, error codes, and time/date stamps.

For more information on the AQ Software reporting capabilities, see the *Reporting* section 5.10.

5.6.3 Print Statistics by Test

To print a set of statistics, first select a test by clicking on the test name (the column header) and then click the **Preview Statistics** button on the Data Review toolbar.

A dialog box will open which allows the user to select a range of cups to view statistics.

This will open an HTML format report in a print preview window. From here the results can be printed or simply viewed. This report can be generated at any time before the tray is accepted.

The report will contain a list of the samples included in the statistics report (standards and controls are excluded) and statistical breakdown of the test results. Statistics given include: number of samples in the set, mean, minimum, maximum, standard deviation, and percent relative standard deviation (%RSD).

For more information on the AQ Software reporting capabilities, see the *Reporting* section 5.10.

5.6.4 Accepting an Entire Tray

To accept an entire tray, click the **Accept All** button on the Data Review toolbar. This will label all valid test results **ACC**. If all test results are valid, this tray will be detected as complete and the prompt to fully archive the tray will appear.

**Note:**

Tray cannot be archived if it is still running on the analyzer.

Once the tray has been accepted, the tray will be removed from the Current Trays list and added to the archive. A series of reports will also be generated. Refer to the section *Tray Archives* (section 5.11.5) for more information.

The tray will not be archived if some points are still marked as Pending (PND) or marked to be Rescheduled (RSC). To continue, either complete the tray or mark PND points as cancelled and remove the RSC flag from samples flagged as RSC.

5.6.5 Deleting an Entire Tray

To Delete an Entire Tray, click on the **Delete All** button on the Data Review toolbar. This deletes the entire tray from the analyzer. A dialog box will appear asking if you are sure you wish to perform this action. If yes is selected, all traces of this tray are removed, it is not archived, and no reports are generated. **THIS STEP IS IRREVERSIBLE**. If no is selected, then no changes are made to the current tray. If yes is selected, an additional dialog box will appear asking if you would like to save the schedule. Click Yes and the schedule will be saved (no data is saved). You can then run the schedule again. Click No and the schedule will not be saved.

**Note:**

Trays that are currently running on the analyzer cannot be deleted.

5.6.6 Reworking an Entire Tray

To rework the entire tray, click the **Rework All** button on the Data Review toolbar. This will rework all the current data points with the most recent calibration for each test.

This is useful if changes have been made to the most recent calibration in **Calibration Review**, e.g., a calibration point removed, or if the tray's scheduled calibration has been re-run.

5.6.7 Accepting an Individual Data Point

To accept an individual data point, click on the point you wish to accept and click the **Accept** button on the Data Review toolbar. This labels the data point as 'ACC'.

If your entire tray is labelled as accepted (ACC), you will be prompted to archive the tray. Refer to the section on *Accepting an Entire Tray* for more details (section 5.6.4).

Multiple data points can be selected by highlighting them with the mouse. An entire test can be selected by clicking on the test's name.

This operation can be reversed by highlighting a cell that has been labelled ACC and clicking the **Accept** button.

5.6.8 Cancel an Individual Data Point

To cancel an Individual Data Point, click on the point you wish to cancel and click on the **Cancel** button on the Acceptance toolbar. The point will be labelled CAN.

A point from a calibration can be removed by cancelling it and then re-working the tray. All of the results will also be updated as they are now applied to a new calibration model.

Although a cancelled point may be accepted, an accepted point cannot be cancelled. To cancel an accepted point the user must first toggle off the ACC by hitting the **Accept** icon again.

Multiple data points can be selected by highlighting them with the mouse. An entire test can be selected by clicking on the test's name at the top of the **Results** column.

This operation can be reversed by highlighting a cell that has been labelled CAN and clicking the **Cancel** button.

5.6.9 Reschedule an Individual Data Point

To reschedule (RSC) an Individual Data Point, select the point to be rescheduled and click on the **Reschedule** button on the Data Review toolbar. The point will be labelled RSC.

Multiple data points can be selected by highlighting them with the mouse. An entire test can be selected by clicking on the test's name.

This operation can be reversed by highlighting a cell that has been labelled RSC and clicking the **Reschedule** button. This operation *cannot* be reversed if the tray is re-run.

To begin analysis of the rescheduled sample, simply run the tray a second time. Only samples marked with RSC or PND will be re-ran.

5.6.10 Rework by Test

If changes have been made to a single test calibration, it may become necessary to rework only a single test, rather than the entire tray. As with Reworking an Entire Tray, **Rework by Test** will recalculate the results of the test based on the most recent calibration.

This function can only be performed on an entire test. To select the test, click the column header.

5.6.11 Select a New Dilution Factor

You can select a new Dilution Factor that will over-ride the default factors set for each test. Click on **Dilution Factor** in the Data Review toolbar and select a dilution factor. The new Dilution Factor will be used for any subsequent dilutions, whether automatic or manual.

5.6.12 Rescheduled with Cup Dilution

If a test result is returned as over-range, the sample can be re-run with a cup dilution (manual dilution performed by the user before placing on the analyzer).

Select the samples that will manually be diluted and then click the **Manual Dilution** button on the Data Review toolbar. An RSC should now appear by the selected samples. The Manual Dilution Factor to be applied will be shown by the **Cup Number**, just like it is in Scheduling mode.

Unless a new dilution factor has been selected by clicking on the **Dilution Factor** button on the Data Review toolbar, the test default dilution factor will be used.

5.6.13 Reschedule with Test Dilution

If a test result is returned as over-range, the sample may be re-run with a dilution. The analyzer can automatically perform this dilution. To reschedule a sample with an automatic dilution, select the sample result (or results) to be diluted and click the **Auto Dilute** button on the Data Review toolbar. An RSC will now appear by any selected samples along with the Auto Dilution Factor to be applied.

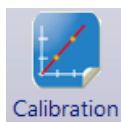
Unless a new dilution factor has been selected by clicking on the **Dilution Factor** button on the Acceptance toolbar, the test default dilution factor will be used.



Note:

The Dilution Factor may automatically be slightly modified from the value chosen to ensure that the analyzer can only draw a sample volume at or above the minimum value permitted and in integer values.

5.7 Calibration Review

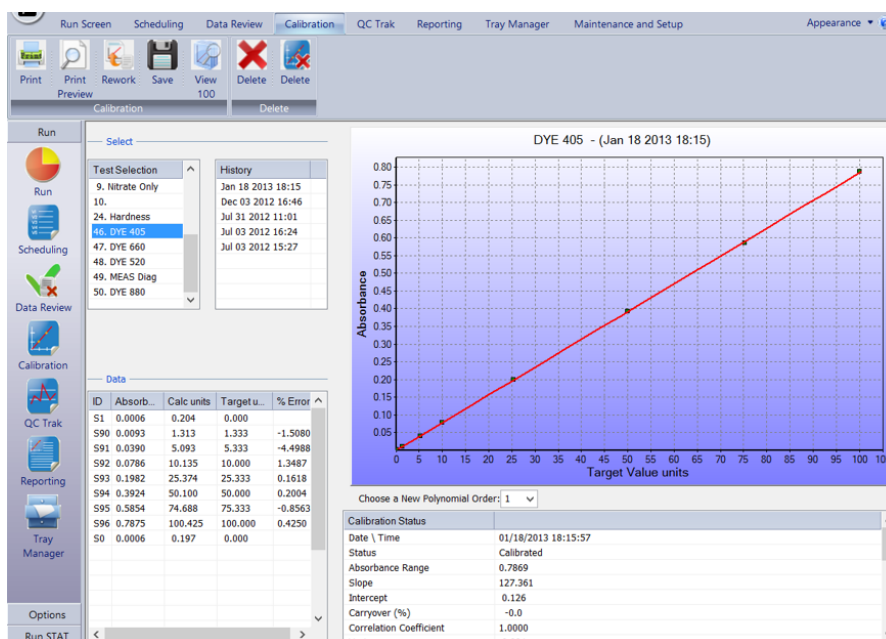


The **Calibration Review** section of the AQ Software allows easy access to all the calibrations performed on the system.

The last twenty calibrations for each test are stored in their entirety and the last one hundred are stored in a compact form.

5.7.1 Selecting a Test

A list of all the currently defined tests appears in the upper left side of the screen under **Test Selection**. Any tests that have previously been defined and run, but subsequently deleted from the system, will not appear here and any calibrations performed are lost.



5.7.2 Selecting a Calibration

Once a test has been selected, any calibrations that have been performed will be listed in the **History** column. Selecting one of the calibrations will load it and display a graph, a **Data** table with results, and a **Calibration Status** table of further information.

5.7.3 Understanding the Graph

The Calibration Review **Graph** plots the raw absorbance values of each standard against the target concentration and draws a best fit line in red. This best fit line is then used to calculate the concentrations for the **Data** table (see above).

The scales on the axes are determined by the standard values and the absorbance range.

5.7.4 Understanding the Results

The **Data** table contains the results and four columns:

ID	This column lists the type of each standard used in the current calibration.
Absorbance	This column contains the actual absorbance value recorded by the analyzer during aspiration.
Calc units	This column contains the calculated concentration of the standard based on the recorded absorbance and the equation of the best fit line generated.
Target units	This column contains the target concentration result of the calibration standard.
% Error	This column contains the percent error between the target concentration and the calculated concentration.

5.8 Understanding the Calibration Status

The **Calibration Status** table contains some specific information about the current calibration. This includes the date and time (**DateTime**) it was completed and the **Polynomial Order** applied to it.

Absorbance Range is the difference between the absorbance of the highest standard and the absorbance of the blank

The **Slope** and **Intercept** are values within the best fit line equation. For a linear calibration, this equation is $y = bx + a$ where b is the slope, a is the intercept, y is the concentration, and x is the absorbance.

Carryover is calculated as follows:

$$\text{Carryover} = \frac{(A - B)}{C}$$

Where A = absorbance of carryover blank (S0)

B = absorbance of blank (S1)

C = absorbance of highest standard

Also included is the **Correlation Coefficient** which is a calculation of how close the best fit line is to the actual standard values.

Variance is calculated as follows:

$$\text{Variance} = \frac{\sum (\text{Actual Values})^2}{\sum (\text{Observed Values})^2} \times 100$$

5.8.1 Print Preview Calibration Report

Once a calibration has been selected, it can be previewed (and subsequently printed) by clicking the **Print Preview** button.

5.8.2 Reworking the Calibration with a Different Polynomial Order

It is possible to rework the current calibration with a different polynomial order. First, alter the number in the dropdown labelled **Choose a Polynomial Order**. Then click the **Rework** button. This will rework the current calibration with the new polynomial order. The polynomial order can be changed and reworked as many times as required. Once the calibration is satisfactory, click the **Save** button. Only now is the calibration saved and added to the top of the calibration list. Any unknown sample this test is run with will now have its result calculated against this calibration.

To place your old calibration at the top of the list again, select it, click **Rework** (keeping the default polynomial order) and then **Save**.

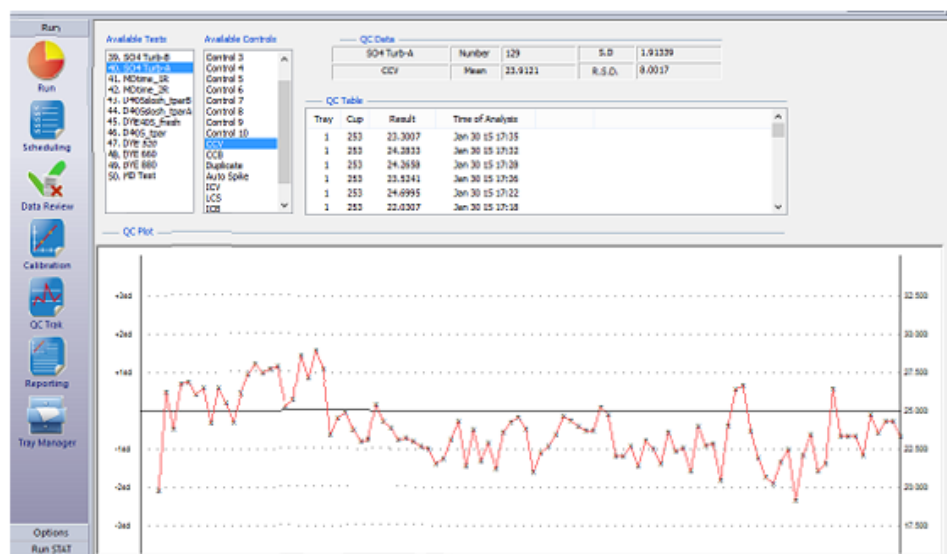
5.8.3 The Last 100 Calibrations

Clicking the **View 100** button will open a dialog box showing simple information on the last 100 calibrations for the currently selected test.

5.9 QC Trak



The **QCTrak** section of the AQ Software allows easy access to all aspects of the QC Pro and Control systems.



All accepted controls are available for review in this section. If the control data you seek is not present make sure you have accepted the tray that contains it.

5.9.1 Select a Test

A list of all the currently defined tests appears in the upper left side of the screen under **Available Tests**. Any tests that have previously been defined and run but subsequently deleted from the system will not appear here and any QC data is lost.

5.9.2 Select a Control

Once a test has been selected, a control can be selected in the **Available Controls** column. The user may choose between a generic Control (i.e. Control 1) or a QCPro Control (i.e. CCV). Once selected, the QC Data and QC Plot for that control will be displayed. If no data exists for the selected control, an error message will be given.



Note:

If the control data you seek is not present make sure you have accepted the tray that contains it.

5.9.3 Selecting a Plot Type

The AQ Software will display three different plot types in QC Trak. Each type gives the user a unique way to monitor the long term stability of the controls.

Last 100

This selection plots the last 100 results of the current control in chronological order.

Daily Means

This selection plots the average value of the selected control for each day it has been run.

Levi Jennings

This selection plots the daily average along with the range of results of the selected control for each day it has been run.

5.9.4 Understanding the QC Table

The QC Table displays different results depending on which Plot Type is selected. When **Last 100** is selected, the **Tray**, **Cup**, **Result** and **Time Of Analysis** of each control plotted on the graph is displayed.

If either of the other two Plot Types are selected (**Daily Means** or **Levi Jennings**) the table displays the **Daily Average** along with the **Number of Replicates**, the **Minimum** and **Maximum**, and the **Date** that the control was run.

5.9.5 Understanding the QC Plot

The QC Plot shows the appropriate graph for the selected Plot Type. One point on this graph corresponds to one row in the QC Table.

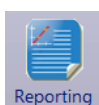
5.9.6 Understanding the QC Data

The **QC Data** table provides a summary of each control type. Select a control type and the data is displayed here. It is the same regardless of Plot Type. The data shown includes the **Number** of times the control has been run on the current test, the overall **Mean**, the overall standard deviation (**S.D.**), and the relative standard deviation (**R.S.D.**).

5.9.7 Print Preview

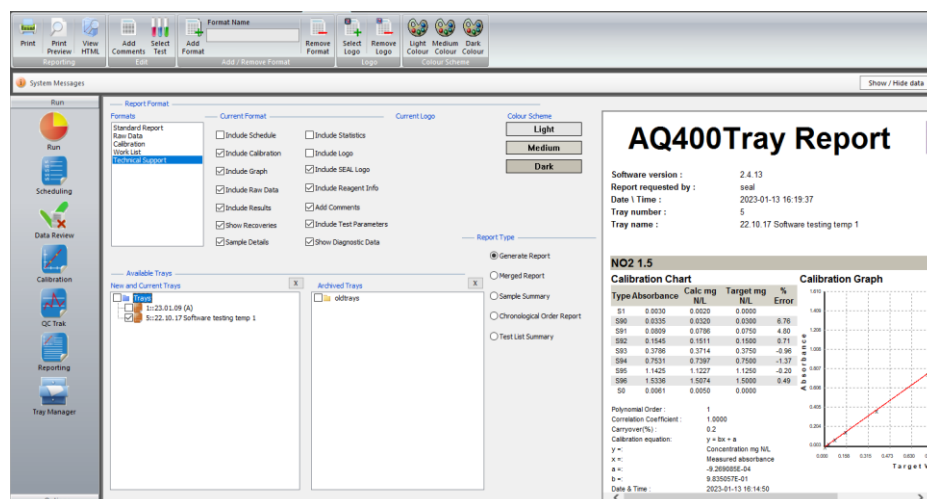
Once a control has been selected and the appropriate QC Plot Type is displayed, a report can be previewed (and subsequently printed or saved) in HTML format by clicking on **Print Preview**.

5.10 Reporting



The AQ Software features an advanced HTML reporting facility. This feature has been integrated throughout the software. When selecting a test, a report preview will appear on the right-hand side of the screen. To open an HTML report that can be saved or printed, select the **View HTML** icon.

The **Reporting** section of the software allows the user to customize reports to their specifications. First, the tray (or trays) desired for the report are selected. Then, the user chooses the parameters to be included in the report. Finally, the report type is selected. The final HTML report can then be printed or saved for later use. A report can be generated for any **New or Current Trays** as well as any **Archived Trays**.



5.10.1 Select a Tray

To select a tray simply locate that tray's name in the **New and Current Trays** or **Archived Trays** columns and tick the box next to the tray's name.

The archived trays contain a folder for each date a tray has been archived. Clicking on one of these folders lists all the trays archived on that day.

The tray you select will be the one used to generate the report.

5.10.2 Select Multiple Trays

The report generator will allow you to select multiple trays across all columns. For more information see *Generate Merged Report* (section 5.10.10) below.

5.10.3 Clear Tray Selections

Each column has a small button on the same level as the title that contains a small cross. Click this **X** to clear all selections in that column. Alternatively, deselect individual trays by unticking the box next to the tray name.

5.10.4 Customize Content

After a tray has been selected, the associated displayed data can be customized. There are pre-defined formats to choose from or a new format can be built.

Select Format

The **Formats** list will initially contain five built-in formats. These formats cannot be removed and all include the SEAL Logo and default color scheme. They are listed below:

Standard Report	This will generate a report that includes the standard tray information, a calibration, reagent information (including batch number and expiration date), and results. If more than one test was run on the selected tray, this information is displayed for each test.
Raw Data	This will generate a report that shows either the raw absorbance data or the Well Result. The software will ask which raw data to display when Print , Print Preview , or View HTML is clicked. Selecting Absorbance displays the raw absorbance for each result on the report. Well Result displays the calculated result before any sample blanking, dilution factors, or spike calculations are taken into account.
Calibration	This will generate a report that shows tray information and the calibration for the selected tray. This includes the graph image and the tabular data.



Note:

This report works regardless of whether a calibration was scheduled on the tray or not. If a calibration was not run with this Tray, then the calibration that was used to calculate the sample results is displayed.

Work List

This will generate a report that consists of tray information and the schedule. This is the equivalent of the report generated in the Scheduler.

**Note:**

This report can be generated on any tray, regardless of whether it has been run or not, and will not output any results or calibration information.

Technical Support

This will generate a report that includes the standard tray information, a calibration, reagent information (including batch number and expiration date), absorbance values, results, test parameters, and diagnostic data. If more than one test was run on the selected tray, this information is displayed for each test.

Any other formats on this list have been generated after the initial software install. Clicking on the format name and examining the ticked selections to its immediate right will show what it will include. For information on how to create your own formats, see the *Add Format* section 5.10.7 below.

Current Format	
☐ Include Schedule	☐ Include Statistics
☒ Include Calibration	☐ Include Logo
☒ Include Graph	☒ Include SEAL Logo
☐ Include Raw Data	☒ Include Reagent Info
☒ Include Results	☒ Add Comments
☒ Show Recoveries	☐ Include Test Parameters
☒ Sample Details	☐ Show Diagnostic Data

Include Schedule

Selecting this option will insert a complete Work List at the top of the generated report. This will appear only once on any report.

Include Calibration

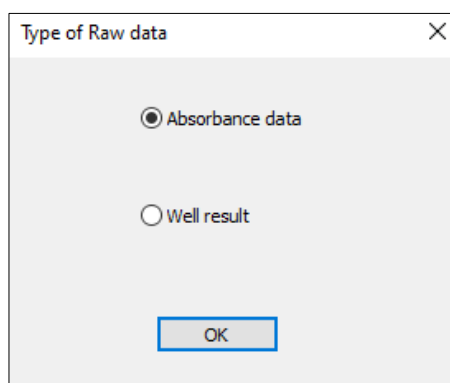
Selecting this option will include the tabular data that relates to the current test's calibration. This data will appear once for every test on the tray. This data includes the calculated, actual and target values for all the standards and the time the calibration was performed. This option is usually used in conjunction with **Include Graph**.

Include Graph

Selecting this option will include a graphical representation of the current test's calibration. This data will appear once for every test on the tray. This option is usually used in conjunction with **Include Calibration**.

Include Raw Data

Selecting this option includes the actual absorbance value recorded during sample aspiration. **Well Result** or **Absorbance** data can be selected when printing raw data on a report.



This value is included up to four times per sample per test (the AQ software will store up to four values for each sample in each test). For example, if a tested sample is rescheduled and a report is generated that sample will have two sets of absorbance values. These will both be included in the report.

Selecting this option will also include other relevant information such as the sample type, ID, the dilution factors used and the time of aspiration. Reagent information will also be included.

Include Results

Selecting this option will include the calculated concentration of sample based on the included calibration. This option can be used in conjunction with **Include Raw Data** to output all previous results for each sample.

Selecting this option will also include other relevant information such as the sample type, ID, the dilution factors used and the time of aspiration. Reagent information will also be included.

Show Recoveries

Selecting this option will include numerical results for relevant QCPro sample types. This includes the %RPD for DUP and SPKD, % Recovery for SPKA and SPKM, and % Reduction Efficiency for REFF.

Include Statistics

Selecting this option will append a Statistics summary to the end of each test section in the report. This summary is exactly the same to that generated in **Data Review**.

When the report is generated (see Generate Report below), a cup range for each Test scheduled on the Tray will be needed, upon which the statistical calculations will be performed.

The summary includes a results list with the minimum and maximum results highlighted. At the bottom of this list is a table that includes the following data: Number of samples, minimum, maximum, average, standard deviation, and relative standard deviation.

Include Logo

Selecting this option will include the current selected logo (see Select Logo, section 5.10.5 below) on the generated report.

Include SEAL logo

Selecting this option will include the SEAL logo on the generated report.

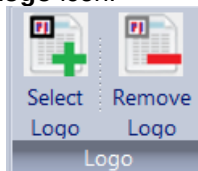
Include Reagent Info

Selecting this option will include Reagent information, such as Name, Batch, Prepared by, and Expiry Date.

Add Comments	Selecting this option will include the comments that were added during the scheduling of the run. Comments can also be added in Data Review or Reporting .
Include Test Parameters	Selecting this option will include the test parameters of the test or tests of the selected tray.
Show Diagnostic Data	Selecting this option will include the T & W values (See section 5.4.7 for more info) for all results of the selected tray.

5.10.5 Select Logo

The logo shown on the report can be changed. The user is allowed to select a file of type .bmp to be the new logo. It is not possible to alter the logo of the built-in types. If a personalized logo is needed, a new format should be created, see **Add Format** section 5.10.7. To add a logo, click on the **Select Logo** icon and then select the file. A logo can be removed by clicking on the **Remove Logo** icon.



Note:

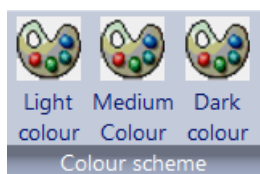
The custom logo can be any size; but for size comparison, the included SEAL logo image is 189 x 53 pixels.

5.10.6 Color Scheme

The HTML reports contain a three-value color scheme. The darkest color is used as the main heading background, the medium color is used as the sub-section heading color and the lightest color is used as the alternating row background color.

The text color is always black so any background color selected should always be light enough to allow easy reading of the foreground text.

To select a new color, click the color icon of the color to be changed. To save a color scheme, you must create and save a new format.



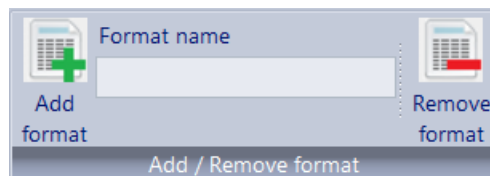
To return the colors to their defaults, simply select a built-in format (one of the first four) from the formats list.

**Note:**

To preserve the background colors when printing the HTML reports, the **Print Background Colors and Images** option needs to be set in the internet browser. This can be set by clicking on the **Tools** menu, choosing the **Options** menu item, and then clicking on the **Advanced** tab. Finally, scroll down to the **Print** sub-sections and tick the appropriate box.

5.10.7 Add Format

Once you have selected your desired report format, you can save this by typing in a name in the **Format name** text box and clicking **Add Format**. Once formats are saved, they cannot be edited. If you wish to edit the format, remove it and then recreate it with your desired settings.



5.10.8 Remove Format

To remove a format, select it from the **Formats** list and then click **Remove Format**.

**Note:**

The five built-in formats at the top of this list cannot be removed.

5.10.9 Generate Report

Once you have selected a tray and a report format, select **Generate Report**. Selecting **View HTML** will generate an HTML version of the report which will open in the default internet browser.

**Note:**

If you select multiple trays, only the first tray will be used to generate the report. If you wish to generate a Merged Report, see the **Generate Merged Report** section 5.10.10 below.

**Note:**

Page breaks are inserted after every test in the HTML reports. If the report is printed, each test will begin on a fresh page.

5.10.10 Generate Merged Report

A **Merged Report** will allow a user to merge multiple trays onto a single report. All of the trays in the merged report must use ONE calibration per test. Up to ten different trays can be merged into one report. These can include New and Current or Archived trays.

For example: If test A and test B were run on 200 samples on four separate trays, these four trays could be merged onto a single report if each tray utilizes a single calibration model for each test, A and B.

If you wish to create a seamless Merged Report, select **Merged Report**. This will create a report that takes all the results from the selected trays and merges them to streamline data run on the same calibration curve.

5.10.11 Generate Sample Summary

A **Sample Summary** report will break the report down sample by sample, rather than by test. The samples will be listed in order of their Cup Number, followed by the results of all the tests run on that sample. This is helpful if the user is interested in knowing all of the test results on a given sample at a glance.

To view a Sample Summary, select a tray and report format, and then select **Sample Summary**.

5.10.12 Generate Chronological Order Report

A **Chronological Order Report** will organize a report according to the time date stamps of the samples. This means that all reruns and auto-dilutions will appear at the end of the report.

To view a Chronological Order Report, select a tray and report format, and then select **Chronological Order Report**.



Note:

If multiple tests are run on a tray, the samples will be displayed chronologically, regardless of test.

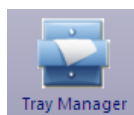
5.10.13 Test List Summary

A **Test List Summary** report will display the results as seen in Data Review.

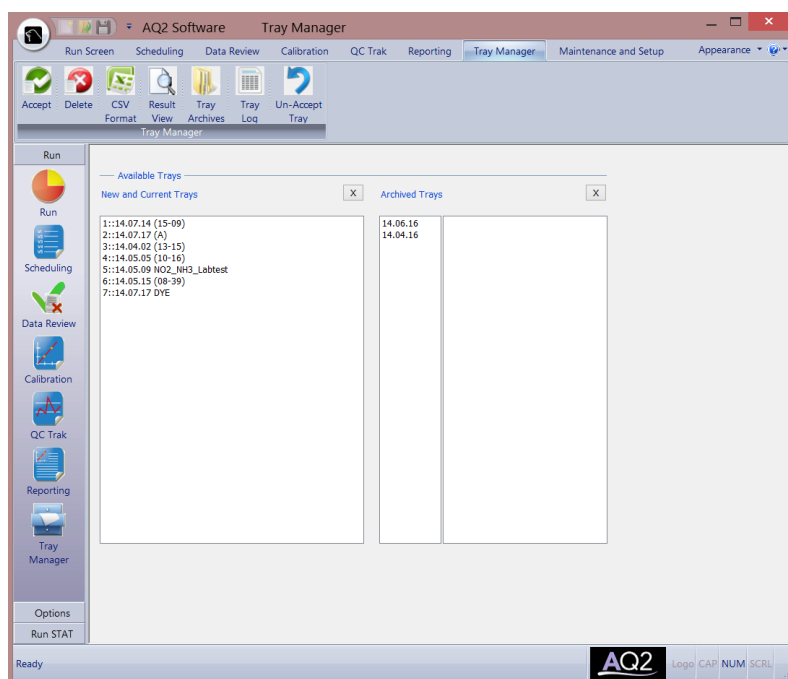
5.10.14 Reports for Archived Trays

When a tray is accepted a Standard Report including Test Parameters is generated and filed under the Audit sub-directory of the AQ folder located on the C:\ drive. These reports are stored in HTML format and stored by date of acceptance. These reports can also be produced by finding the tray in the archived section of the **Reporting** screen and generating a **Standard Report**.

5.11 Tray Manager



This section collects several tray level operations into one area. The user can quickly accept or delete current trays, edit the CSV file output format, search for results of specific samples, access tray archives, view tray logs, or un-accept archived trays.



5.11.1 Accepting Trays

The **Accept** option allows the user to accept trays that have been completed. This operation is analogous to accepting a tray in Data Review.

Select the tray that will be accepted under the **New and Current Trays** column. Then, click the **Accept** button at the top. The tray will go into final acceptance and will be moved over to the **Archived Trays**.

The tray will not be archived if some points are still marked as Pending (PND) or marked to be Rescheduled (RSC) in **Data Review**. To continue, either complete the tray or mark PND points as cancelled and remove the RSC flag from samples flagged as RSC using the tools in **Data Review** (section 5.6).

5.11.2 Deleting Trays

The **Delete** option allows the user to delete trays. Both complete and incomplete trays can be deleted, but trays that have already been Archived (Accepted) cannot. This operation is analogous to deleting a tray in Data Review.

Select the tray that will be deleted under the **New and Current Trays** column. Then, click the **Delete** button at the top. A dialog box will appear asking if you are sure you wish to perform this action. If yes is selected, all traces of this tray are removed, it is **not** archived, and no reports are generated. **THIS STEP IS IRREVERSIBLE**. If no is selected, then no changes are made to the selected tray.

5.11.3 Editing CSV Format

When a tray is accepted, an output file in Comma Separated Values (.CSV) format is created. The default location for storing this file is in the Out sub-directory of the AQ folder located on the C:\ drive (C:\AQ\out). To edit what is included in this .CSV file, select the **CSV Format** icon.

The following dialog box will appear. The first column (**Sample Field**) allows the user to select which data fields are to be included in the file. To include a Sample Field in the .CSV file simply tick the box next to it. Next, the user must decide the **Order** these fields will display. Each field will be a separate column, and will be displayed sequentially, 1-15, left to right. Finally **Heading Text** (column header for exported file) can be chosen for each selected Sample Field. An additional column can be added, called **Total Dil Factor**, which combines the Auto and manual dilution factors and reports in one column of the .CSV file.

There are two other options in this window. Selecting **Calibrate standards against curve** will cause the .CSV file to show concentration values for calibrations standards in the **Results** column instead of the usual absorbance values. Selecting **Split date and time** will cause the **Date and time** column to separate into individual **Date** and **Time** columns in the CSV file.

Lastly, there are two options to **Export archived results on file** and **Export all in chronological order**. The first will export all results that were run with that tray. If chronological order is selected it will export in chronological order.

CSV Output Format

Sample Field	Order	Heading Text
☐ Sample Type	0	
☒ Sample ID	1	Sample ID
☒ Sample Details	2	Sample Details
☐ Test Number	0	
☒ Long Test Name	3	Test
☐ Short Test Name	0	
☒ Units	5	Units
☒ Result	4	Results
☒ Absorbance	6	Absorbance
☒ Time Of Analysis	11	Date and Time
☐ Split date and time		
☒ Operators Initials	8	Operator
☐ Tray Number	0	
☐ Cup Number	0	
☒ Man Dil Factor	9	Man Dil Factor
☒ Auto Dil Factor	10	Auto Dil Factor
☐ Well Result	0	
☒ Recoveries	7	QC Pro result
☐ Total Dil Factor	0	
☐ Error codes	0	
☐ Export archived results on file		
☐ Export all in chronological order		

☐ Calculate standard against curve

Set defaults

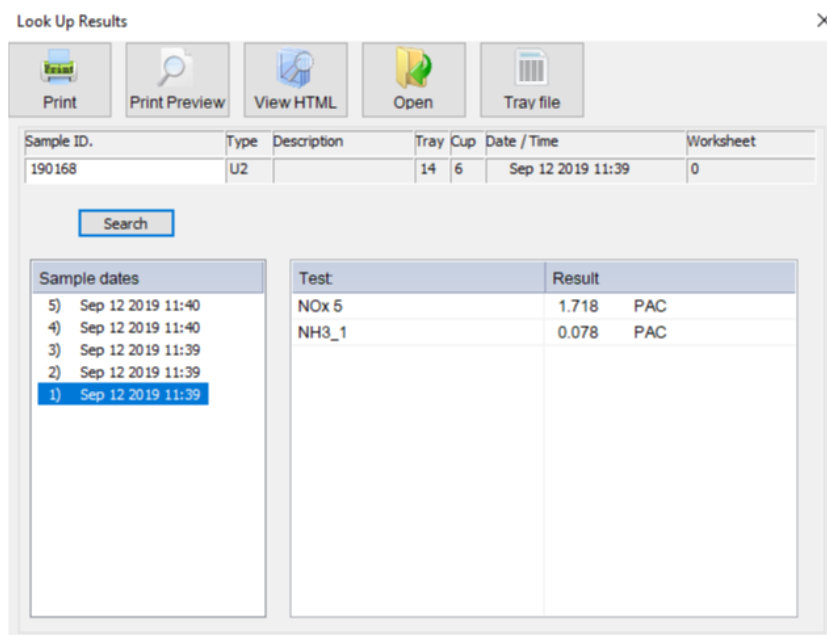
Clear All

OK

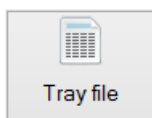
Cancel

5.11.4 Result View

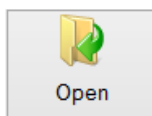
Result view allows the user to search the archived folders for a specific Sample ID. To begin, type a **Sample ID** in the designated box and hit **Search**. A list of each time that Sample ID was used will appear in the box on the left called **Requisitions for Current Sample**. Select a result to see information about that sample. This information includes Sample ID, Sample Type, Sample Details (Description), Tray Number, Cup Number, and Date/Time.



By selecting a sample and clicking the **Tray File** icon, the user can obtain a text file.



By selecting a sample and clicking the **Open** icon, the user can browse through all the Sample IDs used since the software was installed. The IDs are sorted into folders by the first letter of the Sample ID.



The **Print**, **Print Preview**, and **View HTML** buttons function similarly to the same buttons in the **Reporting** tab. **View HTML** will create an HTML report of the selected sample IDs along with their corresponding results.

5.11.5 Tray Archives

Tray Archives allows access to the archived HTML reports that are generated when a tray is accepted. These reports are sorted in directories that represent the day when the tray was accepted. Double-click on a directory to open it. Double-click on a tray name to open the report in your default browser.

5.11.6 Tray Log

Selecting **Tray Log** allows a user to see the software's log of a tray. This log will keep track of everything done on an individual tray. Every result, rerun, dilution, QC, etc. are all tracked in the Log file. For more information on *Tray Logs* see section 7.1.

5.11.7 Un-Accepting a Tray

AQ Software allows a user to **Un-Accept** a tray, bring it out of archives and restore it as a Current Tray. The user can then open the tray in **Data Review** and manipulate it just like any other New or Current Tray. When the unaccepted tray is first opened in **Data Review**, all results will be labelled **ACC**. This label must be removed before anything else can be done with the tray. To remove the label, highlight all samples and click the **Accept** button (not the **Accept All** button).



Note:

For an archived tray to be Un-Accepted, its original Tray Number must be **FREE**. It cannot appear in New and Current Trays.

6 **Advanced Settings**

6.1 Reagent Sets

This part of AQ Software allows the user to view the five defined Reagent Sets, enter Batch Number, Expiration and Preparation Dates, and assign previously defined reagents to separate locations on the Reagent Tray. To bring up this window select **Reagent Sets** from the **Maintenance and Setup** tab.

Reagent Sets

4 NH3 NO2 NO3 CN Reagent set 3 Reagent set 4 Reagent set 5

Reagent Set Name: NH3 NO2 NO3

Pos	Reagent Name	Volume	Batch Number	Expiry Date	Operator	Preparation Date
1						
2	Salicylate	44.28		1/11/2019	seal	12/11/2018
3	DCI	44.68		1/8/2019	seal	1/7/2019
4	Sulfanilamide-NEDD	30.40		2/7/2019	seal	1/7/2019
5	NH4Cl Buffer	34.84				
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						

Print

Delete

OK Cancel

The window shows 5 tabs. Each tab refers to one of five Reagent Sets.



Note: New reagents can only be added through Reagent Parameters. For more information on adding new reagents see section 4.4.

6.1.1 Positioning Reagents

The first column, **Pos**, indicates the position of that Reagent on the Reagent Tray. The AQ300 has 18 positions, the AQ2 has 15 positions, and the AQ400 has 26 positions.

The second column is the **Reagent Name** as defined in the **Reagents** window.

You may also assign a single reagent to multiple locations on a single Reagent Set. This is helpful for long runs when one 40 mL reagent wedge will not provide enough reagent to complete the test. To do this, select the reagent name from the drop down menu in all desired positions for this specific reagent.

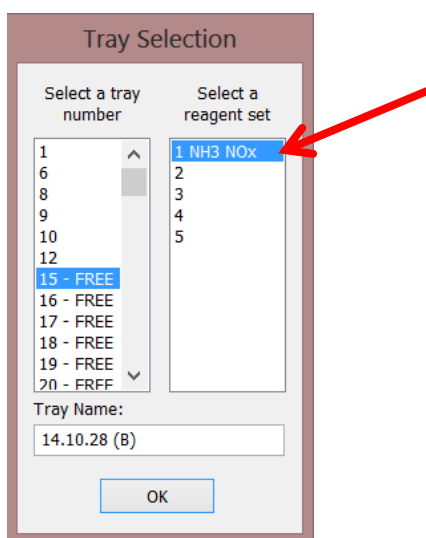
6.1.2 Reagent Volume

The AQ300, AQ2, and AQ400 will actively track Reagent Volumes during a run. The user can manually enter a starting **Volume** on the Reagent Sets page. A full Reagent Wedge holds

45 mL. When the instrument is equipped with an active Reagent Level Sensor, the analyzer will measure the volume of each reagent at the beginning of a run and determine if there is enough of each reagent to complete the tray. This volume will be automatically entered into the **Volume** field in Reagent Sets. The software will automatically subtract the volumes used from the reagent wedge as the run progresses.

6.1.3 Naming the Reagent Set

To name the Reagent Set, simply enter a name in the **Reagent Set Name** box. This name will then appear on the Tray and during Tray Selection in **Scheduling** as shown in the image below.



6.1.4 Additional Reagent Information

AQ Software can also track the lifetime of each reagent. From the Reagent Sets screen, the user can input a **Batch Number**, **Expiration Date**, and **Preparation Date** for each reagent. Changing any of this information will automatically update the **Operator** column with the name of the user currently logged in.

This information will appear on a Standard Report when generated through the **Reporting** section of AQ Software, if selected.



Note:

If a user attempts to run a test with a reagent that is past its expiration date the software will flag the run and not allow it to start until the Reagent Set information is updated with a non-expired reagent.

6.1.5 Saving Changes

Click **OK** to save changes or **Cancel** to revert to the previous settings.

6.2 Test Profiles

Test Profiles allows the user to schedule multiple tests with a single click of the mouse. To begin, select **Test Profiles** from the **Maintenance and Setup** tab.

All of the **Available Tests** will be listed on the left side of the window. To add a test to a **Test Profile** simply select the desired test and hit the **Add** button. Do this with all of the tests to be combined into a single Test Profile and name the Test Profile. You can remove tests from a Profile by selecting the test in the **Tests in Profile** column and clicking the **Remove** button.

Edit Test Profiles



Note:

All of the tests selected for a Test Profile must have their reagents located within the same Reagent Set. For more information on Reagent Sets please see section 6.1.

The AQ Software allows for up to 5 Test Profiles to be saved. The profiles can also be named by editing the **Test Profile** box. This is the name that will appear on the scheduling screen.

To Schedule the Test Profile, select it from the **Profiles** window on the **Scheduling** screen.

6.3 Test Order

The AQ Software allows for multiple tests to be run on a single tray. The default order for these tests to be run is based on the reaction time, from shortest to longest. **Test Order** allows the user to manipulate this sequence. To begin, select **Test Order** from the **Maintenance and Setup** tab.

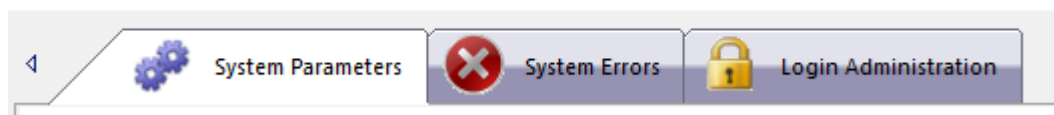
[illegible]

The user can now program the software to run tests in a specific order by assigning each test an order number. Tests with the lowest order number will be run first. If some tests have order numbers and some do not, those with order numbers will be run before those with no number. Tests with no order number will continue to be run in an order based on their reaction time.

6.4 Advanced Parameters

Advanced parameters can be found by selecting **Advanced Parameters** from the **Maintenance and Setup** tab. The three tabs available from the pop-up screen are **System Parameters**, **System Errors** and **Log-in Administration**.

Advanced Parameters



6.4.1 System Parameters

System Parameters are advanced settings which can influence the system's behavior. To enter **System Parameters**, go to **Maintenance and Setup** and select the **Advanced Parameters** icon.

Advanced Parameters

System Parameters System Errors Login Administration

Property	Value
COM Port	COM
Serial Number	
Automatic Maintenance Prompt	Yes
☒ Analyser Parameters	
Auto dilute	Yes
Smart Auto dilution	No
Auto stop	Continue to end of tray
Use multi-rule control evaluation	No
Run In development mode	Yes
☒ Reporting Parameters	
Allow users to choose output file name	Yes
Report results below zero as zero	No
Print software version on reports	Yes
Show serial number on reports	No
Archive the test parameters	Yes
Date format	2019-10-21 10:11:34 (YYYY-MM-DD hh:mm:ss)
Extra audit trail fields	No

Defaults OK Cancel Apply

COM Port

COM Port must be set to the port number of the AQ USB device. The analyzer must be turned on and connected via the correct USB cable. The dropdown box will display all available Com Ports. To confirm the Com Port assigned (typically COM3 or COM4) go to the ports settings of Windows® Device Manager which is accessed via the Control Panel. For more information see section 2.4.

Serial Number

The serial number of the instrument can be input into this box. This can then be shown on all future reports.

Automatic Maintenance Prompt

If yes is selected for Automatic Maintenance Prompt, then the software will prompt the user when opening the software for Daily, Weekly, Monthly, etc. maintenance on the instrument. If selected as no, then there will be no future prompts when logging on or off the software.



Note:

The timing of maintenance prompts is based on the date of installation using the computer date and time.

Analyzer Parameters

Auto Dilute allows the user to select whether the system will prepare and run dilutions on over range samples automatically. If “Yes” is selected, then the system will auto-run dilutions. If No is selected, auto dilution will not occur, no matter what auto-dilution information has been set in **Test Parameters**.

Smart Auto Dilute will allow the software to automatically reschedule samples that read higher than the calibration range with a dilution factor based on the sample's absorbance rather than based on the auto-dilution factor set in **Test Parameters**.

Auto Stop is a system wide control which determines a response should a control fail during a run. If running a generic control from the Controls tab of **Test Parameters** and the control fails, this setting is used to decide on a course of action. If running QC PRO samples, this is used to decide on a course of action should a control fail for a second time. For more information on each of the options, see section 4.6.5.

Use multi-rule control evaluation allows the user to activate the multi-rule evaluation for scheduled controls. This protocol will follow what is entered into **Test Parameters** of the AQ software.

Run in development mode, if set to “Yes”, will set the software to run without being connected to an instrument. Users should not enter Development Mode unless instructed by SEAL staff.

Front shield alarm – AQ400 & AQ300 Only – has 3 possible settings. In all 3 settings the Front Shield Alert Box on the **Run** page will indicate if the front shield is open. (RED – Open | Ouvert and GREEN - Closed | Fermé). With “NO” selected, the software will take no action based upon the front shield. With “Stop in 2 Cycles” selected, the software will not allow a RUN to start with the Front Shield Open and will give an error message. If a run is in progress and the Front Shield is opened, an error will display “Front Shield Open – Stopping in two cycles” after the two cycles the instrument will do an orderly stop. With “Yes” selected, the software will not allow a RUN to start with the Front Shield Open. If the Front Shield is Opened during a RUN, the software will do an immediate stop and issue a notification.

Reporting Parameters

Allow users to choose output file name, if set to “Yes,” will allow the user to save the .CSV file as any name and to any location selected. If set to “No”, the .CSV file will be automatically saved with the current date as the file name and saved into the c:\AQ\out folder.

Report results below zero as zero will report any negative values as zero.

Print software version on reports will show the software version number on any reports that are generated.

Show serial number on reports will allow the software to display the serial number (entered into the **Serial Number** box above) of the analyzer on the report.

Archive Test Parameters allows the user to archive your Test Parameters with the report in html format.

Date Format allows the user to pick from five date formats that would be displayed on your report.

10/28/2014 15:27:29 (MM/DD/YYYY hh:mm:ss)

28-Oct-2014 15:27:29 (DD-MMM-YYYY hh:mm:ss)

28/10/2014 15:27:29 (DD/MM/YYYY hh:mm:ss)

Tue Oct 15:27:29 2014

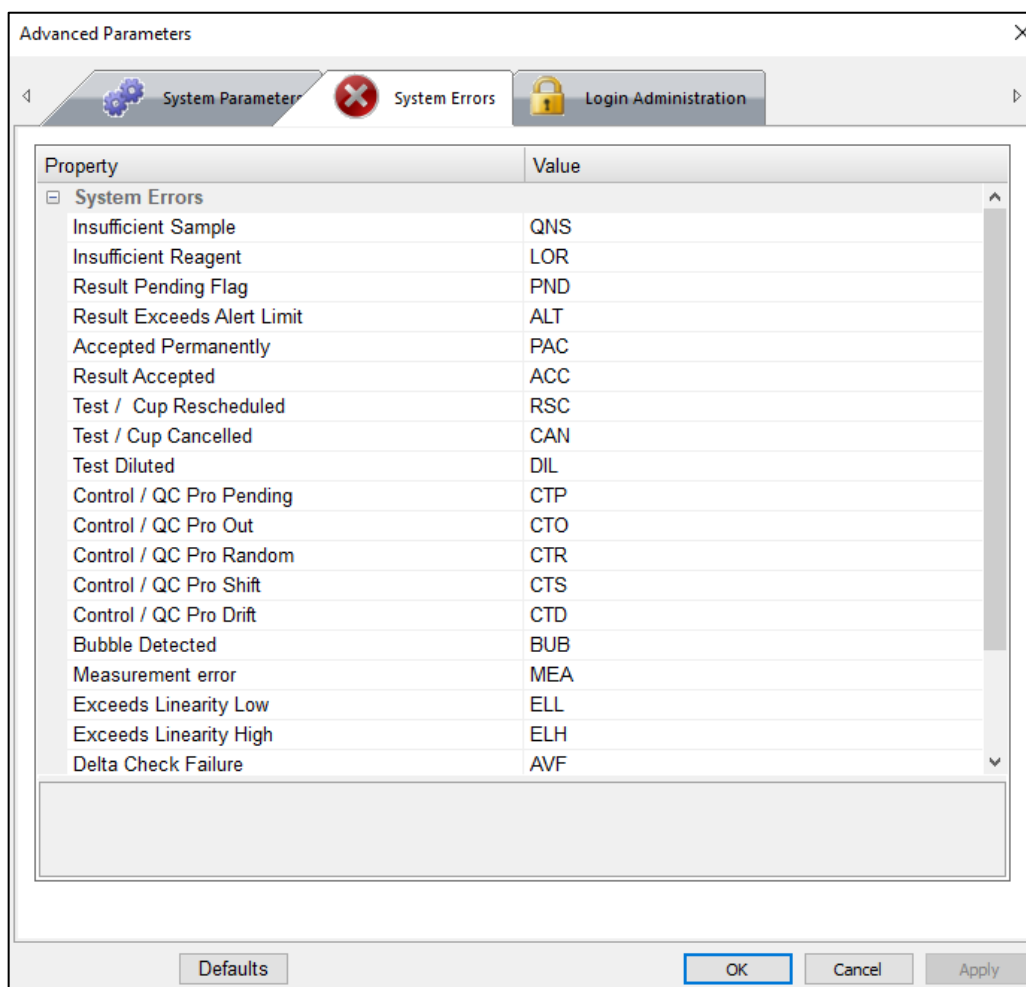
2014-10-28 15:27:29 (YYYY-MM-DD hh:mm:ss)

Extra Audit Trail fields adds Details, QC Pro Results, and QC Pro Decision to the audit trail (found under **Tray Manager – Tray Log**).

To save any changes, click **Apply** and then select **OK** to close out of **System Parameters**.

6.4.2 Edit System Error Codes

System Errors appear in Data Review, the Run screen, and some Reports. They are used to flag results that may have a problem associated with them. These error codes are customizable by the user. The default codes are shown below, with a brief description of each. To edit **Error Codes**, go to **Maintenance and Setup – Advanced Parameters – System Errors**.



Active and Commonly Used Codes

PND **Result Pending Flag** appears in the run screen until an absorbance/concentration has been measured.



Note:

When rescheduling a partial run, the original data will remain until it is replaced by the new determination.

ALT **Result Exceeds Alert** appears when the concentration of the sample or standard is not within the high and low concentration alert limits set by the user when building the method in the Test Parameters menu.

PAC **Accepted Permanently** appears when a result has been accepted and archived.

ACC **Result Accepted** appears when a result has been accepted but not yet archived.

RSC **Test/Cup Rescheduled** appears when a result has been rescheduled.

CAN **Test/Cup Cancelled** appears when a result has been cancelled.

CTP **Control/QC Pro Pending** appears pending the concentration of the controls defined in the analysis.

CTO **Control/QC Pro is Out** appears when the concentration of a control is not within the limits defined by the user when building the method in the **Test Parameters** menu.

MEA	Measurement Error appears typically if there is a loss of communication while reading.
ELL	Exceeds Linearity Low appears if the absorbance is less than the absorbance of the calibration blank or zero point.
ELH	Exceeds Linearity High appears if the absorbance is greater than the top standard absorbance.
AVF	Delta Check Failure appears if a sample has been rescheduled and the new results is greater than the allowable percent difference set in the main parameters tab of Test Parameters .
CAL	Calibration Error appears in the message window on the run screen if there is an error during the calibration.

Inactive or Rarely Used Codes

DIL	Test Diluted appears when a sample has been diluted.
QNS	Insufficient Sample , however this will not appear while the sample level sensor is turned off (default).
LOR	Insufficient Reagent will only appear in the run screen if the reagent level is too low to complete the scheduled analysis.
BUB	Bubble Detected appears if a bubble has been detected in the measurement. It looks for the absorbance between 2 consecutive cups to contain the same pattern of numbers (5 digits in a row must match). If the digits match for the 2 samples, it flags BUB to say the result has a possible problem.
CTR	Control/QC Pro Random appears when three controls during an analysis have randomly failed (only if Multi-rule control evaluation is set to "Yes").
CTS	Control/QC Pro Shift appears when a control fails at the beginning and end of an analysis. The flagged results fail in the same direction either high or low concentration hence shifting the results of the analysis either up or down (only if Multi-Rule Control Evaluation is turned on).
CTD	Control/QC Pro Drift appears when a control fails at the beginning and end of the analysis. The flagged results fail in different directions with one having a high concentration and the other with a low concentration (only if Multi-Rule Control Evaluation is turned on).

6.4.3 Customize Error Codes

To change an error code simply type the desired code into the appropriate field. The code can be a maximum of three characters. When changes are complete, click **OK** to save the new definition. Clicking **Cancel** will restore the previous value. **Defaults** overwrites all user changes with the standard SEAL settings.

6.4.4 Log-In Administration

This utility allows an AQ software administrator to create new users, delete old users from the system, or reset the password for individual users. To enter Log-In Administration, go to **Setup – Advanced Parameters – Log In Administration**.

The screenshot shows the 'Advanced Parameters' dialog box with the 'Log In Administration' tab selected. The dialog has three tabs: 'System Parameters', 'System Error', and 'Log In Administration'. The 'Log In Administration' tab contains a table with two columns: 'Property' and 'Value'. The table has the following rows:

Property	Value
Select User or Enter New User	
Initials	
Authorization	
Failed Logins	0
☐ Reset Password	
☐ Reset Password	
☐ Delete User	
☐ Delete User	

At the bottom of the dialog are four buttons: 'Defaults', 'OK', 'Cancel', and 'Apply'.

Add/Modify User Detail

To add a new user, type their name into the **Select User or Enter new User** field. To modify a user's details, select their name from the drop down list.

User Initials

Each user must have unique initials entered into the **Initials** field. These are added to all sample records as a permanent record of the person who performed each analysis.

Authorization

The authorization level must be set. A **User** may access all basic functions of the software but cannot alter Test Parameters or Advanced Settings. A **Supervisor** can use all functions except Log-In administration. The **Administrator** can use all functions.

Failed Log-Ins

Each user is allowed 9 **Failed Log-Ins** before the system locks them out of the software. Failed Log-Ins tracks how many times a user has entered an incorrect password since their last successful entry. If the number of Failed Log-Ins exceeds 9, the user will be locked off until their password is reset.

Reset Password

Reset Password will reset the selected user's password. After the reset, the password will be the same as the username, character for character. Upon first log-in after the password is reset the user will be prompted to change their password.

Delete User

Delete User removes a user from the system. A deleted user will not be able to enter the software. However, their log-in name and initials remain attached to any trays that had been run while they were logged in.



Note:

Log In Names or Initials can never be used again, even if that entry is deleted.

Log In as New User

Once a new user has been added to the AQ software, exit out of the AQ software and then re-log into the AQ software. Type the new user's log in name into the **Log In Name** Field, or select it from the drop-down menu. The **Password** is the same as the **Log In Name**, character for character. Then select **OK**.

SEAL Analytical

AQ2

Serial Number

Log In

Log In Name

Password

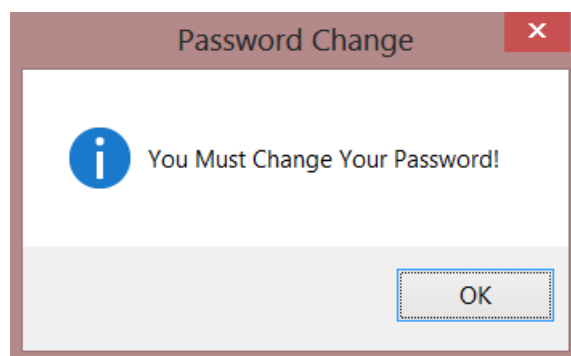
Change password

New Password

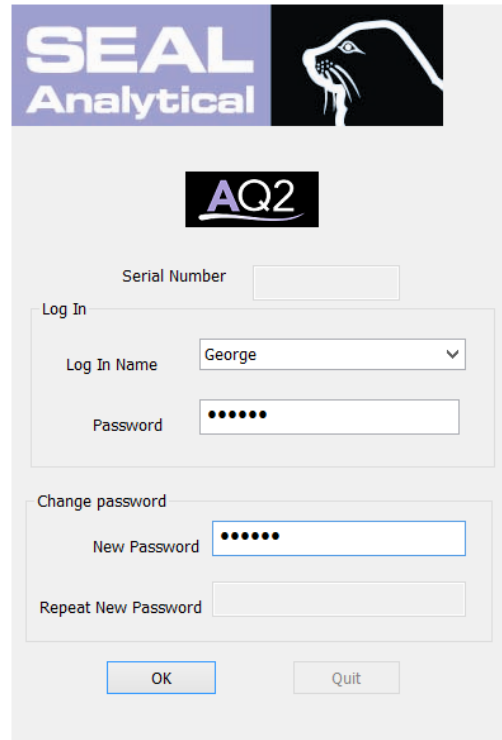
Repeat New Password

OK Quit

The AQ software will now prompt the user to change his/her password.

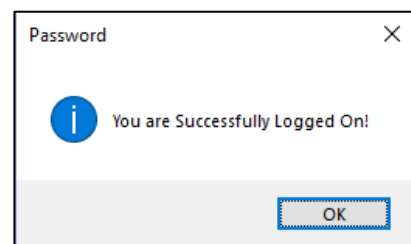
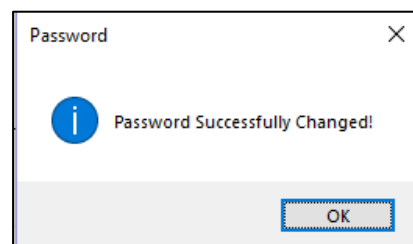


Enter the new password into the **New Password** field. The password must be a minimum of 6 characters and it is case sensitive. Select **OK** and re-enter the new password into the **Repeat New Password** field.



The image shows the SEAL Analytical AQ2 software interface. At the top, there is a logo for SEAL Analytical and a stylized seal head. Below the logo is the text 'AQ2'. The main area contains a 'Log In' section with a 'Serial Number' field, a 'Log In Name' dropdown menu (currently showing 'George'), and a 'Password' field with six dots. Below the 'Log In' section is a 'Change password' section with a 'New Password' field (six dots) and a 'Repeat New Password' field. At the bottom, there are two buttons: 'OK' and 'Quit'.

If done correctly, selecting **OK** will prompt the following 2 screens to appear, click **OK** on both to successfully log onto the AQ software.



7 Extended Operation

7.1 Tray Logs

Tray logs are generated in real time and track everything done to a tray. The user can view these while a tray is running as well as after it has completed.

The logs record significant information as it happens. This includes test results, calibration results, QC results, and auto-dilution results.



Note:

The real-time logs are an ideal way to track everything the analyzer does, and allows you to fully audit all the work done on the instrument.

7.1.1 Viewing Real-Time Logs

There are two ways to get to the Log Viewer from the AQ Software.

Choose the **View Log** option from the **View** menu. This option will load the log for the tray currently loaded in the Run Screen. If there isn't an active tray on the Run Screen nothing will happen.

In **Tray Manager**, select a tray and then click the **Tray Log** button. This will load the log for that tray. This will work for any tray that has been run (or partially run). This includes any archived trays.

In addition, the logs can also be opened as a text file from the logs folder located in the AQ folder on the C:\ drive.

7.1.2 Formatting Options

The **Log Viewer** screen provides several options for the user to customize the output of the log. These are detailed in the following sections.

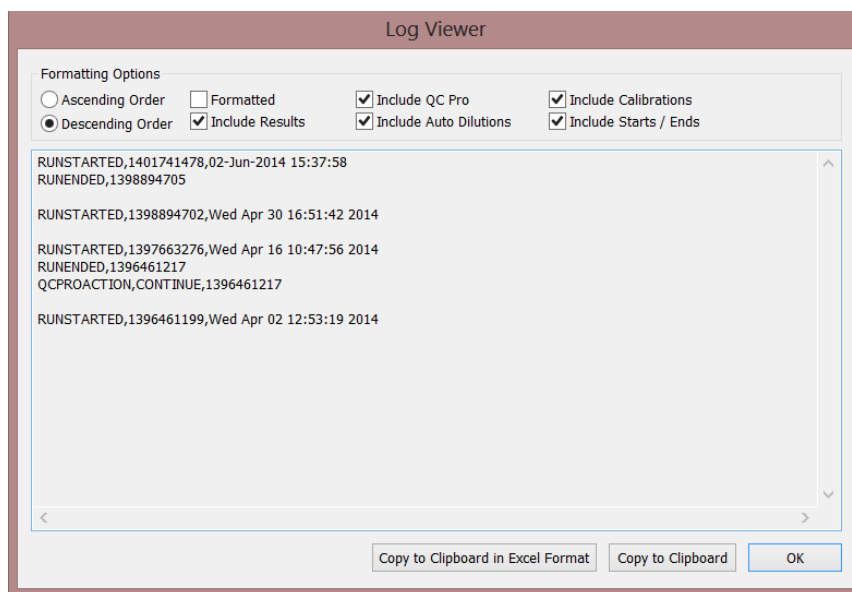


Note:

The options provided in this page are merely formatting options. All data is included in the log file, however the user chooses how it is displayed.

Display Order

These options will display the discrete log entries in either an **Ascending Order** or **Descending Order** of time.



Format Log

Ticking the **Formatted** checkbox will take the raw, comma separated data and reformat it into an organized sample-by-sample list.

Include Results

Ticking this checkbox will include the result entries from a test including the test name, sample ID, and the result. This includes Standards and Controls.

Include QC Pro

Ticking this checkbox will include the further information about Control types. This extra information includes whether the control was out and, if so, what action was taken.

Include Auto Dilutions

If a sample returns an over-range result and this box is ticked, the log will include entries to indicate whether an auto dilution has been scheduled.

Include Calibrations

Ticking this checkbox will include a notification of any calibrations that may occur while a tray is running.

Include Start/End

Ticking this checkbox will include data on when the tray began and ended its run.

7.1.3 Export Options

Copy to Clipboard

Clicking this button copies the contents of the **Log Viewer** to the computer's clipboard.

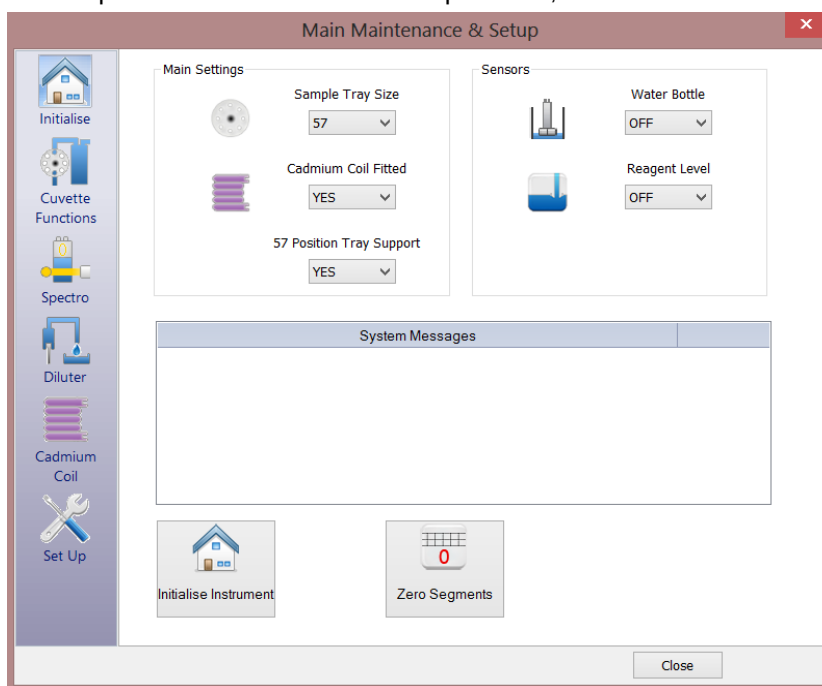
Copy to Clipboard in Excel Format

If you wish to copy unformatted, tabular data into Microsoft Excel you should click this button. During the copy procedure all the commas will be transformed into 'Tab' characters. This will allow each data item to be placed into the correct Excel column.

8 Maintenance and Set Up

8.1 Introduction

To enter software maintenance, first make sure that the instrument is switched on and there is nothing blocking any of the motorized/moving parts. When ready select **Maintenance** from the **Maintenance and Setup** tab. Upon entering this menu the instrument will **Initialize** and all of the motorized parts will move to their home positions, and the screen below will appear.



8.2 Regular Maintenance

Upon start-up, the AQ software will open an html window containing a checklist of maintenance work that needs to be done for the day (to turn this setting off, see section 8.2.1).

These will pop-up with the following frequencies:

- Daily
- Weekly
- Monthly
- Semi Annually

These notifications are cumulative. For example, a weekly notification will also contain that day's daily notification. These notifications are meant to compliment the Maintenance Checklists found in the Customer Support Manual.



Note:

A copy of the Customer Support Manual can be found on the SEAL Analytical Website, www.seal-analytical.com.

8.2.1 Turning Off Regular Maintenance Notifications

If the user does not wish to see these regular notifications, they can be turned off in **System Settings**. This screen can be reached by clicking on the **Advanced Parameters** icon from the **Maintenance and Setup** tab. Select "No" for **Automatic Maintenance Prompts** and click **OK**.

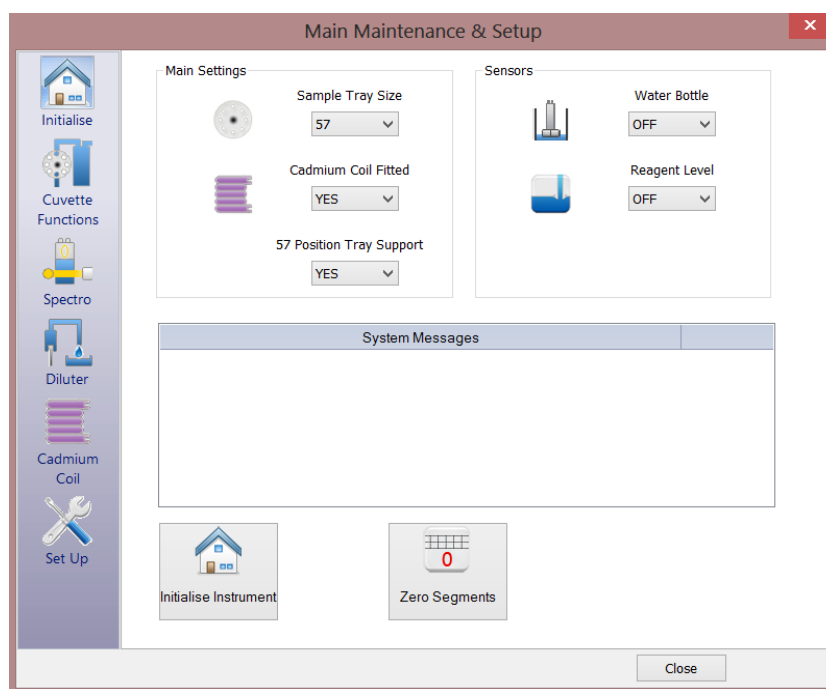
8.2.2 Daily Shutdown Maintenance

Every time the AQ Software is shut down, a prompt will ask if the analyzer is being shut down overnight. This is not ticked by default. If this box gets ticked, the **Daily Evening Maintenance Checklist** will be opened.

8.3 Maintenance

The Software Maintenance is split into six sections, each of which can be accessed via the icon on the left: **Initialise**, **Cuvette Functions**, **Spectro**, **Diluter**, **Cadmium Coil**, and **Set Up**.

8.3.1 Initialize



Sample Tray Size

This allows the user to change the size of sample wheel to be loaded onto the instrument. For the AQ300 the choices are 57 position (2 mL sample cups) and 97 position (1.2 mL sample cups). For the AQ2 they are 57 position (2 mL sample cups) and 100 position (1.2 mL sample cups). For the AQ400 the choices are 80 position (60 – 2 mL cups and 20 – 5 mL cups) and 120 position (100 – 1.2 mL cups and 20 - 2 mL cups).

Cadmium Coil Fitted

This setting should be set to **Yes** if the instrument is equipped to have a Cadmium coil (i.e. has a Cadmium Switching Valve) and should be set to **No** if it is not. This setting can only be edited when logged in as a SEAL Authorized User.

57 Position Tray Support (AQ2 Only)

This option, available on AQ2 only, allows the user to toggle on or off the 57-position tray size from the Sample Tray Size box.

Water Bottle

If set to **On**, the **Water Bottle** sensor will alert the user when the instrument is almost out of water. If the user fails to refill the reservoir in time, the instrument will clean itself out and stop before it runs completely out of water. The alert will go off 10 cycles before stopping. If set to **Off**, no alert will be issued.

Reagent Level (AQ2 and AQ400, some AQ300s)

The **Reagent Level** sensor uses a small sensor on the Sample Probe to read reagent volumes at the beginning of a run.

When set to **Auto**, at the beginning of a run the analyzer will check the reagent volumes for all of the reagents needed for the run. It will then calculate if there is enough of each reagent to complete the test. If not, it will prompt the user to add additional reagent.

If set to **Off**, the user will need to manually put in the initial reagent volume and the software will track the amount of reagent used.

System Messages

While in the Maintenance menu, any messages that the instrument outputs (i.e., Water Baseline readings or Error Messages) will appear in this window.

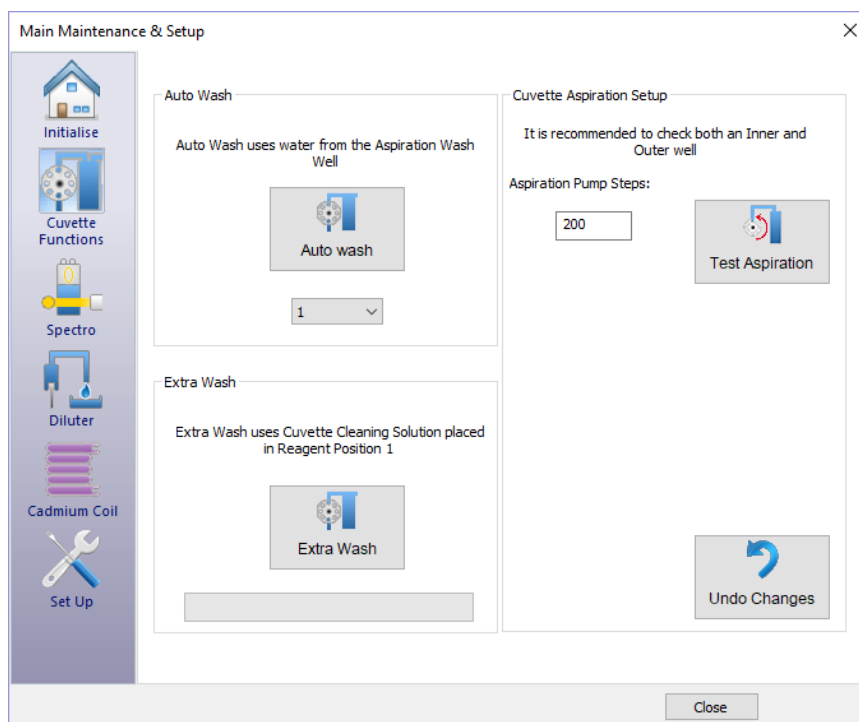
Initialization Button

The **Initialise Instrument** button will send all of the motorized parts to their home position.

Zero Segments

Selecting this icon will prompt the user to replace the used reaction segments. A dialogue box will appear that reads **Replace Used Reaction Segments # through #?** Selecting **Yes** confirms the used segments have been removed. If **No** is selected, the analyzer will continue from the last used reaction well.

8.3.2 Cuvette Functions



Auto Wash

An **Auto Wash** uses water from the Water Bottle to pull high speed flushes through the Cuvette. A range of 1 to 5 washes can be selected using the drop down menu. Click the Auto Wash icon to begin.

Extra Wash

An **Extra Wash** cleans the cuvette with a cleaning solution placed in Reagent Position 1. After pulling the cleaning solution through the cuvette three times, the analyzer will automatically rinse with water from the Water Bottle several times to complete the cycle.

Test Aspiration

A **Test Aspiration** dispenses 500 μL of water into a reaction segment which is then pulled into the cuvette to simulate a reaction.

Aspiration Pump Steps

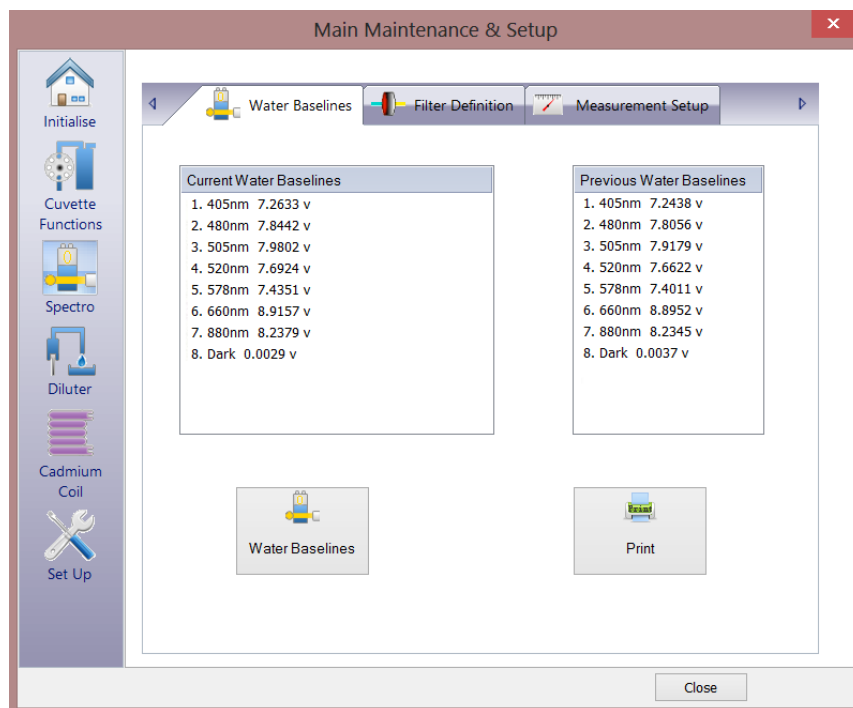
This determines the number of pump steps used to pull the reaction liquid from the Reaction Well to the Cuvette. It should be adjusted so that at least 1-2 inches (3-5 cm) of liquid can be seen entering and exiting the Cuvette. Click the **Test Aspiration** icon to test the current settings.

Undo Changes

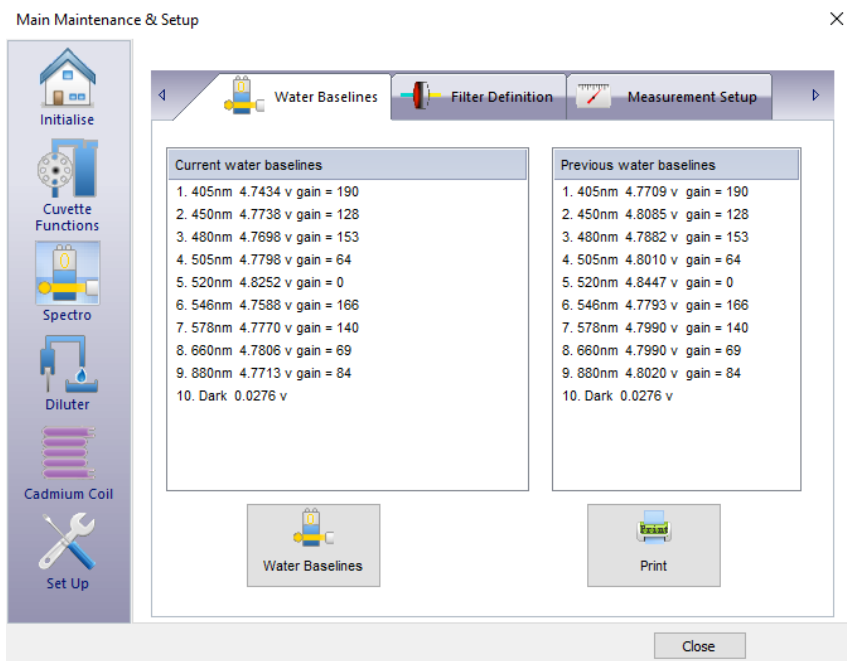
This icon eliminates any changes made and restores the previous settings.

8.3.3 Spectro – Water Baselines

Water Baselines allows the user to view the current and past Water Baseline values, generate new Water Baselines, or print the Water Baselines. To enter this section of **Maintenance**, select the **Spectro** icon from the left and then select the **Water Baselines** tab.



Adjustable or Fixed Offset



Auto-gain Spectro (AQ300 and AQ400 only)

Current Water Baselines

This window displays the most recent Water Baseline values determined by the analyzer. For the AQ2, these values should be between 4.5 and 8.9 volts for filters 1 through 7, and between 0.0025 and 0.0045 volts for filter 8. For AQ400 with adjustable offset, the filters 1 through 9 should be between 3 and 3.9 volts and 0.003 ± 0.001 for filter 10. For AQ300, AQ2 and AQ400 with fixed offset, filters 1 through 9 should be between 3.5 – 4.85 volts and 0.027 volts for filter 10. For AQ300 and AQ400 with auto-gain, filters 1 through 9 should be between 4 – 4.85 volts and 0.027 volts for filter 10. To adjust these settings please see section 8.3.5 **Spectro – Measurement Setup**.

Previous Water Baselines

This window displays the last Water Baseline readings. Comparing previous values to current values can help determine if the instrument is operating correctly.

Water Baselines (Icon)

This icon will take a new set of Water Baseline readings. Current Water Baselines will be updated with these values.



Note:

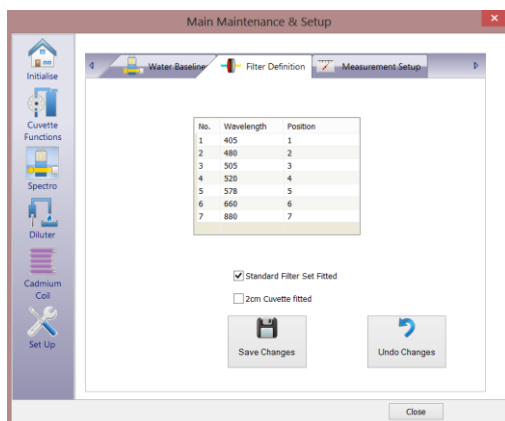
The cuvette needs to be entirely filled with water to take accurate Water Baseline readings.

Print (Icon)

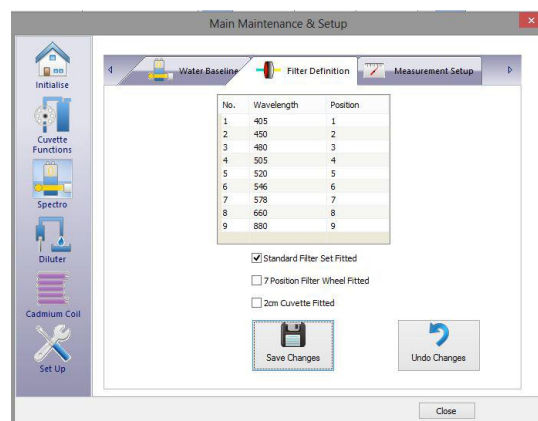
This icon will print off a two column report showing the Current Water Baselines and the most recent set of Previous Water Baselines.

8.3.4 Spectro – Filter Definition

Filter Definition allows the user to customize the filter wheel set up. To enter this section of **Maintenance**, select the **Spectro** icon from the left and then select the **Filter Definition** tab.



AQ2 – 7 Position Filter Set



AQ300 & AQ400 – 9 Position Filter Set

Filter Definitions

This table states the **Wavelength** and **Position** of each filter on the Filter Wheel. This table can be edited by a SEAL Authorized account only. Clicking the **Save Changes** icon saves the new filter definitions to the AQ software memory. Selecting **Undo Changes** eliminates any changes made and restores the previous settings.

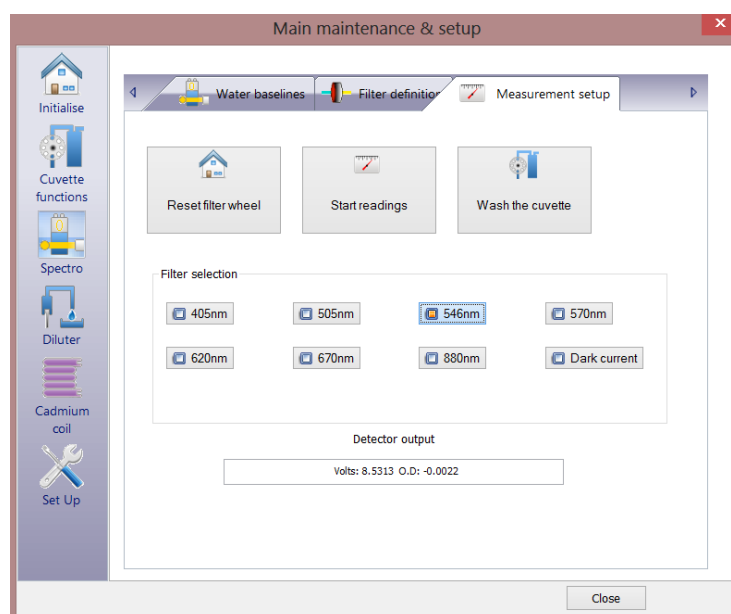


Note:

Changing the wavelength values in the AQ Software does not change the physical location of the filters.

8.3.5 Spectro – Measurement Setup

Measurement Setup allows the user to view live Water Baseline readings. To enter this section of **Maintenance**, select the **Spectro** icon from the left and then select the **Measurement Setup** tab.

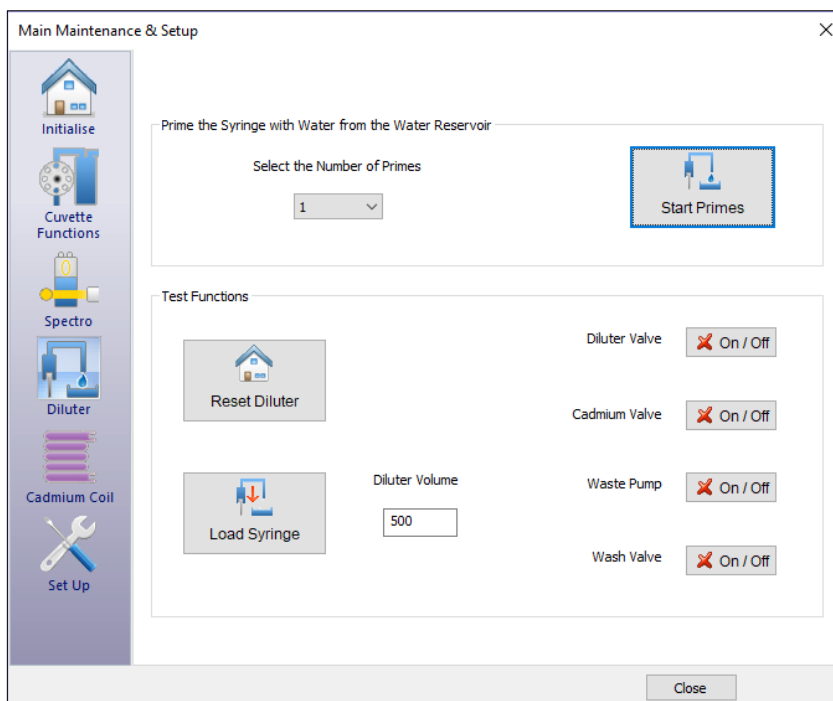


To view live Water Baseline readings, first click the **Wash the Cuvette** icon to pull water into the cuvette from the Water Bottle. Next, click the **Start Readings** icon and select one of the eight (ten for the AQ400 and AQ300) filter icons. The live voltage reading will be displayed in the **Detector Output** window at the bottom of the screen. The **Reset Filter Wheel** icon will reset the filter wheel to its home position (Filter 4 by default). As described previously, for the AQ2, filters 1- 7 should have Water Baseline values between 4.5 and 8.9. Filter 8, the Dark Current, should have a value between 0.0025 and 0.005. For AQ400 with adjustable offset, the filters 1 through 9 should be between 3 and 3.9 volts and 0.0030 ± 0.001 for filter 10. For AQ300, AQ2, and AQ400 with fixed offset, filters 1 through 9 should be between 3.5 – 4.85 volts and 0.027 volts for filter 10. For AQ300 and AQ400 with auto-gain, filters 1 through 9 should be between 4 – 4.85 volts and 0.027 volts for filter 10. A **Lamp On/Off** button is also present for AQ400s and AQ300s. This button can be used to switch the lamp on and off.

These values can be adjusted using the Gain and Offset Potentiometers (if applicable) located near the lamp on the Photometer casing. For more information on how to adjust these settings, please see the Operator Manual, Customer Support Manual, visit the website at www.seal-analytical.com, or call SEAL Technical Support.

8.3.6 Diluter

Diluter allows the user to prime the syringe and test the operation of the syringe, various valves, and the wash pump.



Prime the Syringe with Water from the Water Reservoir

This allows the user to prime the syringe. Priming the syringe will pull water into the fluidic system from the water reservoir. To begin, select the desired number of primes (1 to 10) from the drop down menu and hit the **Start Primes** icon.



Note:

If the analyzer is set up for Cadmium reduction, a set of primes will be done on both the standard sample pathway and the cadmium reduction pathway after the **Start Primes** button is clicked.

Test Functions

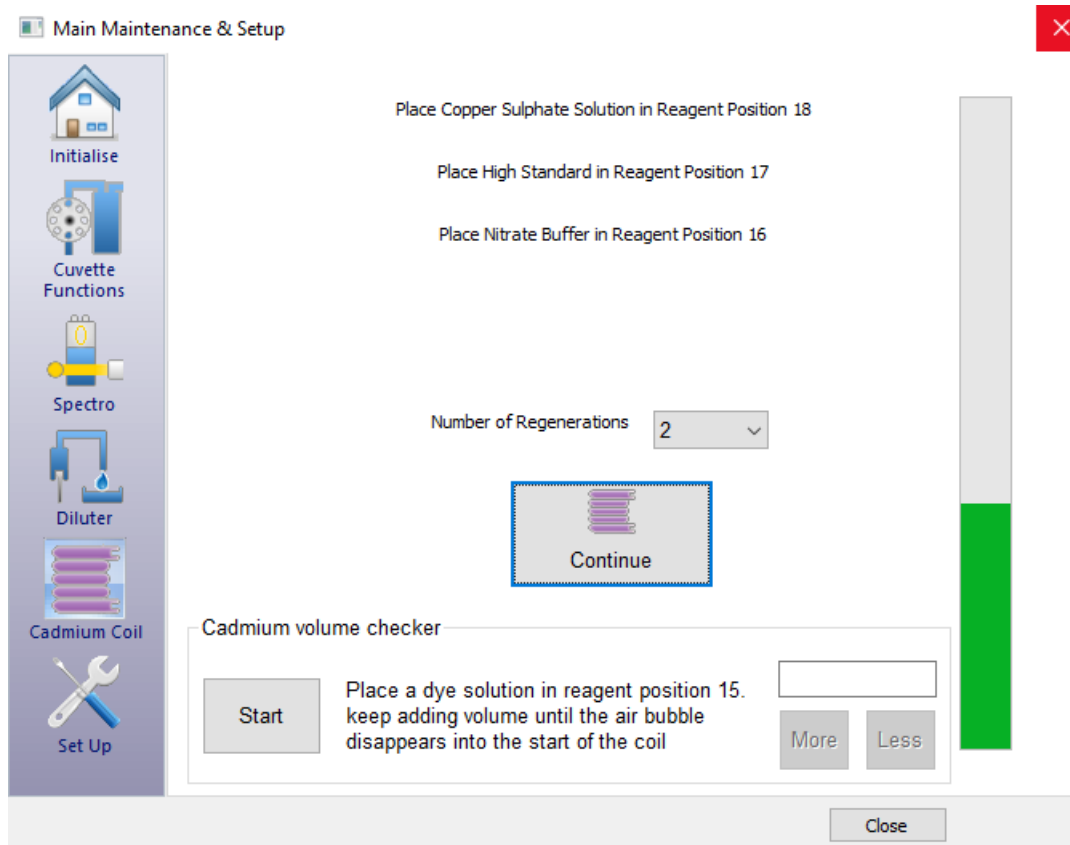
The left side of this menu allows the user to load the syringe with a specific amount of water from the water reservoir. Enter a volume in the **Diluter Volume** box and hit the **Load Syringe** icon. The syringe will load with the requested volume of water. Select the **Reset Diluter** icon to empty the syringe.

The right side of this menu allows the user to toggle various pumps and valves **On** or **Off**. The **Diluter Valve**, **Cadmium Valve**, **Wash Valves** and the **Wash Pump** can all be toggled on or off in this way.

To test the operation of the Probe Wash Assembly, an important test for the AQ300, AQ2 and AQ400, toggle **On** the **Wash Pump** AND the **Wash Valve**. The user can then place their finger under the Probe Wash Assembly and test for suction and water movement through the acrylic unit.

8.3.7 Cadmium Coil

Cadmium Coil allows the user to begin an automatic **Cadmium Coil Regeneration** through the software.



To begin, ensure that the Cadmium Coil is installed on the instrument properly and check to make sure that the three required reagents are in place. The number of automatic regenerations can be chosen as 1 or 2. Select the **Continue** icon to start the sequence.

Cadmium volume checker – This function is used to determine the volume required to reach the cadmium coil. This function can only be used when logged in as a SEAL Authorized User. Users should not use this function unless instructed by SEAL staff.

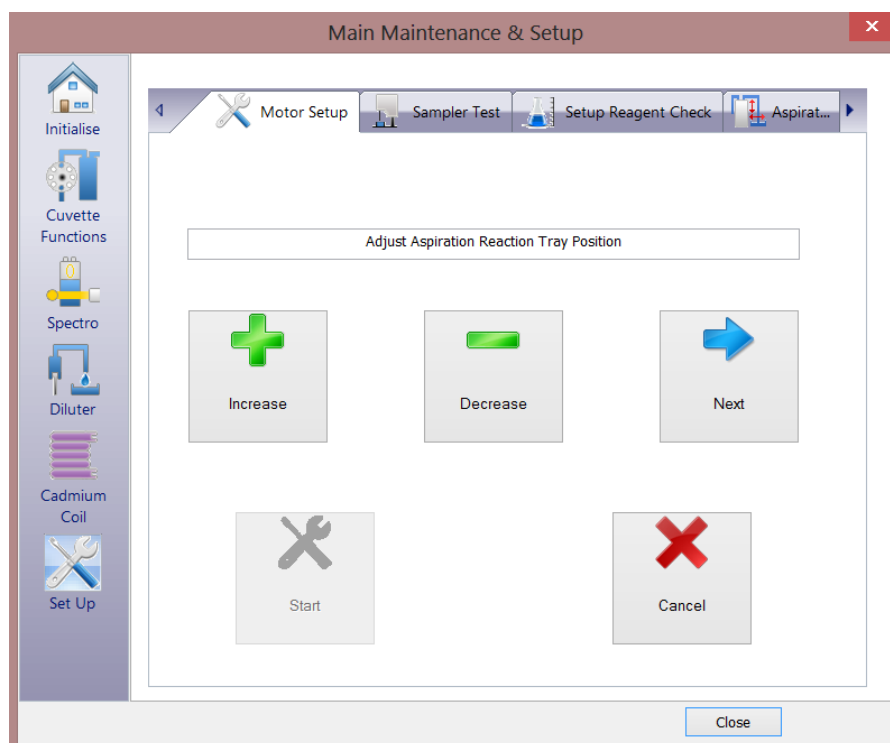


Note:

For more information on correctly installing and maintaining a Cadmium Coil, please refer to the Customer Support Manual, visit the SEAL website at www.seal-analytical.com, or contact SEAL Technical Support.

8.3.8 Set Up – Motor Setup

Motor Setup allows the user to align the Sample and Aspiration Probes. To enter this section of **Maintenance**, select the **Set Up** icon from the left and then select the **Motor Setup** tab. Before starting the **Motor Setup**, make sure the analyzer is fitted with a sample wheel, filled with fresh Reaction Segments, and has a Reagent Wedge in Position 1 for AQ2, Position 1, 21, and 26 for AQ400, or Position 9 for AQ300.



To begin the Motor Set Up, select the **Start** icon. The software will take the user through a series of alignment settings. The current setting will be displayed in the dialogue box. These settings can be adjusted by clicking the **Increase** or **Decrease** buttons (see **Adjustment Settings** below). Click the **Next** icon to continue to the next setting. **Cancel** will exit the Motor Setup without saving any changes. At the end of the Setup, the screen will read **Writing Adjustments to EEPROM**, and the Software will save all of the settings.



Note:

Exiting the Motor Set Up before the “Writing Adjustments to EEPROM” screen will not save any changes to the settings.

Adjustment Settings (AQ400 and AQ2)

Adjust Aspiration Reaction Tray Position – Center the Aspiration Probe over the fifth (center) reaction well by moving the reaction tray left or right. This is the rotation of the Reaction Tray.

Adjust Sampler Over Internal Sample Cup – Position the Sample Probe over the internal sample cup. The probe may not be perfectly centered, but it must be away from the sample wheel walls.

Adjust Sampler Over External Sample Cup – Position the Sample Probe over the external sample cup. The probe may not be perfectly centered, but it must be away from the sample wheel walls.

Adjust Sampler Over Internal Reaction Well – Position the Sample Probe over the internal Reaction Well. The probe should be approx. 4mm from the center wall dividing the 2 rows of reaction wells. The probe will be closer to the center wall than to the outer wall.

Adjust Sampler Over External Reaction Well – Position the Sample Probe over the external Reaction Well. The probe should be approx. 4mm from the center wall dividing the 2 rows of reaction wells. The probe will be closer to the center wall than to the outer wall.

Adjust Reaction Tube Under Sampler – Center the Sample Probe over the fifth (center) reaction well by moving the reaction tray left or right. This is the rotation of the Reaction Tray.

Adjust Sampler Over Wash Well – Center the Sample Probe over the internal sample wash well.

Adjust Sampler Over External Wash Well (AQ2 and AQ400 Only) – The Sample Probe should be adjusted so that it is halfway between the inner well and the outer wall.

Adjust Sampler Over Bottle – Center the Sample Probe over Reagent Wedge in Position 1.

AQ400 Continued

Adjust sampler over reagent 21 – Center the Sample Probe over Reagent Wedge located in position 21 (In the 6 position Reagent Tray).

Adjust sampler over reagent 26 – Center the Sample Probe over Reagent Wedge located in position 26 (In the 6 position Reagent Tray).

Adjust aspirator over internal well – Center the Aspiration Probe over the Internal Well.

Adjust aspirator height in internal well – To set the height correctly, first hit the (+) Increase button the Aspiration probe reaches “Soft Bottom” in the external well – the point where the reaction segment cannot move up and down, but can still move side to side. Check to determine if it is at the bottom between each (+) increase movement. To determine if it is on the bottom, try to lift up on the reaction segment. Next hit the (-) Decrease button three (3) times for the AQ400.

Adjust aspirator over external well – Center the Aspiration Probe over the External Well, by stepping the Horizontal movement of the Aspiration Assembly.

Adjust aspirator height in external well – To set the height correctly, first hit the (+) Increase button until the Aspiration probe reaches “Soft Bottom” in the external well – the point where the reaction segment cannot move up and down, but can still move side to side. Check to determine if it is at the bottom between each (+) increase movement. Next hit the (-) Decrease button three (3) times for the AQ400.

Writing Adjustments to EEPROM – Motor Setup is complete.

AQ2 Continued

Adjust Aspirator Over External Reaction Tube – Position the Aspiration Probe over the external Reaction Well. The Probe should be approx. 4mm from the center wall dividing the 2 rows of reaction wells. The Probe will be closer to the center wall than to the outer wall.

Adjust Aspirator Height in External Tube - To set the height correctly, first hit the (+) Increase button until the Aspiration Probe bottoms out in the external Reaction Tube (listen carefully for the probe slightly tapping the bottom of the reaction well). Next hit the (-) Decrease button two (2) times for the AQ2

Adjust Aspirator Over Internal Reaction Tube – Position the Aspiration Probe over the internal Reaction Well. The Probe should be approx. 4mm from the center wall dividing the 2 rows of reaction wells. The Probe will be closer to the center wall than to the outer wall.

Adjust Aspirator Height in Internal Tube – To set the height correctly, first hit the (+) Increase button until the Aspiration Probe bottoms out in the external Reaction Tube (listen carefully for the probe slightly tapping the bottom of the reaction well). Next hit the (-) Decrease button two (2) times for the AQ2.

Writing Adjustments to EEPROM – Motor Setup is complete.

Adjustment Settings (AQ300)

Adjust aspirator over external reaction well – Position the Aspiration Probe over the external Reaction Well. The probe should be approx. 4 mm from the center wall dividing the 2 rows of reaction wells. The probe will be closer to the center wall than to the outer wall.

Adjust aspirator over internal well – Position the Aspiration Probe over the internal reaction well. The probe should be approx. 4 mm from the center wall dividing the 2 rows of reaction wells. The probe will be closer to the center wall than to the outer wall.

Adjust aspirator height in external well – To set the height correctly, first hit the (+) Increase button until the Aspiration probe reaches “Soft Bottom” in the external well – the point where the reaction segment cannot move up and down, but can still move side to side. Check to determine if it is at the bottom between each (+) increase movement. Next, hit the (-) Decrease button four (4) times.

Adjust aspirator height in internal well – To set the height correctly, first hit the (+) Increase button until the Aspiration probe reaches “Soft Bottom” in the external well – the point where the reaction segment cannot move up and down, but can still move side to side. Check to determine if it is at the bottom between each (+) increase movement. Next, hit the (-) Decrease button four (4) times.

Adjust sampler over reagent 9 – Position the sample probe over the center of the reagent 9 wedge, or as close as the probe can get to the center in its arc of motion.

Adjust reagent tray under sampler – If the sample probe wasn't able to center over the reagent wedge, position the reagent tray under the sample probe to finish centering the sample probe under the center of reagent 9 wedge.

Adjust sampler over external wash well – The sample probe should be adjusted so that it is halfway between the inner well and outer well.

Adjust sampler over external reaction well – To set the alignment correctly, first position the sample probe over the middle ridge of the reagent segment. Then press the (+) Increase button five times.

Adjust reaction well under sampler – Position reaction carousel under the sample probe so that the sample probe is centered (right to left) above the fifth (center) reaction well.

Adjust sampler over internal reaction well – To set the alignment correctly, first position the sample probe over the middle ridge of the reagent segment. Then press the (-) Decrease button five times.

Adjust sampler over external sample cup – Position the sample probe over the external sample cup. The probe may not be perfectly centered, but it must be away from the sampler wheel walls.

Adjust sampler over internal sample cup – Position the sample probe over the internal sample cup. The probe may not be perfectly centered, but it must be away from the sampler wheel walls.

Writing adjustments to EEPROM – Motor setup is complete.

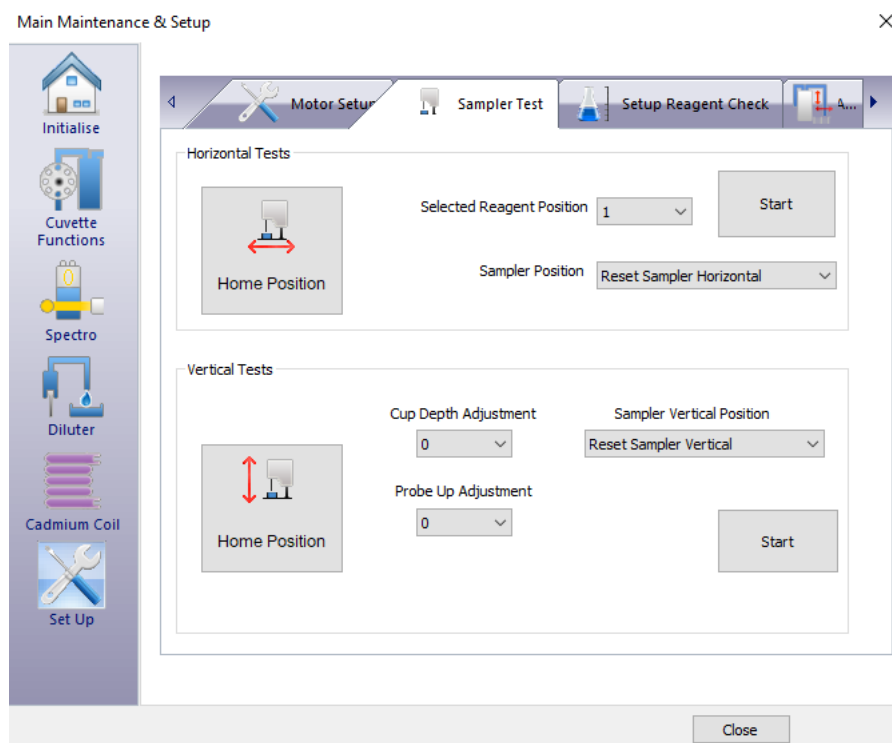
8.3.9 Set Up – Sampler Test

Sampler Test allows the user to test the operation of the Sample Arm. To enter this section of **Maintenance**, select the **Set Up** icon from the left and then select the **Sampler Test** tab.



Caution:

This section of the Maintenance menu should only be used by Advanced Operators.



Horizontal Tests

The **Horizontal Tests** section allows the user to rotate the Sample Arm to the selected location. These locations include the inner and outer Reaction Wells, inner and outer Sample Cups, inner and outer Sampler Wash Wells, and Reagent Bottles. The location can be chosen from the **Sampler Position** drop down menu. The sampler can be sent to any of the Reagent positions by selecting a position (1-15 for AQ2, 1-18 for AQ300 or 1-26 for AQ400) from the **Selected Reagent Position** drop down menu. Hit the **Start** button to move the Sampler to the selected position.

The **Home Position** icon will reset the Sampler Arm to its home position.



Caution:

Make sure that the Sample Probe is in its Vertical Home Position before selecting a Horizontal movement.

Vertical Tests

The **Vertical Tests** section allows the user to lower the Sample Probe to the selected location. These locations include the inner and outer Reaction Wells, inner and outer Sample Cups, Sampler Wash Wells, and Reagent Bottles. This location can be chosen from the **Sampler Vertical Position** drop down menu. Hit the **Start** button to move the Sampler to the selected position.

The **Home Position** icon will reset the Sample Probe to its upmost position.

Cup Depth Adjustment

Cup Depth Adjustment sets the depth of the probe in the Sample Cup. This may need to be adjusted when switching between the 1.2 and 2.0 mL Sample Cups. To test this, use the following steps:

- 1) With a Sample Cup in Cup Position #2 on the Sample Wheel, select **Reset Sampler Vertical** from the drop-down menu.
- 2) Select **Sampler down to Sample Cup** from the **Sampler Vertical Position** drop down menu and hit **Start**. The Sample Probe will descend into the Sample Cup. Check that it is close to the bottom of the cup, but not touching.

If any adjustment is required, select a new value from the **Cup Depth Adjustment** drop down (in steps of 1) and repeat these steps. Select **Vertical Home Position** when complete. In order to save the new **Cup Depth Adjustment**, select **Sampler down to Sample Cup** from the **Sampler Vertical Position** drop down menu and hit **Start**. The new setting will now be saved to the analyzer.

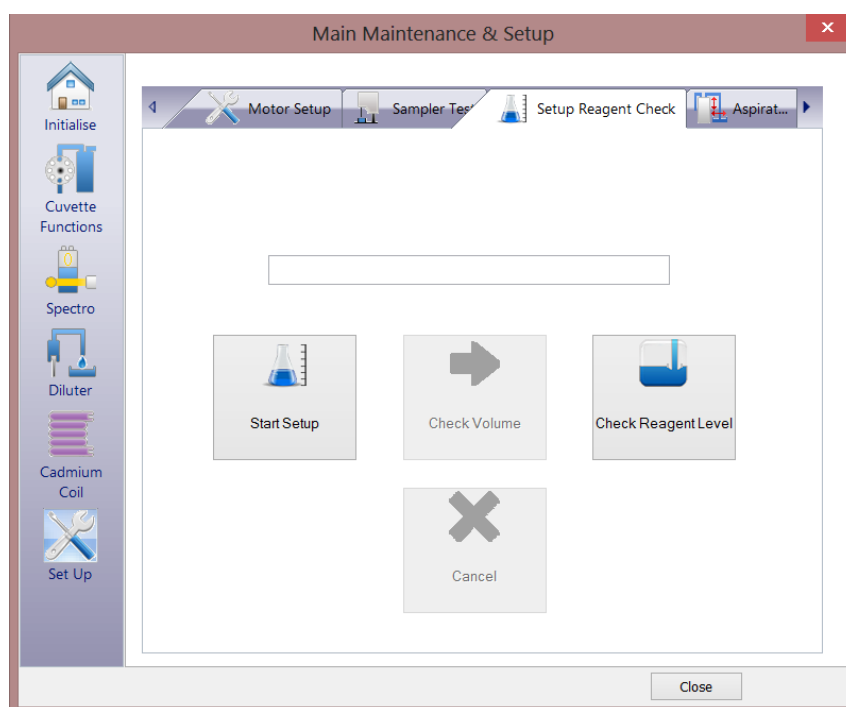
Probe Up Adjustment

To ensure that the Probe Wash Assembly is able to work properly, the Probe Up setting must be adjusted properly. The tip of the Sample Probe should protrude below the Probe Wash Assembly by approximately 0.5-1.0 mm when the Sampler is in the “Up from Cup” vertical position. To test this, select **Sampler Up from Cup** from the **Vertical Position** drop down menu and click **Start**.

If the Probe Up setting needs to be adjusted, select a new setting from the **Probe Up Adjustment** drop down menu (1 step at a time). Test again by selecting **Sampler Up from Cup from the Vertical Position drop down menu**. In order to save the new **Probe Up Adjustment**, select **Sampler Up from Cup** from the **Sampler Vertical Position** drop down menu and hit **Start**. The new setting will now be saved to the analyzer.

8.3.10 Set Up – Setup Reagent Check (AQ2 and AQ400, some AQ300s)

The AQ2, AQ400, and some AQ300s (optional addition to AQ300s) are equipped with a Reagent Level Detector on the Sample Probe. This section of the maintenance menu allows the user to calibrate this detector to calculate reagent volumes accurately. To enter this section, select the **Set Up** icon on the left side of the Maintenance screen followed by the **Setup Reagent Check** tab at the top.



Start Setup

Clicking this icon will allow the user to calibrate the Reagent Level Sensor. Once the user clicks the **Start Setup** icon, they will be prompted to place 5 mL of water into a Reagent Wedge located in Position 8. Once completed, click the **Check Volume** icon. The Sampler will move to Reagent 8 and note the volume. Continue with 10, 20, 30, and 40 mL in the Reagent Wedge (the volume is cumulative). After the final reading, the Software will display **Saved New Calibration**. Hit the **Cancel** button to exit the Calibration mode at any time (changes will not be saved).

Check Reagent Level

This icon allows the user to check the volume of any reagent placed in reagent position 8. Place the reagent in Position 8 and select the **Check Reagent Level** icon. The volume will be displayed in the Dialogue Box.

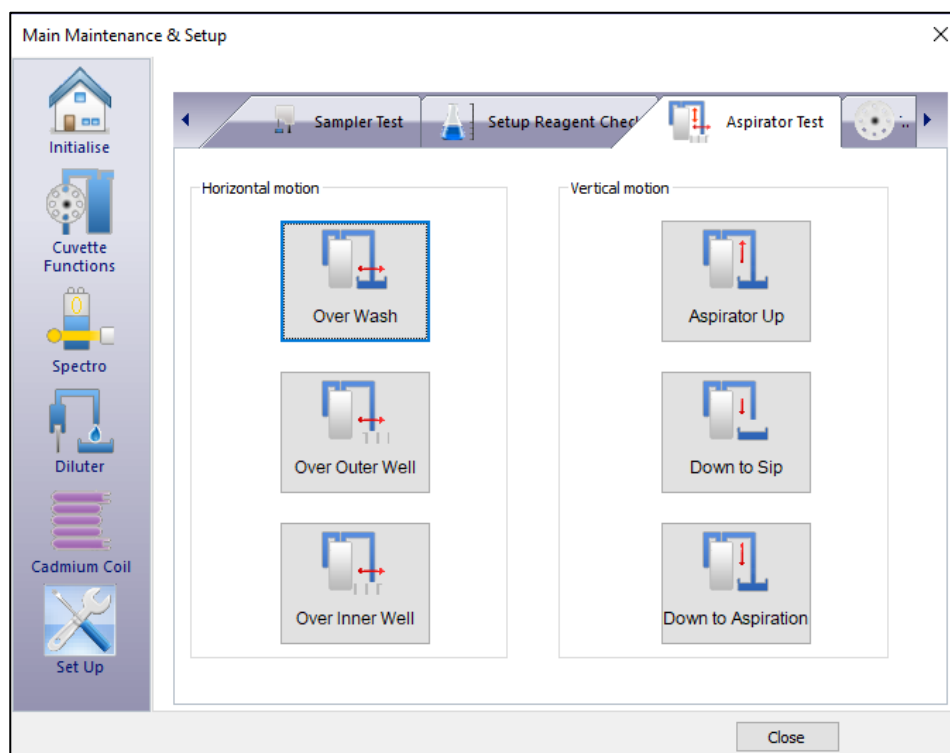
8.3.11 Set Up – Aspirator Test

Aspirator Test allows the user to test the operation of the Aspiration Arm. To enter this section of **Maintenance** select the **Set Up** icon from the left and then select the **Aspirator Test** tab at the top.



Caution:

This section of the Maintenance menu should only be used by Advanced Operators.



Horizontal Motion

This set of icons drives the horizontal positioning of the Aspiration arm. The user can move the Aspiration Probe over the Wash Station, over the Outer Reaction Well, and over the Inner Reaction Well by selecting the designated icon.



Caution:

Make sure that the Aspiration Probe is in its Vertical Up position before selecting a Horizontal movement.

Vertical Motion

This set of icons drives the vertical positioning of the Aspiration Probe. The user can move the Aspiration Probe down to sip or aspirate from wherever it has been positioned horizontally. The **Aspirator Up** icon resets the probes vertical position.

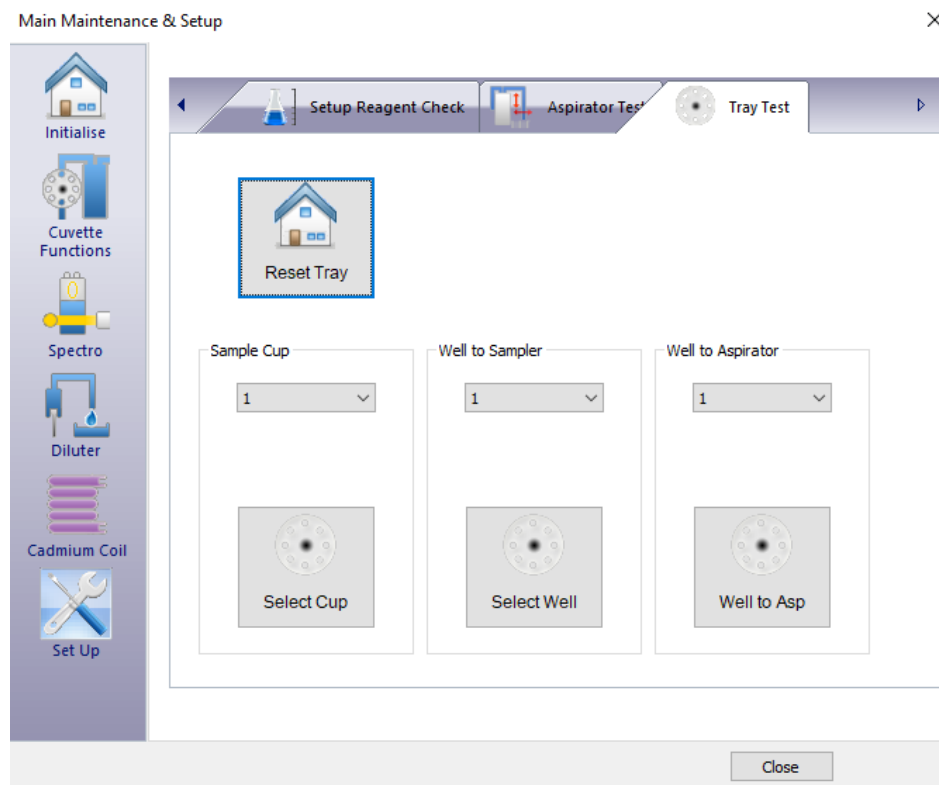
8.3.12 Set Up – Tray Test

Tray Test allows the user to evaluate the operation of the Reaction Tray. To enter this section of **Maintenance** select the **Set Up** icon from the left and then select the **Tray Test** tab at the top.



Caution:

This section of the Maintenance menu should only be used by Advanced Operators.



Reset Tray (Icon)

The **Reset Tray** icon will reset the Reaction Tray to its Home position.

Sample Cup

Select a **Sample Cup** number from the drop-down menu and hit the **Select Cup** icon. The Tray will rotate that Sample Cup into the sampling position.

Well to Sampler

Select a Reaction Well number from the drop-down menu and hit the **Select Well** icon. The Tray will rotate that Reaction Well into the sampling position.

Well to Aspirator

Select a Reaction Well number from the drop down menu and hit the **Well to Asp** icon. The Tray will rotate that Reaction Well into the aspirating position.

9 Operation Modes

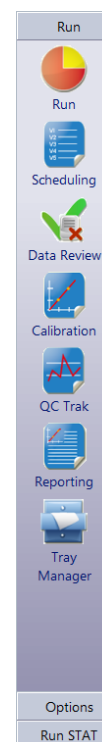
9.1 Introduction

The AQ Software includes three operation modes Standard, Touchscreen, and STAT. These are detailed in the sections below.

The manual up to this section describes how to operate the AQ Software in Standard mode. This section will go on to describe the differences between the three modes, and how to switch to these other Operation Modes.

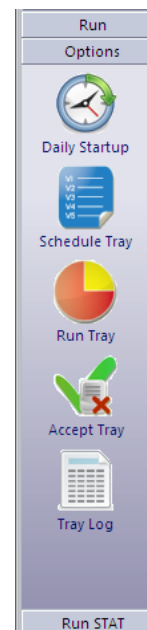
9.1.1 Standard Mode (Run)

Most users will use the AQ Software in Standard mode. This is the traditional view previously described in this manual.



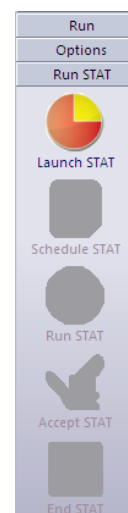
9.1.2 Touch-Screen Mode (Options)

Touch-Screen mode is designed to allow users to interact with the AQ Software through a touchscreen. Touch-Screen mode streamlines the AQ Software so the user can quickly schedule, run and accept trays. Since Touch-Screens are functionally equivalent to computer mice the Touch-Screen mode can be used without a touch-screen as a Simple View mode.



9.1.3 STAT Mode

STAT mode is initialized if the user wants to interrupt a running tray to run urgent samples.

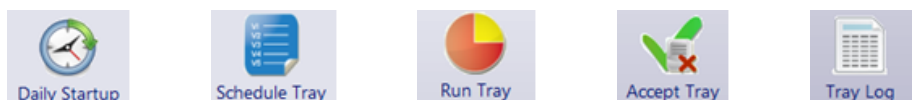


9.2 Touch- Screen Mode

The Touch-Screen mode in the AQ Software is fully-integrated into the existing User Interface Design. To switch to Touch-Screen mode click the **Options** button on the toolbar.

The Touch-Screen mode is a combination of specially designed dialog boxes and alternate ways to perform common tasks. These are all detailed in the following sections. It is important to note that tasks such as parameter entry still require a keyboard.

When the user switches to touch-screen mode the left-hand contents display is altered to the following five options:



9.2.1 Daily Start-Up

The **Daily Startup** button, also located under the **Maintenance and Setup** tab in Standard mode, is also located on the left side-pane in Touch Screen mode.

9.2.2 Schedule Tray

Start New Tray

The first difference in Scheduling is the dialog box used to select the tray. This has been especially designed for use with a touch-screen.

The **Tray Number** is automatically set to the next fully free number and the tray receives the default **Tray Name**. Users will be able to alter the name in **Scheduling**, as usual, if they wish. This will require use of a keyboard.

The **Reagent Set** can be selected by tapping the appropriate button.

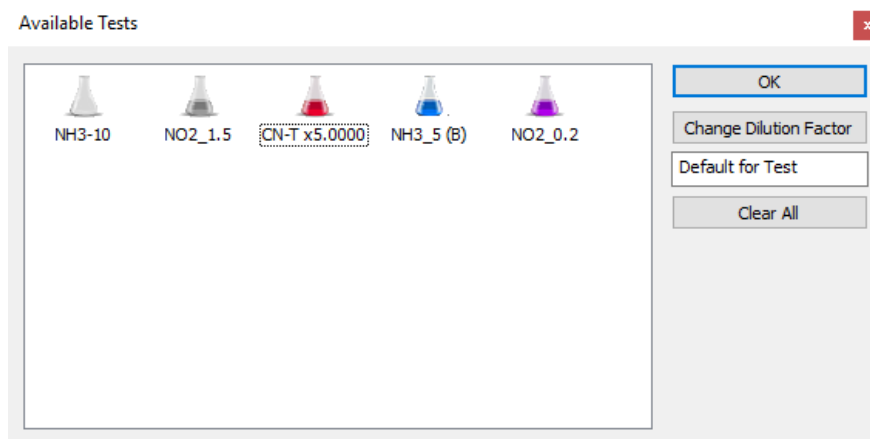
Click **OK** to continue.

Schedule Tests

Since it is too difficult to schedule tests in the standard manner with a touchscreen, an alternative method is provided. Above each selection box (Available Tests, Calculated Tests and Profiles) there is a small button like the one shown below:



This button will open a new dialog box designed to allow test scheduling with a touch-screen.



The user should tap the icon corresponding to the test they wish to schedule. This will change the icon to reflect it has been scheduled. Further tapping will cycle through five options that reflect the five allowable states for each test.

- Flask is **empty**. This means the test is not scheduled.
- Flask is **filled** with **grey liquid**. This means the test is scheduled normally.
- Flask is **filled** with **red liquid**. This means the test is scheduled with an automatic dilution. The dilution factor will be displayed below.
- Flask is **filled** with **blue liquid**. This means the test will be blanked.
- Flask is **filled** with **purple liquid**. This means the sample will be both automatically diluted and blanked.

The user can access the normal dilution factor dialog box by clicking the **Change Dilution Factor** button.

The entire test set can be unselected by clicking **Clear All**.

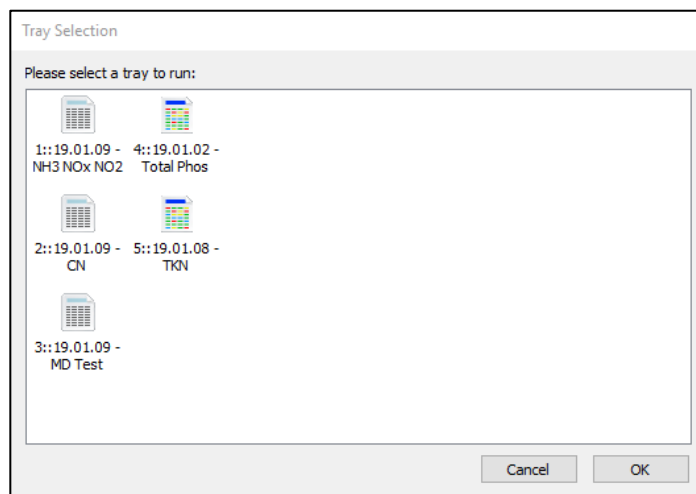
Calculated Tests and **Profiles** are selected in a similar way.

Click **OK** to return to the **Scheduling** screen.

9.2.3 Run Tray

Run a New Tray

When a user runs a tray in Touch-Screen mode they will be presented with a new tray selection dialog box. This has been designed to simplify selection when using a touch-screen.



This dialog box distinguishes between trays that have never been run and trays that have been run but not accepted with two different icons. The icon showing **colored entries** denotes a tray that has already been fully or partially run. An icon with **black entries** denotes a newly scheduled tray.

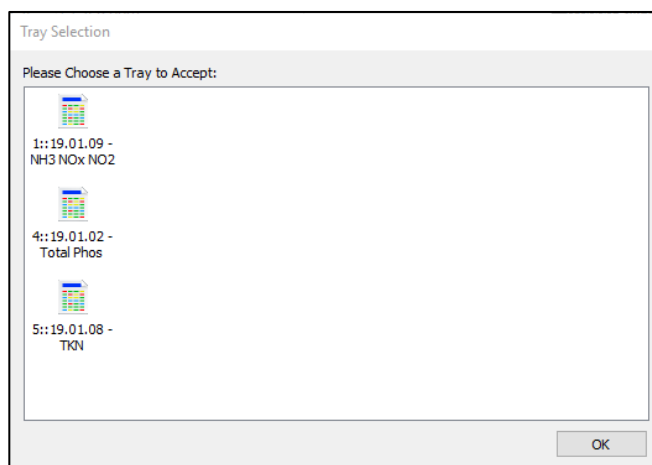
Automatic Reagent Level Sensing

Touch-screen users should set their analyzer to automatically detect the level of their reagents (except on AQ300s without a reagent level sensor). This will allow them to bypass the volume entry screen. It is not possible to enter reagent volumes without a keyboard.

9.2.4 Accept Tray

Accept a Tray

When a user tries to accept a tray in Touch-Screen mode they will be presented with a new tray selection dialog box. This has been designed to simplify selection when using a touch-screen.



This dialog box will not show trays that have never been run, only trays that have been run but not accepted.

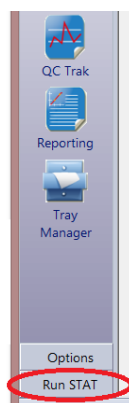
9.2.5 Tray Log

Tapping this button will load the real-time log of the currently running tray. The dialog box will not be displayed if there is no tray running.

For more information on tray logs see the section 7.1 *Tray Logs*.

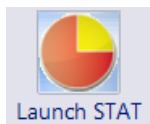
9.3 STAT Mode

The STAT mode in the AQ Software is a controlled user environment that allows the user to interrupt a running tray and run urgent samples. To Switch to STAT mode click the **Run STAT** button on the toolbar.



STAT mode will take the user through the process of stopping the running tray and scheduling the new tray.

9.3.1 Starting STAT Mode

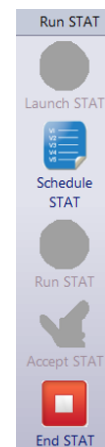


Double-click the icon **Launch STAT** to enter STAT mode. A dialog box will ask you if you are sure you want to continue. If you click **Yes** here, any running trays will start an Orderly Stop. This will continue in the background while you proceed.

9.3.2 Scheduling the STAT Tray

After initializing STAT mode, the user is required to schedule the STAT tray. This is the schedule of the urgent samples. The tray name and number are automatically selected here, as in Touch-Screen mode.

The process of scheduling a tray is the same as normal. Click the Save button on the toolbar when you have finished scheduling the STAT tray.



9.3.3 Waiting for the Running Tray to Stop

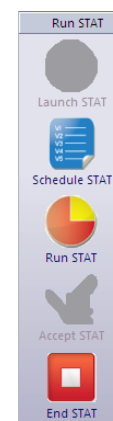
The STAT tray can be run when the following two conditions have been satisfied:

- 1) The currently running tray has completely stopped.
- 2) A scheduled STAT tray has been saved.

Once this is done, the **Run STAT** button will become enabled. Click the Run STAT and a dialog box will appear asking if you wish to run the STAT tray now.

Remove the tray currently on the analyzer and replace it with the tray prepared for the STAT sample cups. Click **Yes** to proceed.

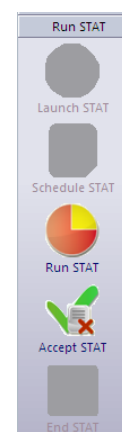
If **No** is clicked, the ability to switch back to Scheduling screen to make further changes to the tray is available. Click the **New** button on the toolbar to run the tray.



9.3.4 Accepting the STAT Tray

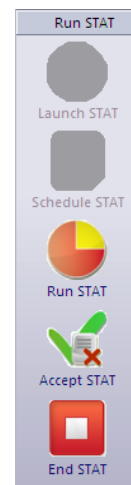
Once the tray has started running, the Acceptance screen is available. The user will not be able to choose a tray to view in the Acceptance screen; the STAT tray will be automatically selected.

The user is free to switch between the Acceptance screen and the Run screen at will. The user should accept the tray when it is completed.



9.3.5 Ending STAT Mode

The user can exit STAT mode at any time by double-clicking the **End STAT** button. In addition, the user will have to manually exit STAT mode when the STAT tray has finished running. Exiting STAT mode will not stop any running trays or restart any stopping trays.



9.4 Manual Read (AQ400 Only)

The AQ400 has the option to perform direct read tests, allowing the user to insert a standard cuvette into the detector for an absorbance reading. The user can then use the AQ software to set up calibration curves, controls, etc.... for automatic data management.

A user must install the Manual Read cuvette onto the AQ400 before using this feature. For more information on installing the Manual Read cuvette, please reference section 5 of the AQ400 Customer Support Manual.

9.4.1 Setting up a Manual Read Test

Just like any other test on the AQ400, a Manual Read test will need a set of Test Parameters entered by the user. To set up the test parameters, go to **Maintenance and Setup** and select the **Tests** icon. If you are unfamiliar with the Test Parameters window, see section 4 for more details.

Click the **New** button to begin a new set of test parameters. Enter the desired Test ID (software will automatically fill in the next available Test ID, but this can be changed if necessary), Test Name, Short Test Name (7 characters or less). Click **OK**.

Most of the fields in Main Parameters tab of the Test Parameters window will be greyed out. First, type in the **Units**, **Decimal Places**, and **Test Type** fields. The **Test Type** field must be set as **Manual Read Test**. Scroll down to the **Wavelength** field and select the desired wavelength of the test from the dropdown. Set the **Polynomial Order** of the calibration curve to the desired setting.

Property	Value
Main Parameters	
Test number	5
Test name	COD
Short test name	COD
Units	mg/L
Decimal places	4
Test type	1. End Point
Calculation	1. End Point
Main Test Parameters	
Sample volume	2. Cadmium Reduction
Water volume	3. Olsen P
Diluent location	4. Sulphate Kinetic
Number of mixes	5. Sample Blank
Cuvette primes	6. Calculated Test
Cuvette washes	7. Manual Read Test
☐ Baseline on wash	
Cadmium reduction time	0
Cadmium flush volume	0
Measurement / Math	
Test type	
Type of chemistry	

Next, navigate to the **Calibration** tab of the Test Parameters window. Input the desired settings for the **All Standards** section of this tab. Only the **Correlation Coefficient Limit** and **Exclude the Blank (S1)** tabs can be filled in here (see section 4.5 for more details).

Scroll down to the **Manual Standards** section and input the desired calibration standards as described in section 4.5.4. Auto-Calibrations are not available for Manual Read tests.

Navigate to the **QC Pro** tab to add in control information. Only the **Controls** section, Duplicates, and Manual Spike/Spike Duplicates fields are available in this tab, as no automated controls are necessary. See section 4.6.12 for more information on how to set up controls.

Once these sections have been filled out, click **OK**. A dialogue box will ask if the parameters should be saved; click **Yes** to save or **No** to cancel all changes.

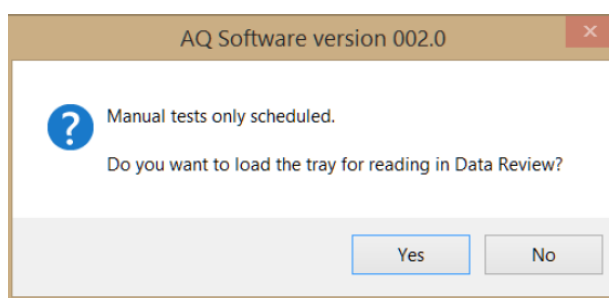
9.4.2 Scheduling a Manual Read Test

Scheduling a manual read test is done in nearly the same manner as any other test. For more details on how to schedule a general test, see section 5.2.

If a calibration curve is to be run, schedule the calibration standards first. Then schedule the desired controls and samples to be analyzed.

To assign the Manual Read test, highlight the controls, standards, and samples to be analyzed and select the Manual Read test from the **Calc'd Test** box towards the top of the screen.

Save the schedule using the **Save** button in the **Scheduling** toolbar. A pop-up box will appear:



Select **Yes** and the tray will be saved. Select **No** if you still wish to make changes to the schedule before opening the tray in Data Review.

9.4.3 Starting a Manual Read Test

To begin the Manual Read Test, double-click on **Data Review** and select the desired tray. When the tray is open, click the test name at the top of the results column to highlight the whole column. Then click the **Manual Read** button located in the Data Review toolbar. A new pop-up screen will appear.

Reading Test: Manual Read Test

Selected Wavelength : 405

	Cup	Type	Sample ID	Sample Details	Manual Read Test
Current sample	2	S2	Standard 2		0.000
Previous sample	1	S1	Standard 1		0.281

Previous Read Next

Water baseline OK Add sample

If a Water Baseline is needed, first insert a clean cuvette containing DI water into the Direct Read cuvette holder. Place the Direct Read Cover over the cuvette holder, then click the **Water Baseline** button in the bottom-left corner of the Manual Read pop-up window. The baseline voltage will be displayed directly above the **Water Baseline** button.

Remove the DI water cuvette and insert the cuvette for the first standard, control, or sample shown in the list. Replace the cover over the direct read cuvette, then hit the **Read** button. The sample will be read and the Direct Read pop-up window will move on to the next standard, control, or sample automatically. The sample that was read will then be shifted down to the **Previous sample** row. If the result is for a calibration standard, the result is reported in absorbance units, if the result is for a control or sample, the result is reported in concentration (as defined in **Test Parameters**).

Continue this process until all results are obtained.

When all of the calibration standards have been analyzed, a pop-up will appear with the message “**Calibrate tests?**” If **Yes**, is selected, the software will create a calibration curve using the current results of the calibration standards. If **No** is selected, nothing will happen and the calibration standards can be re-analyzed.

The **Previous** and **Next** buttons can be used to move from sample to sample. The results can be re-read by using the buttons to move to a sample, inserting the sample into the direct-read cuvette, and clicking **Read** again. Only the final results will be reported.

9.4.4 Other Functions with a Manual Read Test

All other functions in the software relating to test data is the same for both Manual Read tests and generic tests.

Appendix A – Software License Agreement

General Terms and Conditions for the Transfer of Software

1. Subject matter of agreement

- 1.1 You are granted in accordance with this agreement a right to use the programs delivered by SEAL Analytical and the documentation delivered with them. The right of use is non-exclusive and not transferable.
- 1.2 You shall be responsible for installing the programs in accordance with the instructions for installation that form part of the items delivered.
- 1.3 The choice of the program and advice concerning the use intended by you are no more a part of this agreement than instructions, training and other technical assistance. You alone shall bear the risk of selecting the programs and their suitability for the intended use.

2. Performance and function

- 2.1 The performance and function of the programs delivered is determined by the production specifications in force upon the conclusion of the agreement. Any other agreements must be made in writing in order to be valid. Any adjustments to the programs to be made to meet customers' requirements shall be agreed on in a separate agreement. SEAL Analytical does not assume any liability for the function of the programs under special conditions of use (that are not customary in this branch of business).

3. Granting of a license

- 3.1 This transfer agreement entitles you to use a copy of the program acquired from SEAL Analytical at any time on one single computer at the most. If you intend any additional use at the same time, you must obtain an additional license. Even storing the software in a computer in an intermediate storage or fixed storage constitutes use of the software. Storing in a network server to be used solely in other computers does not constitute use. You are responsible for ensuring by suitable working instructions that the software is used only on the scale agreed upon.
- 3.2 Insofar as the software is not provided with a technical copying protection device, you may make a backup copy only for filling purposes or file the original disk as a backup copy if the software is loaded in a fixed storage.
- 3.3 If the software is delivered to you in various media (e.g. 3 1/2 inch and 5 1/4 inch disks), you may use in each case only as many copies of the software as you have concluded contracts.
- 3.4 You may neither rent out nor lend out the software. You may not transfer the rights of use to third parties either.
- 3.5 Reverse engineering, decompiling and disassembling of the software shall only be permitted if this is absolutely

essential to obtain the necessary information to bring about the interoperability of an independent computer program with other programs and this information has neither been published nor is otherwise ease of access and you have not received this information from SEAL Analytical on request. Only on these conditions will SEAL Analytical agree to your decompiling within the framework of § 69e Copyright Act (UrhG). In this case you shall inform SEAL Analytical without undue delay which programs you will decompile. Our rights under § 69e UrhG shall not be affected. For access to this decompiling information SEAL Analytical shall receive an appropriate remuneration from you.

- 3.6 Reproducing written documents, e.g. user manuals and other parts of the documentation, is not permitted.
- 3.7 You shall urge all users of the software by standing instructions to observe the proprietary rights of SEAL Analytical and to comply with the above restrictions of the license.

4. Warranty

- 4.1 SEAL Analytical warrants that the programs delivered have the functional and performance characteristics contained in the product specifications in force upon conclusion of the agreement.
- 4.2 The warranty claims shall expire six months after the date of the delivery of the programs to you.
- 4.3 If during this period the programs prove to be defective, which impairs the value and fitness not just insignificantly, SEAL Analytical shall remedy the defects without undue delay without any charge after receipt of the notice of defects. A requirement of the rectification of defects is that the effects of the defects are reproducible, have been sufficiently described by you and the defects were notified to SEAL Analytical without undue delay. Warranty claims shall be satisfied by rectification of defects, which at the option of SEAL Analytical may also be performed by delivering a new program version free of charge or, until delivery of such a program version, in the form of a temporary correction of defects.
- 4.4 If SEAL Analytical has let lapse a reasonable extension for the rectification of defects through own fault, you may demand a rescission of the agreement. This right shall also exist in other cases of failure to rectify defects.
- 4.5 SEAL Analytical shall be liable for further claims, in particular for compensation of damage that has not been caused to the programs themselves, only in accordance with item 5.

5. Liability of SEAL Analytical

- 5.1 SEAL Analytical shall be liable - on whatever legal grounds - in accordance with legal provisions and the provisions of this agreement only for damage caused intentionally.
- 5.2 SEAL Analytical shall also be liable in case of gross negligence of the proprietor or managerial employees as well as in case of culpable infringement of essential contractual duties.
- 5.3 In case of culpable infringement of essential contractual duties SEAL Analytical shall be liable - except in cases of intent and gross negligence of the proprietor and managerial employees - only for the damage that is typical of the agreement and reasonably foreseeable.
- 5.4 Not affected thereby is liability under the Product Liability Act for personal injury and material damage to privately used objects and liability in the event that expressly warranted characteristics are lacking, if the purpose of the warranty was to cover the customer against the damage occurred.
- 5.5 SEAL Analytical shall not be liable for the recovery of data.

6. Rights of third parties

- 6.1 You shall inform SEAL Analytical without undue delay if you are held liable by third parties for alleged infringements of proprietary rights. In such case SEAL Analytical may, at its option, either procure at its expense the right of use of the programs or freely arrange these proprietary rights or replace them by an alternative comparably efficient program. SEAL Analytical may also request you to defend yourself against the infringement of the proprietary rights. Insofar as you leave it to SEAL Analytical to take the necessary defensive measures and conduct any negotiations about a settlement, SEAL Analytical shall assume the costs imposed on you and amounts of compensation.
- 6.2 If it is not possible by reasonable means to take the measures described in 6.1 each contracting party may terminate this transfer agreement without notice. In this case SEAL Analytical shall be liable for the damage caused to you by the notice of termination, our liability being limited in terms of amount to the charge for the use agreed on for one year or the single royalty of the

subject matter of the agreement. Further, item 5 shall apply.

- 6.3 We shall no longer be liable insofar as claims of third parties are based on the fact that the program delivered was changed by you yourself or used under other conditions than the specified conditions of use.

7. Return of the program

- 7.1 If the program delivered with this agreement is exchanged in whole or in part within the framework of the warranty or on account of infringement of rights of third parties, you shall be obligated to destroy the first edition of the program, furnishing proof of such destruction, or to return the same to the licensor.

8. Termination of the agreement

- 8.1 You may terminate the agreement, insofar as it is of unlimited duration, in whole or in part with six months' notice to the end of the month. If a single royalty was paid for the delivery of the programs, this shall not be reimbursed.
- 8.2 The right to terminate this agreement without notice in accordance with this agreement and for cause shall not be affected thereby.
- 8.3 Upon termination of the agreement you shall be obligated to return to SEAL Analytical the original of the program concerned as well as well copies and partial copies and changed copies or copies connected with other program materials, or to destroy the same, furnishing proof of such destruction. The same applies to program documentation and other documents delivered. Retention of a file copy for security purposes requires a separate written agreement.

9. Written form

- 9.1 Subsidiary agreements and amendments must be made in writing; this also applies to any amendment of this clause.

10. Jurisdiction

- 10.1 The courts of Hamburg shall have jurisdiction over all disputes arising from this contractual relationship.
- 10.2 German law applies.